H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics.

Michael Jhon Anas, Prince John B. Bundalan, John Lloyd Cantallopez, Jacinto E.Ronquillo Jr. and Reyniel Josh Rojo

College of Information Technology Education, PHINMA University of Iloilo, Philippines

Abstract — In order to address issues with manual sales recording, inventory monitoring, reporting, and multi-branch operations, this study describes the creation of H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics for High Island V. A Point of Sale (POS), inventory control, management dashboard, role-based access control, audit trail, automated reporting, and Simple Linear Regression predictive sales forecasting are all integrated into the system. H2Ops offers timely and structured information for tracking business activities across branches by centralizing operational data in a cloud-based platform. The study is important because it offers a digital solution that can enhance decision-making, data accuracy, and operational efficiency. The system facilitates more efficient sales and inventory management by automating sales analysis and reporting and decreasing reliance on manual record-keeping. All things considered, H2Ops shows how modernization and better management of multi-branch water refilling station operations can be supported by a cloud-based information system.

Formatted: Line spacing: Multiple 1.15 li

Formatted: Font: 9 pt

CHAPTER 1

INTRODUCTION

~~I.~~ ~~Introduction~~

1.1 Background of the Study

Water is fundamentally recognized as the most vital substance for the sustenance of human life, serving as an irreplaceable resource for biological health, domestic sanitation, and industrial progress. On a global scale, the discourse surrounding water has shifted from a matter of mere access to a matter of quality and safety. According to the (United Nations, 2015) Sustainable Development Goal 6 (SDG 6), ensuring the availability and sustainable management of water and sanitation for all is a critical global priority [1]. However, despite these international mandates, billions of people continue to face challenges regarding the safety of their drinking water.

In many developing nations, the rapid pace of urbanization has outstripped the capacity of municipal water systems. Public infrastructure, often aged and poorly maintained, is frequently prone to contamination from groundwater seepage, pipe leakages, and inadequate filtration. As a result, the "bottled water" and "refilled water" industry has seen an unprecedented surge. What was once considered a luxury product for the elite has become a daily necessity for the common citizen. This global trend toward purified water consumption marks a significant psychological shift in how modern societies view public utilities versus private commodities.

In the specific context of the Philippines, the water refilling station (WRS) industry has become a ubiquitous part of the urban and rural landscape. The history of water consumption in the country is marked by a deep-seated public skepticism toward tap water. During the late 20th century, outbreaks of waterborne diseases such as cholera, typhoid, and various gastrointestinal infections led to a national realization that the traditional "piped-in" water provided by local utilities was often insufficient for direct consumption without additional boiling or treatment.

—Consequently, the early 2000s saw the birth of the local water refilling industry. These stations, typically small-to-medium enterprises (SMEs), operate by sourcing water from municipal lines and subjecting it to a rigorous multi-stage purification process—including sediment filtration, carbon block treatment, reverse osmosis, and ultraviolet (UV) sterilization. The Philippine Department of Health (DOH, 2017) regulates these establishments through the Philippine National Standards for Drinking Water (PNSDW) [2]. Today, the WRS industry is not just a commercial sector; it is a critical pillar of the country's public health infrastructure, providing millions of households with an affordable alternative to expensive branded bottled water.

—High Island V represents the modern evolution of the Philippine water refilling station. Starting as a single-unit operation, the company has successfully expanded its reach into multiple branches, strategically positioned to serve diverse customer bases. High Island V distinguishes itself by offering a dual-product model: standard purified water for general consumption and alkaline water for customers seeking specific health and pH-balancing benefits.

—As High Island V transitioned from a single-location business to a multi-branch enterprise, the complexity of its daily operations increased exponentially. In the early stages of a small business, a "hands-on" approach by the owner is often sufficient to monitor sales and inventory. However, when a business expands across different cities and provinces, the physical presence of the owner at every location becomes impossible. High Island V now faces the "growing pains" common to successful SMEs: the challenge of maintaining consistent operational standards and financial transparency across geographically dispersed locations.

—Despite its impressive commercial growth, High Island V's administrative foundation remains rooted in traditional, manual methods. Currently, the company relies heavily on paper-based record-keeping. Sales are recorded in physical logbooks, receipts are handwritten, and inventory levels are checked through manual counting at the end of every

Formatted: Font: Bold
Formatted: Centered, Line spacing: Multiple 1.15 li
Formatted: Font: 13 pt, Bold
Formatted: Centered, Line spacing: Multiple 1.15 li, No bullets or numbering
Formatted: Font: Bold
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Font: 10 pt

business day. While these methods served the company well in its infancy, they have become a significant liability in the digital age.

—Manual record-keeping is inherently prone to human error. In the heat of a busy afternoon shift, staff members may miscalculate a total, forget to record a delivery, or lose a physical receipt. Furthermore, paper records are fragile; they are susceptible to physical damage from the very environment they exist in—water. A single spill can erase a whole week's worth of financial data. From a management perspective, the biggest drawback of this manual system is the lack of "real-time" visibility. The owners of High Island V cannot instantly see the performance of a specific branch. Instead, they must wait for physical reports to be compiled and sent, leading to a "management lag" where problems are often discovered weeks after they occur.

—One of the most pressing issues in the water refilling industry is the management of reusable containers, commonly referred to as "gallons." These containers represent a significant capital investment for the business. In a manual system, tracking which customer has which gallon, or how many empty containers are currently in a delivery truck, is a logistical nightmare. High Island V has experienced instances of "inventory shrinkage," where containers are lost or go unrecorded, leading to financial losses.

—Moreover, without a centralized digital system, "financial leakage" becomes a constant risk. Without an automated audit trail, it is difficult to verify that every gallon filled at the station corresponds to a recorded sale. Discrepancies between the amount of water processed and the amount of revenue generated are hard to track manually. This lack of accountability not only affects the bottom line but also creates an environment of uncertainty for both the management and the employees.

—The Fourth Industrial Revolution (Industry 4.0) has brought about a democratization of technology. Sophisticated Enterprise Resource Planning (ERP) systems, which were once only affordable for multinational corporations, are now being adapted for SMEs through cloud computing and web-based applications [3]. For a business like High Island V, digital transformation is no longer an option—it is a survival mechanism.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

-Cloud-based systems offer the advantage of accessibility and scalability. By moving data from a physical notebook to a secure online server, a business owner can monitor operations from anywhere in the world using a smartphone or a laptop. This shift from "local" to "cloud" allows for a centralized management style that is essential for multi-branch operations. It eliminates the need for expensive on-site servers and IT staff, as the infrastructure is maintained by cloud providers, allowing the business to focus on its core competency: providing clean water.

—In response to these identified challenges, the researchers propose the development of H2Ops: A Responsive Cloud-Based Operations Management System. This system is specifically designed to address the unique workflows of a water refilling station. Unlike generic point-of-sale (POS) systems, H2Ops will integrate sales, inventory, and delivery tracking into a single, cohesive platform tailored for High Island V.

—The choice of technology for H2Ops—Python and the Flask framework—is strategic. Python is renowned for its readability and efficiency, making it an ideal choice for building robust business logic. Flask, a micro-web framework, allows for the creation of a lightweight yet powerful application that can be easily scaled as High Island V continues to grow. By utilizing a "responsive" design, the system will be equally functional on a desktop computer at the shop counter and on a mobile phone used by a delivery driver on the road.

—A critical component of H2Ops is its emphasis on security and data integrity. In a manual system, anyone with access to the logbook can alter records. H2Ops will implement Role-Based Access Control (RBAC), ensuring that only authorized personnel can view sensitive financial data or modify inventory records. Every transaction will create a digital footprint, providing the management with a clear audit trail [4].

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Beyond just "keeping records," H2Ops aims to provide High Island V with "business intelligence." By collecting data over time, the system can generate reports that reveal patterns—such as peak sales hours, the most profitable delivery

routes, and seasonal fluctuations in water demand. Instead of relying on "gut feelings" or guesswork, the owners of High Island V will be empowered to make data-driven decisions. They will know exactly when to order new filters, which branch needs more delivery staff, and where to open their next location.

—The journey of High Island V from a small local station to a multi-branch enterprise is a success story of Filipino entrepreneurship. However, to sustain this success in an increasingly competitive and digital world, the company must modernize its internal engine. The transition from paper to the cloud represents a significant milestone in the company's history.

—Through the implementation of H2Ops, the researchers aim to demonstrate that technology can be a powerful equalizer for SMEs. This project is not just about writing code; it is about providing High Island V with the tools to ensure that they can continue their mission of providing safe, clean, and affordable water to their community with greater efficiency and transparency. H2Ops will bridge the gap between traditional service and modern management, securing the future of High Island V in the digital age.

1.2 Statement of the Problem

General Problem Statement

The operational integrity of essential utility services, specifically in the water purification and distribution sector, serves as the primary engine for community health and business sustainability. For a multi-branch enterprise like High Island V, the management of sales transactions, inventory levels, and cross-regional reporting represents a complex triad of functions that must operate with absolute precision. In the context of the Philippine water refilling industry, where demand is high and profit margins depend on volume and efficiency, the ability to maintain real-time, accurate records is not merely a modern convenience—it is a fundamental requirement for survival and growth.

——Currently, High Island V operates within a landscape of "Management Blindness" caused by a reliance on decentralized, manual record-keeping. While the company has successfully expanded its physical footprint across various city and provincial locations, its administrative infrastructure has remained stagnant, anchored to paper-based logs and disconnected digital files. This reliance on legacy systems creates significant gaps in how the business monitors its financial health, manages its physical assets (such as water containers and filtration components), and protects its data from internal and external risks.

—The manual processes currently employed by High Island V introduce multiple points of failure that compromise the establishment's operational integrity. Handwritten sales entries are prone to mathematical inaccuracies, illegibility, and physical degradation. Inventory counts, often performed in a hurried environment, fail to capture the dynamic movement of stock—particularly the reusable gallon containers that constitute a significant portion of the company's capital. When management attempts to generate a performance report, they are forced to engage in a laborious process of reconciling scattered notebooks from different branches, leading to summaries that are often outdated before they are even finished.

Furthermore, the absence of an integrated, cloud-based system prevents the owners from exercising effective oversight over remote branches. In the current setup, an owner must physically visit a provincial branch or rely on trust-based reports to understand daily performance. This lack of real-time visibility prevents proactive management; instead, the business remains in a reactive state, only addressing problems like stockouts, mathematical errors, or financial discrepancies long after they have occurred. These challenges, if left unaddressed, pose a growing risk to High Island V's ability to remain competitive, manage its resources efficiently, and scale its operations in a digital-first economy.

Specific Problem 1: Inefficiency in Manual Record-Keeping and Transaction Processing

Formatted: Font: 13 pt, Bold

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Line spacing: Multiple 1.15 li

The most immediate operational hurdle at High Island V is the total reliance on manual methods for recording daily transactions. This practice consumes an enormous amount of staff time and creates a recurring cycle of errors that affect both walk-in retail sales and delivery operations. This "analog" approach is characteristic of many growing SMEs (Small and Medium Enterprises) that have outpaced their original management methods but have yet to adopt automated solutions. At each branch, sales are currently recorded in handwritten logbooks [3].

The researchers observed that during peak hours—typically during the morning rush and early evening when families return home—staff members are forced to multitask between filling containers, managing delivery riders, and handling cash. In this high-pressure environment, math errors in price calculation or total summation become inevitable. For instance, a staff member may miscalculate the price difference between a standard purified refill and a premium alkaline refill, or fail to account for a bulk discount accurately. These "micro-errors" lead to consistent, small-scale financial leakages that, when aggregated across multiple branches over a month, accumulate into significant financial losses.

Furthermore, the physical environment of a water refilling station is fundamentally hostile to paper records. The constant presence of water, high humidity, and wet hands means that logbooks are frequently subject to splashes or spills. Illegible handwriting caused by rushed entries or ink smudging can make end-of-day reconciliation impossible. In the worst-case scenario, a single accident or a misplaced notebook could erase weeks of financial history in an instant, leaving the business with no way to verify its accounts or track its growth.

To definitively resolve these operational vulnerabilities, H2Ops (A Responsive Cloud-Based Operations Management System) replaces fragile paper ledgers with a streamlined digital Point-of-Sale (POS) and sales management module. By implementing automated backend calculation logic within an intuitive c H2Ops automatically computes subtotals, itemized pricing (such as differentiating purified refills, alkaline refills, and container caps), and payment totals in real time—effectively eradicating human arithmetic mistakes and micro-financial leakages during high-pressure peak hours.

Furthermore, by persisting all transaction records within a secure, cloud-hosted relational database, H2Ops completely insulates vital business records from physical water spills, ink smudges, or lost logbooks while automatically attributing every sale to the authenticated cashier (user_id). This guarantees instantaneous, error-free end-of-day sales reconciliation and ensures that historical transaction data remains securely preserved and easily accessible across all branches.

Specific Problem 2: Decentralized Data and the Geographic Management Gap

High Island V's successful expansion into both city and provincial branches has introduced a "data silo" problem. In the current operational model, each branch operates as an independent island, with its critical data trapped within the physical walls of that specific location. Because there is no centralized, online database, the management lacks a unified, "bird's-eye view" of the company's total performance.

This decentralization creates a significant "Information Lag." In a competitive market, time is a critical resource. However, if an owner needs to know which branch is currently the most profitable or which location is experiencing a decline in sales, they cannot simply log into a dashboard. Instead, they must wait for branch managers to manually transcribe their paper logs into text messages, photos, or basic spreadsheets. These are then sent via messaging apps and manually compiled at the main office.

This process is not only slow but also introduces opportunities for data to be altered, lost in transit, or misrepresented during the transcription process. The inability to synchronize data across geographic locations prevents the company from operating as a single, cohesive unit. This geographic gap makes it difficult for the owners to implement uniform pricing strategies or allocate resources to the branches that need them most, effectively stalling the company's ability to scale.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Line spacing: Multiple 1.15 li

To bridge this geographic management gap and dismantle isolated data silos, H2Ops establishes a centralized, multi-tenant cloud database architecture (PostgreSQL via SQLAlchemy) that securely unifies all city and provincial branches under a single operational platform. By partitioning localized data through branch_id relational entities while providing a real-time global administrative dashboard, H2Ops completely eliminates the "Information Lag" associated with manual messaging and basic spreadsheet aggregation. Business owners and administrators can log into the system from any web-enabled device to immediately access an omnipresent, "bird's-eye view" of enterprise revenue, compare branch profitability, and monitor real-time sales metrics without waiting for end-of-day reports. Enforced by Role-Based Access Control (RBAC) middleware—which restricts local branch staff to their assigned location while granting administrators global oversight—H2Ops guarantees absolute data integrity, prevents unauthorized record alteration in transit, and empowers management to make agile, data-driven decisions that foster seamless geographical expansion [4].

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Specific Problem 3: Inventory Integrity and the Complexity of Container Tracking

In a water refilling business, the water is the product, but the primary circulating asset is the container—the heavy-duty, blue 5-gallon containers. High Island V faces a constant struggle with "Inventory Shrinkage," which refers to the loss of containers, seals, and caps that cannot be accounted for at the end of a business day.

The current manual system fails to maintain a "real-time" link between a completed sale and its corresponding inventory deduction. A major logistical challenge occurs during delivery operations. When a delivery rider takes twenty full containers to a residential area and returns with eighteen empties, the manual reconciliation of those missing containers is often neglected during busy shifts. Over time, the company loses track of hundreds of containers spread across its customer base.

Without an automated system that monitors every cap, seal, and gallon used, the business frequently suffers from "emergency stockouts." Management often realizes they are out of essential supplies—like alkaline filters or tamper-evident seals—only when the shelf is already empty. This forces them to either turn away customers, resulting in lost revenue and damaged reputation, or pay premium prices for last-minute, local supplies, which slashes their profit margins.

Specific Problem 4: Absence of Data Visualization and Decision Support

The raw data currently collected in High Island V's logbooks is "flat." It consists of long, chronological lists of numbers and names that provide no immediate insight into business trends or health. Management currently lacks the analytical tools required to turn these thousands of entries into "actionable intelligence." In the current state, owners and managers must spend hours, if not days, manually calculating totals to see if the business is improving.

Without automated reporting and data visualization—such as bar charts showing monthly growth or pie charts showing product distribution—the owners cannot easily identify patterns. For example, they may not notice that alkaline water sales are booming in the city branch but failing in the provincial branch until it is too late to adjust their marketing. This lack of visual decision support means that management must rely on "intuition" or "gut feelings" rather than hard evidence. In an increasingly professionalized industry, relying on guesswork puts High Island V at a strategic disadvantage. Larger franchises use sophisticated Point of Sale (POS) analytics to optimize their operations; without similar tools, High Island V struggles to compete on the same level of efficiency.

Specific Problem 5: Lack of Predictive Analytics for Sales Forecasting

A critical deficiency in High Island V's current operation is its inability to anticipate future demand. The business is currently "reactive" and purely historical; it knows what it sold yesterday, but it has no systematic, mathematical way to predict what it will sell next week. This makes the scheduling of staff and the ordering of supplies a guessing game.

-This study identifies a major need for a mathematical approach to forecasting. Without a model like Simple Linear Regression to analyze the last 30 days of sales data, the business cannot accurately predict the upcoming 7 days of demand [5]. This leads to two costly extremes. First, "Over-preparing," where too many staff members are scheduled and too much water is processed, leading to wasted labor and electricity. Second, "Under-preparing," where the station is overwhelmed by unexpected demand, leading to long wait times, exhausted staff, and lost revenue. The absence of a predictive module prevents High Island V from optimizing its resource allocation and maintaining a consistent service level.

Specific Problem 6: Security Vulnerabilities and Lack of Accountability

-Finally, the current manual system poses significant security and accountability risks. Physical logbooks offer zero data security. Anyone with physical access to the counter—whether an employee, a delivery rider, or an inquisitive customer—can potentially view the company's total earnings, customer contact lists, and sensitive expense records.

-Furthermore, there is no "Audit Trail" in a paper-based system. If a record is erased, crossed out, or a total is changed, there is no way for the owner to verify who made the change, when it was made, or if it was done for a legitimate reason. This lack of a secure, Role-Based Access Control (RBAC) system—where front-desk staff only see the sales screen while managers see the financial reports—creates an environment of low accountability.

-As the business grows, it also faces the challenge of protecting its digital future. The current manual setup lacks modern security features like One-Time Passwords (OTP) or encrypted user roles. This leaves the company's sensitive financial data vulnerable to tampering, unauthorized viewing, or internal theft, all of which could be mitigated by a robust, cloud-based management system with built-in security protocols.

1.3 Objectives of the Study

1.3.1 General Objective

The main objective of this study is to develop and evaluate H2Ops: A Responsive Cloud-Based Operations Management System with Automated Sales Analytics and Predictive Forecasting, a comprehensive operations management platform that will fundamentally improve operational decision-making at High Island V by providing efficient, accurate, and timely monitoring of sales, inventory, multi-branch performance, and overall operational productivity through sophisticated data visualization, automated report generation, intelligent stock alert mechanisms, and machine-driven sales forecasting using Simple Linear Regression. This objective encompasses both the rigorous technical development of a fully functional, robust software system and the empirical, evidence-based evaluation of its practical effectiveness, usability, and transformative impact within a real-world multi-branch water refilling business environment.

The achievement of this general objective necessitates a comprehensive, multi-dimensional approach that systematically addresses several interconnected dimensions of system development, deployment, and assessment. The technical dimension involves designing and building a software system that reliably captures, processes, stores, secures, and presents operational data with high fidelity, minimal latency, and robust error handling across multiple geographically dispersed branches. This includes architecting scalable cloud-based database schemas using MySQL, implementing responsive user interfaces through Flask and Bootstrap, ensuring data integrity through atomic transaction management, and establishing secure access controls through Role-Based Access Control (RBAC) and Email-based One-Time Password (OTP) authentication that protect sensitive business information from unauthorized access, data tampering, or accidental loss. The technical dimension further requires the system to handle real-time synchronization between city and provincial branches, ensuring that every sale, refill, and delivery transaction is reflected in the central database within seconds of occurrence, regardless of the originating location.

Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: (Default) Times New Roman, 13 pt, Bold
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

The operational dimension involves ensuring that the system effectively supports, enhances, and seamlessly integrates with High Island V's daily business processes, workflow patterns, and operational rhythms. This requires a deep understanding of how the water refilling business functions during peak demand periods (such as summer months and holidays) versus off-peak periods, how inventory flows from raw water sources through purification, filtration, and final container distribution, how sales transactions are processed across walk-in customers and home/office delivery channels, how reusable gallon containers circulate between the station and customers, and how management currently makes decisions with limited and often delayed information. The system must not merely digitize existing manual processes but optimize and streamline them, eliminating bottlenecks at the point of sale, reducing the cognitive load on staff during high-pressure service hours, and enabling operational practices that were previously impractical or impossible with paper-based records, such as instantly comparing the performance of all branches at the touch of a button.

The human dimension involves facilitating smooth user adoption, minimizing resistance to technological change, and ensuring that the system is accessible, intuitive, and genuinely valuable to its intended users, including the owners (Super Admins), branch managers, front-desk staff, and delivery riders. This dimension recognizes that technology adoption in small and medium enterprises (SMEs) often fails not because of technical deficiencies, but because of poor alignment with user needs, inadequate training, or disruption of established work practices.

The study, therefore, emphasizes participatory design approaches by involving High Island V staff during the development cycle, iterative user feedback during testing phases, and comprehensive training support to ensure that even employees with minimal computer literacy can confidently operate the system. The system interface is intentionally designed to be simple enough that a new employee can be trained to use it within a single shift, while still providing the depth required by management.

The evaluative dimension involves systematically assessing the system's performance against clearly defined criteria encompassing functional correctness, usability, reliability, performance under concurrent multi-branch load, security against unauthorized access, accuracy of the predictive forecasting model, and overall business impact—and documenting the results with methodological rigor for the benefit of both the immediate stakeholders at High Island V and the broader research community interested in information systems adoption within Philippine SMEs, particularly within the water refilling industry.

The evaluation will utilize established software quality standards, including the ISO/IEC 25010 framework, to ensure that the assessment is objective, replicable, and academically valid. By transitioning High Island V from its current reliance on manual, paper-based, and informal record-keeping methods to a centralized, technology-driven, cloud-hosted, data-centric platform, this study aims to address the persistent operational gaps that have historically limited management's capacity to oversee daily activities with accuracy, completeness, and efficiency.

These gaps include the inability to track real-time inventory levels of water containers, caps, seals, and filtration components; the difficulty of identifying sales trends and consumer demand patterns across different branches; the labor-intensive nature of manual report compilation that consumes hours of management time each week; the high risk of human error in handwritten data recording and arithmetic calculations; the absence of organized historical data needed for comparative analysis and demand forecasting; and the lack of a secure audit trail capable of identifying who performed which transaction at which branch and at what specific time.

This transition represents not merely a superficial change in tools or a simple automation of existing manual practices, but a fundamental transformation in how High Island V conceptualizes, collects, manages, analyzes, and acts upon its operational information. The shift from fragmented, paper-based, decentralized records scattered across multiple branch notebooks, handwritten receipts, and the informal memory of staff to a unified, digital, cloud-hosted centralized system enables entirely new possibilities for data integration across all functional domains, real-time access from multiple locations and devices, advanced analytical insight through aggregation and visualization, and evidence-based decision-making that was previously unavailable to the company due to information constraints.

Owners will be able to monitor sales while traveling, branch managers will receive instant alerts when stock falls below critical thresholds, and the staff will be able to process transactions faster and with greater accuracy than was ever possible with pen and paper. Furthermore, the introduction of Simple Linear Regression as a predictive forecasting tool represents a significant leap forward in operational planning.

Unlike traditional descriptive systems that only report on past events, the predictive component of H2Ops will analyze a rolling window of the last 30 days of sales data to generate a statistically grounded 7-day demand forecast. This forecasting capability will allow High Island V to optimize its production schedules, manage its inventory of consumables (such as caps and seals) more precisely, and allocate staff and delivery resources in alignment with anticipated demand rather than relying on guesswork or institutional intuition. This level of predictive sophistication, traditionally reserved for large corporations with dedicated data science teams, will be made accessible to a local SME through the H2Ops platform.

The successful development, deployment, and sustained operation of H2Ops is expected to serve as a practical, sustainable, and replicable solution to the long-standing operational challenges of High Island V, while simultaneously contributing to the broader academic and professional understanding of how small water refilling enterprises in the Philippines can strategically benefit from the adoption of cloud-based operations management systems, business intelligence tools, and predictive analytics. The study thus deliberately positions itself at the intersection of practical problem-solving and rigorous academic knowledge generation, seeking to produce both immediate, tangible operational benefits for High Island V and lasting, generalizable contributions to the literature on information systems implementation, technology doption, and digital transformation in small business contexts within developing economies, where resource constraints, limited technical expertise, and organizational capacity limitations present unique challenges that demand thoughtfully designed, contextually appropriate technological solutions.

Ultimately, this general objective is anchored on the conviction that even a small, locally owned water refilling business such as High Island V deserves access to the same caliber of digital tools, data-driven insights, and operational intelligence that large multinational corporations enjoy. Through H2Ops, this study aims to demonstrate that with careful design, appropriate technology selection, and a deep understanding of the operational realities faced by the business, it is possible to deliver a transformative management platform that not only solves immediate problems but also lays the foundation for the future expansion, modernization, and sustained competitive advantage of High Island V in an increasingly digital marketplace.

Specific Objective

To facilitate the successful realization of the general objective, the following specific objectives have been formulated to serve as the technical and operational roadmap for the development of H2Ops:

Specific Objective 1: Design and Implementation of a Centralized, Cloud-Synchronized Database Architecture

The first specific objective is to architect and deploy a robust, centralized relational database management system (RDBMS) using MySQL, hosted on a secure cloud-based server environment. This objective addresses the fundamental problem of "Data Silos" currently experienced by High Island V's geographically dispersed branches. The technical implementation involves designing a normalized database schema capable of handling high-frequency transaction data from multiple concurrent sources (city and provincial branches) while ensuring ACID (Atomicity, Consistency, Isolation, Durability) properties.

The realization of this objective requires the development of a synchronization logic that allows each branch to update a unified master ledger in real-time. By transitioning from localized, paper-based records to a cloud-hosted MySQL environment, the system will eliminate the "Information Lag" that currently hinders management. This centralized architecture is the foundational layer upon which all other system modules—such as sales and inventory—will be built. It ensures that the owners of High Island V possess a "single source of truth," where a sale recorded in a provincial branch is immediately visible and accounted for in the company's total financial health. This objective also

encompasses the design of data backup and recovery protocols to ensure that the business's historical data is protected against physical disasters or hardware failures that would otherwise destroy traditional paper logbooks.

Specific Objective 2: Development of an Automated Point of Sale (POS) and Transaction Processing Module

The second specific objective is to engineer a user-friendly, responsive Point of Sale (POS) module using the Flask framework and Python. This module is specifically designed to overhaul the manual transaction process, which is currently the primary source of financial leakage and human error at High Island V. The technical focus of this objective is on creating a streamlined "Front-End" interface that is optimized for speed and accuracy, even during high-pressure peak hours.

This POS module will feature a standardized product catalog for both purified and alkaline water, with built-in logic to handle various transaction types, including walk-in retail, home/office delivery, and bulk container cleaning services. By automating the calculation of totals, taxes, and customer discounts, the system will eliminate the need for manual math, thereby preventing the "micro-losses" associated with arithmetic mistakes. Furthermore, the module will include a digital receipt generation feature, which provides customers with professional, accurate billing while simultaneously creating a permanent digital record of the sale. The "Responsiveness" of this objective ensures that staff can process transactions using any device—be it a desktop computer at the main counter or a mobile tablet for delivery tracking—thereby increasing the overall throughput and service quality of the station.

Specific Objective 3: Engineering a Real-Time Inventory Tracking and Asset Management System

The third specific objective is to develop a dynamic, real-time inventory management and asset tracking module that bridges the gap between sales and stock consumption. In the water refilling industry, the loss of reusable gallon containers and the depletion of consumable supplies (caps, seals, and minerals) represent a major operational risk. This objective aims to automate the "Inventory Reconciliation" process by linking every sale directly to the stock levels in the database. Technically, this objective involves implementing database triggers and stored procedures that automatically deduct the appropriate volume of water and quantity of supplies whenever a transaction is finalized in the POS module. For example, if a customer purchases five refills, the system will instantly decrement the "Water Volume," "Caps," and "Seals" counts in the inventory table. Beyond simple tracking, this objective includes the development of an "Intelligent Alert System" that notifies managers when stock levels reach a pre-defined "Reorder Point." This proactive feature is designed to prevent "Emergency Stockouts," ensuring that High Island V never has to turn away a customer due to a lack of supplies. Additionally, the system will include a "Container Tracking" sub-module to monitor the movement of blue gallons between the station and delivery customers, significantly reducing "Inventory Shrinkage" and protecting the company's capital investments.

Specific Objective 4: Implementation of an Analytics Dashboard for Visual Decision Support

The fourth specific objective is to design and implement a web-based management dashboard that transforms raw transaction data into visual, actionable business intelligence using Bootstrap and Chart.js. Currently, High Island V's data is "flat"—it exists as a series of numbers in a notebook that are difficult to interpret. This objective seeks to provide management with a powerful tool for data visualization. The technical development of the dashboard involves creating queries that aggregate data by day, week, month, and branch location. These aggregates will be presented through interactive line graphs (for revenue trends), bar charts (for branch performance comparisons), and pie charts (for product popularity). This visual approach allows the owners to identify "Peak Demand Hours" and "Underperforming Branches" at a single glance. By achieving this objective, the researchers aim to move High Island V away from "Intuitive Management" (guessing based on feelings) toward "Data-Driven Management" (deciding based on facts). The dashboard will be accessible remotely, allowing the owners to monitor the heartbeat of their business from any location with an internet connection, effectively bridging the geographic gap between the city and provincial operations.

Specific Objective 5: Integration of Predictive Sales Forecasting using Simple Linear Regression

The fifth specific objective is to incorporate a sophisticated "Predictive Engine" into the H2Ops system using the Simple Linear Regression algorithm. This objective represents the transition of High Island V from a purely historical business model to a proactive, forward-looking enterprise. The goal is to provide management with an accurate 7-day sales forecast to assist in resource planning and logistics.

The technical implementation of this objective involves developing a Python script that pulls the last 30 days of historical sales data (the Independent Variable,

X

X) and calculates the trend line to predict the upcoming week's sales volume (the Dependent Variable,

Y

Y). The system will utilize the mathematical formula

Y

=

a

+

b

X

Y=a+bX to generate these forecasts. By automating this complex calculation, the system will provide reports that suggest how many staff members should be scheduled and how many supplies should be ordered for the next seven days. This helps High Island V avoid the twin costs of "Over-stocking" (wasted capital and electricity) and "Under-stocking" (lost revenue and customer disappointment). The inclusion of this machine-driven forecasting tool provides a level of operational sophistication that is rarely seen in small-scale water refilling stations, giving High Island V a significant competitive edge.

Specific Objective 6: Establishing Robust Security Protocols and User Accountability Frameworks

The sixth specific objective is to establish a comprehensive security framework that ensures data integrity, confidentiality, and accountability across the entire organization. In a manual system, anyone can read or alter a logbook; this objective seeks to secure the business's digital assets through modern software security principles. The technical implementation will focus on Role-Based Access Control (RBAC), which creates a hierarchy of permissions within the system. "Front-Desk Staff" will be restricted to the sales entry screen, "Branch Managers" can access their specific branch reports, and the "Super Admin" (Owner) has full access to all branches, financial forecasting, and user management tools [4]. Furthermore, to protect against unauthorized remote access, the system will implement Multi-Factor Authentication (MFA), specifically through Email-based One-Time Passwords (OTP). This ensures that even if a password is compromised, the sensitive financial data of High Island V remains secure. Finally, this objective includes the creation of a "Digital Audit Trail"—a hidden log that records every "Create, Update, or Delete" action performed in the system, along with the timestamp and the identity of the user. This ensures total accountability and provides the owners with a tool to identify and investigate any financial discrepancies or data tampering.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

1.4 Scope and Delimitation

Formatted: Font: (Default) Times New Roman, 13 pt, Bold

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

This study focuses on the design, development, and comprehensive evaluation of H2Ops: A Responsive Cloud-Based Operations Management System, a specialized software solution engineered to modernize and optimize the multi-branch business environment of High Island V. The scope of the study encompasses the end-to-end software development lifecycle—including requirements elicitation, architectural design, back-end and front-end development, and rigorous user acceptance testing—within the specific operational context of the water refilling industry. The research is primarily concerned with addressing the systemic deficiencies of manual, paper-based record-keeping and replacing them with a centralized, data-driven, and cloud-hosted platform that enables the management of High Island V to monitor, analyze, and forecast operational performance in real time across geographically dispersed locations. The scope of this study is meticulously defined by several critical dimensions that establish the boundaries of the research effort and the intended applicability of the system's outcomes. Defining these scope dimensions is essential for properly interpreting the study's findings, establishing its technical limits, and assessing its practical relevance to other small and medium enterprises (SMEs) in the utility and service sectors.

Operational Scope

The operational scope of the study encompasses four primary business domains that are directly aligned with the day-to-day management needs of High Island V: Sales and Transaction Monitoring, Real-Time Inventory and Asset Tracking, Multi-Branch Centralized Oversight, and Predictive Demand Forecasting. These domains represent the core operational functions that the H2Ops system aims to modernize. Sales and Transaction Monitoring encompasses the complete lifecycle of recording, processing, and auditing customer orders across two distinct channels: walk-in retail and home/office delivery. This includes the development of a digital Point of Sale (POS) interface that standardizes pricing for purified and alkaline water, automates the calculation of totals and applicable discounts, and records payment methods (Cash or Digital Wallet references). More specifically, the system captures a "transactional footprint" for every customer interaction, recording the exact timestamp, the specific branch location, the identity of the attending staff member, and the quantity of water sold. This granular level of detail allows management to reconstruct the operational history of any given business day with absolute precision, identifying peak hours and high-demand periods that were previously invisible in manual logbooks.

Real-Time Inventory and Asset Tracking involves the monitoring of stock quantities and the movement of physical assets, specifically the 5-gallon reusable containers (blue gallons), seals, caps, and filtration consumables. The scope includes the development of a "deduction-on-sale" logic, where the system automatically updates stock levels the moment a transaction is finalized. A significant portion of this scope is dedicated to the "Container Cycle"—tracking the number of containers sent out for delivery versus the number returned. This ensures that the company's capital assets are accounted for, reducing "inventory shrinkage" and providing alerts when consumable supplies reach a critical reorder threshold. Multi-Branch Centralized Oversight addresses the geographic disconnect between High Island V's city and provincial branches. The scope includes the synchronization of data from multiple remote terminals into a single cloud-hosted master database. This allows the owners to monitor the heartbeat of the entire business from a single interface, eliminating the need for physical travel or manual report transcription.

System Feature Scope

To address the operational requirements of High Island V, the proposed H2Ops incorporates several interconnected functional modules designed to centralize business information, automate routine operations, and provide useful sales insights. The modules are designed to work together within a single cloud-based platform, allowing authorized users to record, monitor, retrieve, and analyze business information more efficiently. The system

focuses on the major operational activities of the water refilling station, particularly sales management, inventory monitoring, customer and delivery management, reporting, and sales analytics.

Sales Management Module: The Sales Management Module serves as the primary component for recording daily sales transactions. It allows authorized staff to enter customer orders, select products, specify quantities, record payments, and save completed transactions into the centralized database. The system automatically calculates transaction totals to minimize manual computation and reduce errors in sales recording. Each completed transaction is stored with relevant information such as the transaction date, products purchased, quantity, amount, and user responsible for the transaction.

The recorded sales information also serves as the primary source of data for the system's reporting and analytics functions. By centralizing transaction records, H2Ops allows administrators and managers to review sales performance without manually consolidating records from different sources. The module is intended to make daily transaction recording faster, more organized, and easier to retrieve when preparing business reports.

Inventory Management Module: The Inventory Management Module is designed to help High Island V monitor the availability and movement of products and other operational supplies. Authorized users can record newly received stocks, update inventory quantities, and monitor current stock levels. The system can automatically deduct applicable inventory quantities based on recorded sales transactions, reducing the need for employees to manually update stock records after every transaction.

The module also provides stock-status information to help management identify items that may require replenishment. Inventory records include information such as product name, quantity, stock movement, and adjustment history. Authorized users may also make manual adjustments when necessary, such as when damaged, lost, returned, or incorrectly recorded items are identified. These functions help maintain more consistent inventory records and support better planning of stock replenishment.

Customer and Delivery Management Module: The Customer and Delivery Management Module is designed to organize information related to customers and water delivery transactions. Customer records may include basic information necessary for processing and monitoring orders, while delivery records allow staff to document customer requests, delivery details, order status, and related transaction information.

This module provides a centralized location for managing delivery-related records instead of relying solely on handwritten notes or separate records. It helps authorized users monitor pending and completed deliveries and retrieve customer information when needed. However, the module is primarily intended for recording and monitoring deliveries and does not include advanced GPS tracking, automated route optimization, or autonomous delivery management.

Automated Sales Analytics Module: The Automated Sales Analytics Module is one of the primary features of H2Ops and is designed to transform recorded sales data into meaningful business information. The module processes historical transaction records and generates summaries, comparisons, and sales trends that can help owners and managers understand business performance.

The analytics may include total sales, sales by product, sales by date or period, transaction volume, and other relevant performance indicators supported by the available data. Instead of manually calculating and comparing records, users can access automatically generated analytical information through the system. These insights can assist management in identifying frequently purchased products, observing changes in sales performance, and making more informed operational decisions.

The analytics component is limited to the information available within H2Ops and does not guarantee future business outcomes. More advanced artificial intelligence or machine-learning forecasting is outside the current scope unless specifically implemented and validated as part of the study.

Management Dashboard: The Management Dashboard provides a centralized visual summary of important operational information. It presents key indicators, charts, graphs, and summarized reports that allow authorized users to quickly understand the current status of the business. Depending on the user's access level, the dashboard may display information such as total sales, transaction counts, inventory status, delivery activities, and sales trends.

The dashboard is designed to reduce the time required to interpret large amounts of operational data. Instead of reviewing multiple records individually, managers can view summarized information in a single interface. For High Island V, this feature is particularly useful for monitoring business performance and identifying operational areas that may require attention.

Role-Based Access Control (RBAC) Module: The Role-Based Access Control (RBAC) Module manages user permissions within H2Ops. Different access levels are assigned according to the responsibilities of each user. This prevents users from accessing functions or information that are not necessary for their assigned roles and helps protect sensitive business information [4].

Formatted: Font: 10 pt

Formatted: Font: 10 pt

The system may provide different levels of access for the Business Owner/Administrator, Branch Manager, and Staff. The Business Owner or Administrator has broader access to business reports, user management, and system settings. Branch Managers can access information related to their assigned branch, while Staff primarily access operational functions such as sales recording and delivery transactions. This structure supports accountability and reduces the possibility of unauthorized changes to important records.

Audit Trail and Reporting Module: The Reporting and Audit Trail Module is responsible for generating organized business reports from information stored in the system. H2Ops can generate daily, weekly, and monthly summaries depending on the reporting requirements of the business. Reports may include sales performance, inventory information, transaction summaries, and other operational data available within the system.

The audit trail records important system activities performed by authorized users, particularly significant changes to business records. Information such as the user responsible for an action, the date and time of the activity, and the type of modification can be recorded where supported by the system. This provides greater accountability and makes it easier for administrators to review changes made to important records.

User Scope

The intended users of H2Ops are categorized according to their responsibilities within High Island V. Each user role is assigned appropriate permissions to ensure that users can perform their required tasks while protecting information that should remain restricted.

The Super Admin (Business Owner): The Business Owner or Administrator has the highest level of access to the system. This user is responsible for overseeing the overall operation of H2Ops and reviewing business information across the organization. The administrator can manage user accounts, monitor sales performance, review inventory information, access business reports, and view sales analytics.

For a multi-branch operation, the Business Owner/Administrator can use the centralized system to compare and monitor information from different branches, depending on the implemented configuration. This provides management with a broader view of business performance without requiring separate manual reports from each location.

The Branch Manager: The Branch Manager is responsible for managing the daily operations of an assigned branch. This user can monitor branch sales, inventory, customer orders, and delivery information relevant to their location. The manager may also review branch-level reports and verify recorded transactions to ensure that information is accurate and complete.

The Branch Manager's access is restricted according to the assigned branch to prevent unnecessary access to information belonging to other locations. This supports proper separation of responsibilities while still allowing managers to obtain the information necessary to manage their branch effectively.

The Front-Desk Staff: This user group has the most limited scope, focusing exclusively on operational execution. Their primary interaction is with the POS module for processing sales and recording container returns. Their interface is streamlined to prevent them from accessing high-level financial reports or system configuration settings, thereby reducing the risk of accidental data corruption or unauthorized information viewing.

Technical Scope

The system is developed as a Responsive Web-Based Platform, utilizing a modern technology stack selected for its scalability, security, and cost-effectiveness. The Back-End is powered by Python and the Flask micro-framework, which handles the system's logic, mathematical forecasting, and server-side processing. The Database layer utilizes MySQL hosted on a cloud-based server, ensuring that data is encrypted, backed up, and accessible 24/7. The Front-End is developed using HTML5, CSS3, and JavaScript, with Bootstrap as the primary CSS framework to ensure "Responsive Design." This means the system will automatically adjust its layout to fit any device, including desktop computers at the branch counter, tablets used for inventory checks, and smartphones used by owners for remote monitoring. The decision to utilize a cloud-based architecture was a strategic choice to minimize High Island V's investment in on-site hardware. By hosting the system in the cloud, the business avoids the need for local servers or specialized maintenance, as the software can be accessed via a standard web browser on any internet-connected device.

Delimitations

While H2Ops is carefully designed to address the core operational, administrative, and analytical needs of High Island V, the study acknowledges several important limitations that precisely define the boundaries of its scope. These delimitations serve a critical function in academic research: they clarify what the study explicitly does not attempt to accomplish, establish realistic expectations for the system's capabilities and practical applicability, and prevent the scope from expanding beyond what can be feasibly developed, tested, and evaluated within the constraints of an undergraduate research project. By articulating these boundaries clearly, the researchers ensure that the evaluation of H2Ops is conducted against a well-defined, achievable set of objectives rather than against an unbounded and aspirational wish list of features.

Functional Scope Delimitations

The H2Ops system is architected and developed as a comprehensive Operations Management System specifically tailored to the water refilling industry. However, within that broad category, several functional domains are deliberately excluded from the development scope.

Exclusion of Hardware Automation and Internet of Things Integration

The system does not incorporate any interface with the physical water purification machinery, including but not limited to flow meters, reverse osmosis sensors, ultraviolet sterilization units, or conductivity testers. The water filtration and purification process at High Island V involves sophisticated industrial equipment that monitors parameters such as total dissolved solids (TDS), pH levels, and microbial contamination. These machines operate as independent, specialized systems with their own sensor arrays and control panels.

The decision to exclude hardware automation from the H2Ops scope reflects several practical and technical realities. First, interfacing with industrial water purification equipment requires specialized knowledge of Programmable Logic Controllers (PLCs), industrial communication protocols such as Modbus or Profibus, and potentially hazardous electrical integrations that fall far outside the expertise and safety authorization of the researchers. Second, such integration would require the procurement of industrial-grade sensors, signal converters, and data acquisition modules that would dramatically increase the project's budget beyond the resources available for an undergraduate thesis. Third,

the water refilling equipment at High Island V varies in age and technical specification across branches, meaning that any hardware integration developed for one location would not be directly transferable to others without significant modification.

Consequently, the volume of water processed and the quality metrics of the purification process must be entered into H2Ops manually by staff members based on physical observations of the equipment gauges and meter readings. While this introduces a minor manual step into the workflow, it is a pragmatic trade-off that ensures the system remains within a manageable development scope while still delivering substantial operational value through its sales, inventory, and forecasting modules.

Exclusion of Payment Gateway and Real-Time Financial Processing

The H2Ops system does not include active integration with third-party payment gateways, merchant services, or financial processing Application Programming Interfaces (APIs) such as those offered by banks, PayMaya, or Stripe. While the system records and tracks payments received from customers—including cash payments and digital wallet references such as GCash transaction codes—it does not process these payments in real time through a direct financial API connection.

This delimitation is rooted in both technical complexity and regulatory considerations. Integrating with payment gateways requires the business to undergo a merchant onboarding process with a payment service provider, negotiate transaction fees, and maintain compliance with Payment Card Industry Data Security Standards (PCI-DSS). For a small business like High Island V, these requirements represent a significant administrative and financial commitment that falls outside the scope of the current research. Furthermore, the development timeline and testing resources available for an undergraduate project are insufficient to validate the security and reliability of live financial transactions.

In practical terms, this means that when a customer pays via GCash, the staff member must confirm the receipt of funds through their personal or business GCash application, note the reference number, and manually enter that reference number into H2Ops as part of the transaction record. While this is a minor manual step, it ensures that all transaction data—regardless of payment method—is centralized within the H2Ops database, maintaining the integrity of the sales records and financial summaries.

Exclusion of Accounting, Tax Computation, and Payroll Processing

The system is explicitly scoped as an Operations Management System and does not extend into the domain of formal financial accounting. H2Ops does not include modules for bookkeeping, chart of accounts management, accounts payable or receivable tracking, tax computation and remittance, or automated payroll calculation for employees. While the system generates reports on revenue, expenses, and profit margins at a summary level, it does not produce audited financial statements, BIR (Bureau of Internal Revenue)-compliant tax forms, or government remittance reports.

This delimitation is based on the understanding that financial accounting in the Philippines involves highly specialized regulatory compliance requirements under the National Internal Revenue Code, BIR Revenue Regulations, and local ordinances. Producing compliant financial documents requires expertise in taxation and accounting that is beyond the scope of a software engineering thesis. Furthermore, High Island V likely already utilizes separate accounting software or engages an external accountant for these functions; H2Ops is designed to complement rather than replace these existing arrangements by providing accurate, real-time operational data that can be exported to external accounting systems if needed.

The exclusion of payroll processing is particularly important to note. H2Ops tracks employee attendance and work hours through its User Management module, generating data that is valuable for scheduling and productivity analysis. However, it does not calculate wages, deductions, benefits contributions (SSS, PhilHealth, Pag-IBIG), or generate

payslips. These functions require integration with government databases and compliance systems that are outside the technical and legal scope of this study.

Exclusion of Customer Relationship Management and Marketing Automation

The functional scope of H2Ops does not include active Customer Relationship Management (CRM) tools such as loyalty point systems, automated SMS or email marketing campaigns, customer preference profiling, or social media integration. While the system records basic transaction histories that could theoretically support customer profiling, it does not actively solicit, store, or analyze personal customer information beyond what is operationally necessary.

This delimitation reflects both privacy considerations and scope management. The Data Privacy Act of 2012 (Republic Act No. 10173) imposes significant obligations on entities that collect, process, and store personal information of customers. By limiting the system's data collection to purely operational metrics—transaction amounts, product types, timestamps, and branch locations—the researchers avoid triggering the compliance requirements, consent protocols, and security measures that would be necessary for a full CRM implementation. This approach protects both the business and the research project from potential legal complications while still delivering meaningful operational intelligence.

Technology Scope Delimitations

The H2Ops system is developed as a cloud-hosted, web-based application with specific technical characteristics that define its operational boundaries.

Exclusion of Offline Functionality and Desktop Application Installation

The system does not include an offline-first mode, a local desktop application, or a standalone mobile application. All functionality of H2Ops requires a stable internet connection to communicate with the cloud-hosted database server. If a branch experiences a network outage, staff members will be unable to process new sales or access inventory records through the H2Ops interface until connectivity is restored. This technical delimitation is a direct consequence of the cloud-based architecture selected for the project. While a fully offline-capable system would provide resilience against network interruptions, its development would require significantly more complex synchronization logic, local data storage management, and conflict resolution algorithms—components that would expand the project

The researchers have determined that the benefits of cloud hosting—centralized data management, automatic backups, and remote accessibility—outweigh the risk of temporary network-related downtime, particularly given that High Island V's operations are conducted in urban and peri-urban areas with generally reliable internet infrastructure. To mitigate the impact of this limitation, the researchers will document a "Manual Backup Procedure" as part of the system's operational manual. In the event of a network outage, branch staff will be instructed to maintain temporary paper records of transactions, which can be manually encoded into H2Ops once connectivity is restored. While this is not an ideal long-term solution, it ensures business continuity during rare offline periods.

Exclusion of Multi-Tenant Architecture and Franchise Support

H2Ops is developed as a single-tenant system optimized for the specific operational workflows, pricing structures, and organizational hierarchy of High Island V. The system does not include multi-tenant architecture that would allow it to serve multiple independent businesses simultaneously, nor does it support a franchise model where different franchisees could maintain separate branding, pricing, and configurations within a shared infrastructure.

This delimitation is important for managing expectations. While the researchers believe that the H2Ops system could potentially be adapted for use by other water refilling stations in the future, the current implementation is tightly coupled to the specific requirements, terminology, and business rules of High Island V. Significant modifications—including rebranding, pricing adjustments, and workflow customization—would require access to the source code and technical expertise that may not be available to external users. Future research could explore the development of a

configurable, multi-tenant SaaS (Software as a Service) platform that generalizes the H2Ops model for broader industry applicability.

Data Scope Delimitations

The data scope of the study is carefully bounded to include operational data necessary for the system's core functions while excluding sensitive or personally identifiable information.

Exclusion of Customer Personal Data

H2Ops is designed to process operational transactions rather than personal customer profiles. The system does not require or store personal customer information such as names, addresses, contact numbers, or email addresses in its transaction records. For walk-in retail customers who pay in cash, the transaction record consists solely of the product sold, quantity, total amount, timestamp, and the identity of the processing staff member. For delivery customers, the system records a delivery address and contact number sufficient for logistics purposes, but these details are not linked to a persistent customer profile or used for marketing analytics.

This data minimization approach serves multiple purposes. It reduces the system's exposure to the Data Privacy Act compliance requirements, minimizes the potential impact of a data breach on customer privacy, simplifies the database schema and reduces storage requirements, and aligns with the operational reality that High Island V primarily serves anonymous walk-in customers rather than registered account holders.

Exclusion of Historical Data Migration

The study does not include the migration of historical paper records into the H2Ops database. The operational data scope begins from the date of system deployment forward; no attempt is made to retroactively encode years of handwritten logbook entries into the system. This delimitation is necessary because the manual records, despite their incompleteness and inaccuracies, are often fragmented, illegible, or missing entirely for certain periods. Attempting to manually encode this historical data would be prohibitively time-consuming, would introduce its own errors, and would not materially contribute to the evaluation of the system's operational effectiveness.

Security Scope Delimitations

The security architecture of H2Ops addresses fundamental data protection and access control requirements appropriate to the scale and risk profile of a small business operation, but it does not extend to enterprise-grade security infrastructure or formal compliance certifications.

Exclusion of Third-Party Security Auditing and Penetration Testing

The system is not subjected to formal security auditing by independent cybersecurity firms, penetration testing by ethical hackers, or vulnerability assessments against OWASP (Open Web Application Security Project) standards. The security measures implemented—including password hashing with bcrypt, SQL injection prevention through parameterized queries, and session token management—are based on established best practices and standard web security guidelines, but they have not been validated through professional security testing.

This delimitation reflects the resource constraints of an undergraduate research project. Professional security assessments require specialized expertise, significant time investment, and potentially expensive tooling that are not available within the current scope. The researchers have designed the security architecture with due diligence, documenting the rationale for each security measure and identifying potential areas for future hardening. However, before H2Ops is deployed in a production environment handling sensitive financial data, a formal security audit by qualified professionals is strongly recommended.

Exclusion of Biometric and Hardware Token Authentication

While H2Ops implements Email-based One-Time Password (OTP) as a secondary authentication factor, the system does not include biometric authentication (fingerprint, facial recognition) or hardware token-based authentication (YubiKey, RSA tokens). The multi-factor authentication is limited to something the user knows (password) and something the user has access to (email inbox), which the researchers consider a reasonable balance between security and practical usability for the target user population.

Evaluation Scope Delimitations

The evaluation of H2Ops is bounded by the practical constraints of an academic research timeline and available resources.

Exclusion of Longitudinal Business Impact Assessment

The study does not include a longitudinal assessment of the system's long-term impact on business performance metrics such as revenue growth, profitability trends, customer retention rates, or employee turnover over an extended period of months or years. The evaluation is conducted during a defined implementation period—typically four to six weeks—which is sufficient to assess functional correctness, usability, and initial user satisfaction but insufficient to measure sustained business transformation.

This delimitation is a pragmatic acknowledgment that demonstrating statistically significant business impact requires controlled experimental designs, large sample sizes, and extended timeframes that exceed the scope of an undergraduate thesis. The researchers will instead focus on measuring proxy indicators of business improvement, such as the reduction in time required to generate reports, the accuracy of inventory counts compared to manual methods, and the subjective assessment of system utility by management and staff.

Exclusion of Comparative Analysis Against Commercial Alternatives

The study does not include a formal comparative analysis of H2Ops against commercially available Point of Sale (POS) or Operations Management systems in the Philippine market. While such a comparison would provide valuable context for evaluating the system's competitive positioning and value proposition, conducting a rigorous, fair, and statistically valid comparison would require access to proprietary systems, licensing costs, and extensive testing that fall outside the resources and timeframe of this project.

The researchers acknowledge that several commercial POS systems—such as those offered by Oracle Hospitality, Square, and local providers—are widely available and may offer features not included in H2Ops. The decision to develop a custom system rather than adopt an off-the-shelf solution is driven by the specific, tailored requirements of High Island V and the educational objectives of the research project, rather than any claim of technical superiority over existing commercial products.

Exclusion of Stress Testing Under Extreme Loads

The evaluation does not include stress testing or load testing conducted under artificially extreme conditions designed to identify system breaking points. While the system will be tested under normal operational loads that simulate peak business hours, it will not be subjected to stress tests that simulate, for example, a sudden surge of 500 concurrent transactions or a denial-of-service attack.

This delimitation is based on the assessment that the probability of such extreme conditions occurring in High Island V's actual operational environment is minimal. As a regional water refilling station with a moderate customer base, the system is not expected to encounter enterprise-scale transaction volumes that would require advanced performance optimization. Future deployment in larger multi-branch scenarios or higher-volume operations may benefit from formal performance testing, which the researchers recommend as an area for future investigation.

Rationale for Scope Boundaries

The scope delimitations described above are not arbitrary limitations imposed by resource constraints alone; each boundary reflects a deliberate and reasoned decision based on the interplay of technical feasibility, practical utility, academic appropriateness, and strategic focus.

From a technical feasibility perspective, certain features—such as hardware automation integration, offline functionality, and multi-tenant architecture—would require expertise, development time, and infrastructure that substantially exceed the capabilities of an undergraduate software engineering project.

The researchers have prioritized features that are achievable with available tools, frameworks, and skill levels while still delivering meaningful operational value to High Island V. From a practical utility perspective, the study recognizes that the most immediate and impactful improvements for High Island V lie in the areas of data accuracy, operational visibility, and decision-support—domains that are well within the proposed scope. Features such as predictive analytics (Simple Linear Regression) and real-time dashboards address fundamental information deficiencies that the management currently experiences. By contrast, features such as automated payroll processing or CRM integration, while potentially valuable, address secondary needs that are less urgent and could be addressed in future development phases.

From an academic appropriateness perspective, the scope boundaries ensure that the research project remains focused, manageable, and evaluable within the standard timeframe and requirements of an undergraduate thesis. An overly ambitious scope that attempts to address every possible operational need would dilute the research's depth, extend development timelines beyond acceptable limits, and complicate the evaluation framework to the point where meaningful conclusions become difficult to draw.

Future Expansion and Extensibility

While the current implementation of H2Ops is bounded by the scope dimensions described above, the system is architecturally designed with modularity, clean separation of concerns, and extensibility in mind. The layered architecture—separating the database layer, business logic layer (Flask/Python), and presentation layer (HTML/Bootstrap)—creates natural boundaries for incremental development and future enhancement.

Future expansion possibilities that are acknowledged but not pursued within the current study include: Mobile Application Development for native iOS and Android access, enabling delivery riders and field staff to interact with the system on mobile devices with optimized interfaces; IoT Integration with water purification sensors to enable real-time monitoring of water quality parameters directly within the H2Ops dashboard; SMS-Based Alerts utilizing Philippine telecom APIs to send low-stock notifications and delivery updates via text message; Advanced Analytics and Machine Learning using more sophisticated forecasting models such as ARIMA or gradient boosting algorithms for longer-range demand prediction; Supplier Portal Integration enabling automated purchase order generation and delivery scheduling with upstream suppliers; and Multi-Branch Financial Consolidation providing a consolidated profit-and-loss view across all locations with transfer pricing and allocation logic.

By documenting the current scope clearly and designing the system with principled architectural foundations, this study establishes a platform upon which subsequent researchers or developers can build without requiring fundamental restructuring of the existing codebase or data model. The researchers view H2Ops Version 1.0 as the foundational first step in a longer-term digital transformation journey for High Island V—one that addresses immediate operational pain points while preserving the flexibility to evolve and expand as the business grows and its needs become more sophisticated.

1.5 Significance of the Study

Formatted: Font: (Default) Times New Roman, 13

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The development, implementation, and rigorous evaluation of the **H2Ops: A Responsive Cloud-Based Operations Management System** are expected to yield significant benefits for several distinct groups. Each beneficiary group derives specific, tangible value from the transition of High Island V from a manual, paper-based infrastructure to a modern, data-driven digital ecosystem. Understanding the multi-layered impact of this research illuminates its broader relevance beyond the immediate technical objectives, positioning the study as a catalyst for operational transformation, academic contribution, and industry-wide modernization.

Beneficiaries: Business Owners and Management (Administrators)

The primary beneficiaries of this study are the administrators and owners of High Island V. In the current operational landscape, the management operates under a condition of "Information Asymmetry," where the lack of real-time data from geographically dispersed branches creates a significant barrier to effective oversight. The implementation of H2Ops fundamentally transforms the management role from one of reactive firefighting to proactive, strategic leadership. By providing a centralized, cloud-hosted platform, H2Ops equips management with a "single source of truth." The system eliminates the need for manual report transcription and physical travel between city and provincial branches to verify sales and inventory levels.

This benefit addresses the core management need for reliable, timely, and accessible business intelligence. With the Management Dashboard, owners can monitor live performance metrics, allowing them to identify underperforming branches or sudden demand spikes as they occur. The shift from fragmented, handwritten logbooks to an automated reporting engine ensures that financial summaries are generated with 100% accuracy, providing a clear and honest view of the company's profitability, expense ratios, and growth trajectories.

Furthermore, the integration of Simple Linear Regression for sales forecasting provides management with a level of "Predictive Intelligence" that was previously unattainable. Instead of ordering supplies or scheduling staff based on institutional intuition or guesswork, the owners can now rely on statistically grounded forecasts. This leads to optimized resource allocation—ensuring that the business is never over-staffed during quiet periods nor under-supplied during peak demand weeks. The ultimate significance for management lies in the preservation of capital through the reduction of inventory shrinkage and the elimination of financial leakages caused by manual math errors. As High Island V seeks to scale and open additional branches, H2Ops provides the scalable administrative framework necessary to support that expansion without increasing the management's cognitive or physical burden.

Beneficiaries: End Users (Branch Staff)

The system is significantly beneficial to the front-line staff who manage the day-to-day operations at each branch of High Island V. Currently, these employees must navigate the high-pressure environment of a busy water refilling station while manually recording every transaction and tracking inventory in paper logbooks—a process that is both tedious and prone to error. With the introduction of the Automated Point of Sale (POS) Module, staff members are relieved of the burden of manual calculations.

The system's intuitive interface simplifies the transaction process, allowing staff to focus more on customer service and quality control rather than paperwork. This reduction in manual recording tasks eliminates the stress associated with end-of-day reconciliation, where discrepancies between cash on hand and logbook entries often lead to conflict and confusion. The standardization of digital workflows ensures that every employee, regardless of their technical background, follows a consistent, guided process for sales and inventory intake.

The Real-Time Inventory and Asset Tracking feature is particularly significant for delivery personnel. By automating the deduction of water containers, seals, and caps, the system provides staff with clear visibility into stock availability. They can confidently inform customers about product availability without needing to perform physical checks. Moreover, the digital accountability provided by the system protects staff members; by maintaining an objective record

of every transaction, the system prevents unfair blame for missing stock or cash discrepancies, fostering a more transparent and equitable work environment. The exposure to a modern, cloud-based management system also serves as a professional development opportunity for the staff, enhancing their digital literacy and preparing them for the increasingly technology-driven demands of the modern service industry.

Beneficiaries: Future Researchers and the Academic Community

This study serves as a valuable reference and foundation for future researchers, developers, and IT students who are interested in the digital transformation of Small and Medium Enterprises (SMEs), particularly within the Philippine context. The detailed documentation of the H2Ops system development—using Python, Flask, and MySQL—provides a replicable model for how micro-frameworks can be utilized to build robust, scalable business applications without the high overhead costs of enterprise software.

This research contributes to the academic literature on "Information Systems in Resource-Constrained Environments." While much research is focused on large-scale corporate ERPs, there is a significant gap in studies addressing the unique operational challenges of local utility-service businesses like water refilling stations. Future researchers can build upon this study by exploring more advanced analytical models, such as integrating IoT sensors for water quality monitoring or using more complex machine learning algorithms for long-term demand forecasting.

The study's methodological approach—which combines a developmental research design with the application of specific mathematical models (Simple Linear Regression)—offers a roadmap for conducting technology research that has high practical utility. By documenting the challenges and successes of implementing a cloud-based system in a local Filipino business, this study provides empirical data that can inform future studies on technology adoption, user resistance, and the socio-technical impacts of digitalization in provincial areas. It demonstrates how academic theories can be translated into functional tools that solve real-world community problems.

Beneficiaries: The Small Business Community and the Water Refilling Industry

Beyond High Island V, this study holds broader significance for the thousands of water refilling stations and small service-oriented businesses operating across the Philippines. The water refilling industry is a vital part of the nation's public health infrastructure, yet it remains one of the least digitized sectors. This study demonstrates that appropriately designed, affordable technology can solve the chronic problems of inventory loss and management opacity that many of these businesses face.

The success of H2Ops provides a "Proof of Concept" for other MSMEs (Micro, Small, and Medium Enterprises), showing that they do not need to be part of a massive franchise to benefit from sophisticated management tools. This research supports the national agenda for MSME Digitalization, as outlined by the Department of Trade and Industry (DTI). By documenting a successful digital transition, this study encourages other local entrepreneurs to move away from inefficient manual methods, thereby increasing the overall competitiveness and resilience of the local economy.

Furthermore, there is a significant public health dimension to the system's impact. By enabling better tracking of filtration consumable inventory (such as when filters and minerals were last replaced, the system indirectly helps ensure that the water provided to the community consistently meets safety standards. A more efficient, data-driven water refilling industry leads to more stable pricing, better service reliability, and higher safety standards for the millions of Filipinos who rely on these stations for their daily hydration. Thus, the significance of the study extends from the individual business owner to the community at large, promoting a more modern, efficient, and safe utility service environment.

1.6 Definition of Terms

Formatted: Font: (Default) Times New Roman, 13

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

For clarity and consistency throughout this study, the following terms are defined both technically and operationally as they are applied within the framework of this research:

Audit Trail refers to the chronological, digital record maintained by the H2Ops system that logs every action performed by users, including the creation, modification, or deletion of sales and inventory data. Operationally, the audit trail provides High Island V with a security mechanism to ensure accountability and to investigate any financial discrepancies or unauthorized data tampering.

Business Analytics refers to the systematic technical processes and practices used by the H2Ops system to collect, process, and analyze transaction data from various High Island V branches. In this study, business analytics is specifically applied to the transformation of raw sales logs into visual insights, such as identifying peak sales periods and determining which branch location is performing most efficiently.

Cloud-Based System refers to a software architecture where data and applications are hosted on remote internet servers rather than on local branch computers. For High Island V, this enables the owners to access the management dashboard from any location with internet connectivity, ensuring that branch data is synchronized and backed up in real-time without the need for on-site server maintenance.

Dashboard refers to the centralized visual interface within the H2Ops system that presents key performance indicators (KPIs) through the use of interactive charts and graphs. The dashboard is designed to provide the owners of High Island V with an at-a-glance summary of total revenue, inventory health, and branch-to-branch comparisons, facilitating rapid and evidence-based decision-making. **H2Ops** is the official name of the operations management system developed in this study. The name is a portmanteau of "H2O" (the chemical formula for water) and "Ops" (short for operations), reflecting the system's primary purpose of optimizing the operational workflows of the High Island V water refilling enterprise.Inventory Tracking refers to the automated process within the H2Ops system that monitors the levels of water, containers, and consumable supplies. Unlike manual counting, inventory tracking in this study occurs instantly whenever a sale is finalized, providing management with live updates on stock availability and low-stock warnings.

Manual Record-Keeping refers to the legacy operational practice currently used by High Island V, involving the use of handwritten entries in paper logbooks and disconnected spreadsheets. This practice is identified in the study as the primary cause of mathematical errors, data loss, and management delays.

Multi-Factor Authentication (MFA) refers to a security protocol implemented in H2Ops that requires users to provide two forms of identification before gaining access to the system. In this study, this is operationalized through a password and an Email One-Time Password (OTP), ensuring that sensitive financial data is protected even if a password is compromised.

Point of Sale (POS) refers to the specific module within the H2Ops system where transactions are processed. Operationally, the POS serves as the digital interface used by branch staff to

record purified and alkaline water sales, apply discounts, and generate digital receipts for walk-in and delivery customers.

Predictive Forecasting refers to the advanced capability of the H2Ops system to anticipate future business trends based on historical data. In the context of this study, predictive forecasting uses past sales figures to generate a 7-day demand report, allowing High Island V to prepare the necessary staff and supplies before the busy period begins.

Responsive Web Design refers to a technical approach to front-end development that ensures the H2Ops system interface automatically adjusts its layout to fit any screen size. Operationally, this allows High Island V management to use the system effectively on a desktop computer, a tablet, or a mobile smartphone without losing functionality.

Role-Based Access Control (RBAC) refers to the security framework within H2Ops that assigns specific permissions to users based on their job titles. For example, a "Staff" user may only access the POS screen, while an "Admin" user can access profit reports and predictive forecasting tools, ensuring that sensitive information is only available to authorized personnel.

Water Refilling Station (WRS) refers to the specific type of SME (Small and Medium Enterprise) that purifies raw water and sells it in reusable containers. In this study, High Island V serves as the primary WRS model used to test the effectiveness of the H2Ops system in a real-world Philippine business environment.

1.7 Conceptual Framework

Formatted: Font: (Default) Times New Roman, 13

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

This study is guided by a comprehensive conceptual framework that integrates theoretical perspectives from information systems, small business management, and technology adoption to structure the research design, system development, and evaluation approach for the H2Ops system. The conceptual framework posits that the operational challenges currently faced by High Island V—stemming from manual record-keeping, geographic information opacity (data silos), inefficient multi-branch reporting, inadequate container/asset monitoring, and the total absence of predictive analytical capabilities—can be systematically addressed through the development and implementation of a purpose-built, cloud-based operations management system.

The framework further posits that the effectiveness and ultimate success of such a system depend not merely on its technical coding or sophisticated algorithms, but also on its deep alignment with the operational context, the technical capabilities of the staff, and the specific decision-making needs of the water refilling industry. The framework identifies three interconnected and mutually reinforcing dimensions that influence system success, ensuring that H2Ops is viewed not just as a software product, but as a socio-technical intervention within a living business environment.

The Technical Dimension

The Technical Dimension encompasses the system's functional capabilities, architectural reliability, performance efficiency, and data security protocols. This dimension addresses the fundamental question of whether the system technically performs its intended functions correctly and securely. For H2Ops, technical success requires that the system accurately captures and processes every refill and delivery transaction without data loss. It must maintain absolute data integrity across the MySQL cloud database, ensuring that sales recorded at a provincial branch are perfectly synchronized with the central office.

Furthermore, this dimension includes the rigor of the Simple Linear Regression algorithm; the system must correctly execute the mathematical logic required to transform 30 days of historical data into a reliable 7-day forecast. Technical excellence also demands that the system responds promptly to user requests even under the load of multiple concurrent branch logins, operates reliably without system crashes, and utilizes robust security measures—such as Email OTP and role-based encryption—to protect sensitive financial information from unauthorized access, accidental corruption, or external cyber threats.

The Operational Dimension

The Operational Dimension encompasses the system's integration with the existing workflows of the water refilling station, its support for daily physical tasks, and its overall contribution to business efficiency and financial health. This dimension addresses whether the system fits the unique operational "rhythm" of High Island V. Operational success requires that the H2Ops system supports, rather than disrupts, the high-pressure environment of the refill counter and the logistical complexity of the delivery channel.

The system must streamline the core tasks of transaction processing and inventory intake, reducing the time staff previously spent on manual math and logbook entry. It must enable proactive management responses through real-time "Low-Stock Alerts" and generate measurable improvements in key business outcomes, such as the reduction of "Inventory Shrinkage" (lost blue gallons) and the elimination of the "Information Lag" between the city and provincial branches. In this dimension, the system is evaluated on its ability to transform raw data into a strategic asset that allows the owners to manage multiple locations with the same precision as a single shop.

The Human Dimension

The Human Dimension encompasses user acceptance, interface usability, training requirements, and the system's psychological impact on staff workload and job satisfaction. This dimension recognizes that a system is only effective if its intended users—from the owner to the front-desk staff—can and will utilize it. Human success requires that the H2Ops responsive web interface is intuitive, legible, and learnable, even for staff members with limited technical background or those accustomed to traditional pen-and-paper methods for decades.

The system's functions must align with the user's mental models of a water station workflow, ensuring that the Role-Based Access Control (RBAC) provides each user with exactly the tools they need without overwhelming them with unnecessary complexity. Successful adoption depends on the users perceiving the system not as a monitoring burden, but as a valuable tool that simplifies their work, reduces the stress of reconciliation errors, and provides clear guidance on their daily responsibilities. The framework recognizes that technology adoption in a family-owned SME like High Island V requires careful management of the transition from analog to digital to minimize resistance and maximize long-term engagement.

Synergy and Theoretical Foundations

The framework hypothesizes that successful system implementation requires a balanced and positive outcome across all three dimensions, and that a deficiency in any single dimension can compromise the overall effectiveness of the H2Ops project. A technically perfect forecasting algorithm will provide no value if the staff finds the interface too difficult to input daily sales (Human failure). Conversely, a highly user-friendly interface that fails to synchronize data correctly to the cloud server (Technical failure) will lead to management making decisions based on incorrect information, thereby undermining business growth (Operational failure).

This three-dimensional framework is informed by established theoretical perspectives on information systems success. The Technology Acceptance Model (TAM) serves as a foundational lens, suggesting that "Perceived Usefulness" and "Perceived Ease of Use" are the primary drivers of whether High Island V staff will embrace H2Ops. Additionally, the Socio-Technical Systems Perspective guides the study, emphasizing that effective operations management must achieve joint optimization of both the technical components (the Flask code and MySQL database and the social components (the staff's daily habits and the owner's management style).

1.8 Summary

This chapter has established the foundational framework for the study by presenting a comprehensive background on the critical role of the water refilling industry in the Philippines and the evolving operational challenges faced by multi-branch enterprises like High Island V. It has identified the specific systemic deficiencies—including manual record-keeping errors, geographic data silos, inventory shrinkage, and the absence of predictive capabilities—that necessitate the development of H2Ops: A Responsive Cloud-Based Operations Management System. By articulating the general and specific objectives, the study has set a clear technical and mathematical roadmap, specifically highlighting the integration of Simple Linear Regression to transition the business from a reactive to a proactive management model.

Formatted: Font: (Default) Times New Roman, 13

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Furthermore, the chapter delineated the precise scope and boundaries of the research, ensuring that the system remains focused on core operational needs while maintaining a secure and scalable architecture. The significance of the study was discussed in relation to its diverse beneficiaries, ranging from business owners and branch staff to the broader academic and small business communities. Finally, the conceptual framework was presented to illustrate the intersection of technical, operational, and human dimensions, while the definition of terms provided the necessary linguistic consistency for the research. Together, these sections provide a rigorous justification for the project and a structured approach for its implementation.

Formatted: Font: (Default) +Body (Calibri), 12 pt, Font color: Red

Formatted: Left, Line spacing: Multiple 1.15 li

Formatted: Font: Font color: Accent 1

Formatted: Centered, Line spacing: Multiple 1.15 li

Formatted: Centered, Line spacing: Multiple 1.16 li

Formatted: Centered

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Formatted: Font: 13 pt

Formatted: Centered, Line spacing: 1.5 lines

Formatted: Line spacing: Multiple 1.15 li

2.1 Foreign Studies

Formatted: Font: 13 pt, Bold

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Cloud-based operations management systems and business analytics have become essential technologies for improving business efficiency and supporting data-driven decision-making across various industries. The increasing adoption of cloud computing enables organizations to centralize operational data, automate business processes, and provide real-time access to information regardless of location. At the same time, business analytics has evolved from basic reporting tools into advanced systems capable of transforming large volumes of operational data into meaningful insights that support planning, performance evaluation, and strategic decision-making. Organizations that integrate cloud technologies with business analytics are better equipped to improve productivity, streamline operations, and respond effectively to changing business demands.

According to *Holsapple, Lee-Post, and Pakath* (2014), business analytics serves as a comprehensive approach that enables organizations to transform raw data into meaningful information that supports effective decision-making [6]. In their study, A Unified Foundation for Business Analytics, the researchers developed a holistic framework that unifies various concepts of business analytics by identifying its key dimensions, perspectives, and applications across different industries. They emphasized that business analytics extends beyond simple data management and statistical analysis, as it also involves systematic reasoning, organizational processes, technology utilization, and strategic decision-making. The study introduced a six-class taxonomy and an inclusive framework that provides researchers

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

and practitioners with a better understanding of how business analytics can improve organizational performance and guide future business initiatives.

-The findings of *Holsapple et al.* (2014) are highly relevant to the present study because they highlight the importance of integrating business analytics into organizational operations to support informed and data-driven decision-making. Similar to the framework presented in this study, H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics is designed not only to store operational data but also to transform daily sales transactions into meaningful business information through automated reports, dashboards, and sales analytics. The system enables water refilling station owners and managers to monitor sales performance, evaluate operational trends, and make timely business decisions based on accurate and organized data. Furthermore, the cloud-based architecture of H2Ops supports centralized data management, making business information more accessible and useful for monitoring multiple branches. The study of Holsapple et al. strengthens the theoretical foundation of this research by demonstrating that business analytics plays a vital role in improving operational efficiency, supporting strategic planning, and enhancing organizational decision-making through the effective use of information systems.

In a similar to *Delen and Ram* (2018), business analytics is a comprehensive approach that enables organizations to transform raw data into actionable insights that support faster and more effective decision-making [7]. In their study, Research Challenges and Opportunities in Business Analytics, the researchers examined the different definitions, challenges, and opportunities associated with business analytics as an emerging field of research and practice. They emphasized that while business analytics has been defined from various perspectives, its primary purpose is to convert organizational data into meaningful information that improves business performance and supports informed decision-making. The study also highlighted that business analytics encompasses descriptive, diagnostic, predictive, and prescriptive analytics, allowing organizations to analyze historical data, understand current business conditions, anticipate future trends, and recommend appropriate actions. Furthermore, the researchers stressed that business analytics has become increasingly important across industries as organizations continue to rely on data-driven strategies to remain competitive and improve operational efficiency.

The findings of *Delen and Ram* (2018) are highly relevant to the present study because they emphasize the value of business analytics in supporting organizational decision-making through the effective use of data. Similar to the concepts discussed in their study, H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics is designed not only to automate operational processes but also to transform daily sales transactions into meaningful business information through automated reports, dashboards, and sales analytics. The system enables water refilling station owners and managers to monitor sales performance, identify business trends, evaluate operational activities, and make informed decisions based on accurate and centralized data. In addition, the cloud-based architecture of H2Ops provides real-time access to operational information, allowing business owners to efficiently monitor their business performance and improve overall management. The study of Delen and Ram strengthens the theoretical foundation of this research by demonstrating that business analytics is an essential component of modern information systems that enhances operational efficiency, supports strategic planning, and enables organizations to make data-driven decisions for continuous business improvement.

Singh and Wajgi (2016) titled "Data Analysis and Visualization of Sales Data", presented at the World Conference on Futuristic Trends in Research and Innovation for Social Welfare, discussed the growing challenges faced by businesses in handling large volumes of sales data. The researchers explained that due to the continuous increase of data from different sources such as online transactions, databases, and digital platforms, organizations often struggle to effectively analyze and interpret this information [8]. As data becomes more complex and voluminous, traditional methods of data processing are no longer efficient or reliable.

The study emphasized that without proper tools, businesses may find it difficult to extract meaningful insights from their data, which can negatively affect decision-making and overall performance. The researchers pointed out that

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

manual analysis and basic reporting techniques are not capable of handling large datasets or presenting data in a clear and understandable way. Because of this limitation, important patterns and trends in sales performance may be overlooked.

-To address these issues, the study proposed the development of a system focused on data analysis and visualization. The system is designed to convert raw sales data into graphical representations such as charts and reports, making it easier for users to understand business performance. Through visualization, users can quickly identify trends, monitor progress, and make informed decisions. The study concluded that integrating data analytics and visualization tools can significantly improve business operations, enhance revenue generation, and support strategic planning.

Tim Mather, Subra Kumaraswamy, and Shahed Latif (2009) in their book "Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance", cloud computing provides organizations with significant benefits in terms of accessibility, scalability, and cost reduction [9]. However, the authors emphasized that businesses must also address security and privacy concerns when storing and managing data in cloud environments. As organizations increasingly rely on cloud-based services, protecting sensitive information becomes a critical aspect of successful cloud adoption.

The authors discussed key cloud security principles, including confidentiality, integrity, and availability of data. They explained that organizations should implement proper security measures such as identity and access management (IAM), authentication, authorization, and auditing mechanisms to ensure that only authorized users can access cloud resources. The study also highlighted the importance of compliance with security standards and regulations to maintain the privacy and protection of business information.

Furthermore, the book emphasized that cloud security frameworks help organizations reduce risks associated with unauthorized access, data breaches, and system vulnerabilities. By implementing effective security practices, businesses can increase trust in cloud services and ensure the reliability of their information systems. This literature is relevant to the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics because the system stores operational and sales data in a cloud environment. Similar to the study, H2Ops requires secure user authentication, controlled access, and proper data management to protect business information. The concepts discussed by Mather, Kumaraswamy, and Latif support the importance of integrating security measures into cloud-based systems to ensure data confidentiality, reliability, and safe access for administrators and staff.

Darya Plinere and Arkady Borisov (2015) in their study "Case Study on Inventory Management Improvement" published in Information Technology and Management Science, inventory management is one of the most critical aspects of supply chain and business operations [10]. The researchers explained that companies need to maintain sufficient inventory levels to meet customer demand while avoiding excessive stock that can increase storage costs and tie up financial resources. Poor inventory management can lead to overstocking, stock shortages, and inefficient business operations.

The study presented a case analysis of an assembling company and proposed the use of inventory management techniques combined with automated systems to improve inventory control. The researchers emphasized that implementing inventory management systems can help organizations reduce unnecessary stock levels, optimize resource utilization, and improve operational efficiency. They also suggested the use of automated agent systems to support inventory monitoring and management processes.

-Furthermore, the study highlighted that automation in inventory management helps businesses maintain accurate records, reduce human errors, and make faster decisions regarding stock replenishment and inventory control. By improving inventory visibility and monitoring, organizations can better respond to customer demands while minimizing operational costs. This literature is related to the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics because the system includes operational

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

monitoring and management features that help businesses track sales and resources efficiently. Similar to the study, H2Ops aims to improve business operations through automation, accurate record management, and real-time monitoring, enabling water refilling station owners to manage their resources more effectively and support better decision-making.

Sanjay Mathrani, Anuradha Mathrani, and Dennis Viehland (2013) in their study "Using Enterprise Systems to Realize Digital Business Strategies" published in the Journal of Enterprise Information Management, enterprise systems (ES) play a significant role in helping organizations achieve their digital business strategies [11]. The researchers explained that organizations invest in enterprise systems to integrate and share information from different departments and sources, allowing them to improve operational efficiency and organizational effectiveness.

-The study examined how enterprise system data are transformed into useful knowledge that supports business decision-making and strategic planning. Through interviews with enterprise system vendors, consultants, and organizations that implemented enterprise systems, the researchers found that integrated information systems help businesses collect, process, and analyze data more effectively. This enables managers to gain valuable insights that support operational improvements and business growth.

-Furthermore, the study emphasized that enterprise systems provide real-time access to business information, allowing organizations to make informed decisions based on accurate and updated data. The researchers concluded that the successful use of enterprise systems contributes to better knowledge management, improved business processes, and the realization of organizational goals through digital transformation.

-This literature is related to the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics because the system also focuses on integrating operational and sales data into a centralized platform. Similar to the study, H2Ops transforms business data into meaningful information through automated analytics, reports, and dashboards that support decision-making, operational monitoring, and business strategy development for water refilling station owners.

Peter G.W. Keen (1980) in his study "Decision Support Systems: A Research Perspective", Decision Support Systems (DSS) are computer-based systems designed to assist managers and decision-makers in analyzing information and making better business decisions [12]. The study explains that DSS combines data, analytical tools, and technology to support the decision-making process rather than replace human judgment. By providing access to relevant information and analytical models, DSS enables managers to evaluate alternatives, solve problems, and make informed decisions more effectively. Keen emphasized that the primary purpose of a DSS is to improve the quality of decision-making by providing timely and accurate information. The system allows users to access data, analyze business situations, and generate insights that support planning and management activities. Furthermore, DSS helps organizations respond more efficiently to operational challenges and changing business conditions.

The study also highlighted that decision support systems bridge the gap between traditional Management Information Systems (MIS) and management science techniques by integrating data processing and analytical capabilities. Through this integration, organizations can transform raw data into meaningful information that supports strategic and operational decision-making

This literature is relevant to the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics because the system provides automated reports, sales analytics, and real-time operational data that support managerial decision-making. Similar to a Decision Support System, H2Ops helps water refilling station owners analyze sales performance, monitor branch operations, identify business trends, and make data-driven decisions that improve efficiency and overall business performance.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Line spacing: Multiple 1.15 li

2.2 Local Studies

The increasing adoption of information systems, cloud computing, and business analytics has encouraged many businesses in the Philippines to embrace digital transformation as a means of improving operational efficiency and maintaining competitiveness. Local studies have shown that many small and medium-sized enterprises (SMEs), including water refilling stations, still rely on manual processes for recording transactions, monitoring inventory, managing customer information, and generating reports. These traditional practices often result in delayed reporting, inaccurate records, and inefficient business operations, highlighting the need for modern information systems that can automate business processes and improve decision-making.

According to *Rey, Villaluz, and Rey* (2024), digital transformation has become a significant factor in improving business operations and economic growth in the Philippines [13]. In their study, the researchers examined the key drivers of digital transformation, including government initiatives, technological advancements, changing consumer behavior, and the increasing adoption of digital technologies by businesses. The study found that digital transformation enables organizations to improve operational efficiency, enhance service delivery, and remain competitive in an increasingly technology-driven environment. However, the researchers also identified several challenges affecting digital transformation, such as limited digital infrastructure, cybersecurity concerns, insufficient digital literacy, and economic constraints. Despite these challenges, the study concluded that collaboration among policymakers, businesses, educational institutions, and other stakeholders is essential to promote digital innovation and maximize the benefits of the digital economy.

The findings of *Rey, Villaluz, and Rey* (2024) are highly relevant to the present study because they emphasize the importance of adopting digital technologies to improve business operations and organizational efficiency. Similar to the concepts discussed in their study, H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics supports the digital transformation of water refilling station businesses by replacing manual processes with a cloud-based management system. The system automates sales recording, inventory monitoring, customer management, and report generation while providing real-time access to business information through automated sales analytics. By utilizing cloud computing and digital technologies, H2Ops enables business owners to improve operational efficiency, reduce manual errors, and make informed decisions based on accurate and centralized data. The study of Rey et al. (2024) strengthens the foundation of this research by demonstrating that digital transformation is essential for enhancing business performance, increasing productivity, and supporting the sustainable growth of organizations through the effective use of information technology.

A study conducted by *Harold Lance A. Espinosa, Kim Ashley O. Maambong, Lisa Marie B. Taylaran, Jamaica A. Bagongon, Apple Grace D. Formentera, Glendell R. Jadraque* (2022) titled "Information Systems Development and Technology Plan for Water Refilling Station" examined the operational challenges encountered by the NADAP Water Refilling Station. The researchers found that the business heavily relied on manual processes for recording transactions, monitoring inventory, and managing daily operations [14]. These traditional methods often resulted in delays, inaccuracies, and difficulties in accessing important business information, ultimately affecting the overall efficiency and productivity of the organization.

The study emphasized that the absence of a well-structured information system limits an organization's ability to effectively manage its operations and achieve both short-term and long-term business goals. Manual record-keeping increases the risk of human error, data loss, and inconsistencies in reports, making it difficult for management to monitor performance and make informed decisions. As business operations grow, these challenges become more significant and can negatively impact service quality and operational effectiveness. Through surveys, interviews, and business process analysis, the researchers identified several areas that required technological improvement, including transaction processing, inventory management, and cloud computing integration.

Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

The findings revealed that implementing an information system could automate routine tasks, improve data accuracy, streamline business processes, and provide better access to information. The researchers highlighted that modern information technologies can help organizations improve operational efficiency while reducing the workload associated with manual record management.

Furthermore, the study proposed the development of a technology plan that incorporates information systems to support business operations. The proposed solution aimed to centralize data management, facilitate faster transaction processing, and enhance organizational control. By integrating digital technologies into business processes, water refilling stations can improve service delivery, strengthen operational monitoring, and increase overall productivity.

The researchers concluded that adopting information systems is essential for modern water refilling stations seeking to improve efficiency, accuracy, and business performance. The implementation of digital solutions enables organizations to better manage resources, monitor transactions, and support data-driven decision-making. As a result, information systems play a crucial role in helping water refilling stations remain competitive and responsive to customer needs.

This study is highly relevant to the present research because the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics is designed to address many of the same operational challenges identified by *Espinosa et al.* (2022). Similar to their findings, the proposed system seeks to automate transaction management, centralize business data, improve operational monitoring, and provide real-time reports that support efficient business management and decision-making.

Junrie B. Matias and Alexander A. Hernandez (2021) in their study titled "Cloud Computing Adoption Intention by MSMEs in the Philippines," cloud computing has emerged as a transformative technology that enables businesses to access information technology resources through the internet without the need for extensive investments in hardware, software, and maintenance [15]. Despite its potential benefits, cloud computing adoption among Micro, Small, and Medium Enterprises (MSMEs) in the Philippines remains relatively low. The researchers sought to identify the factors influencing the intention of MSMEs to adopt cloud computing using the Technology–Organization–Environment (TOE) framework.

The study found that organizational and environmental factors significantly influence the adoption of cloud computing among Philippine MSMEs. Specifically, competitive pressure and regulatory support were identified as key drivers that encourage businesses to adopt cloud-based technologies. Businesses are more likely to embrace cloud computing when they perceive that competitors are utilizing similar technologies and when government policies and regulations support technological innovation and digital transformation.

Furthermore, the researchers emphasized that cloud computing provides several advantages to MSMEs, including reduced operational costs, improved flexibility, enhanced scalability, and greater accessibility to business information. By utilizing cloud-based services, organizations can access computing resources and applications without maintaining costly information technology infrastructure. These benefits make cloud computing an attractive option for businesses seeking to improve operational efficiency and competitiveness.

The study also highlighted the importance of understanding the organizational readiness of MSMEs when implementing cloud technologies. Factors such as management support, technological capability, and awareness of cloud computing benefits play crucial roles in the adoption process. Businesses that are prepared to embrace technological innovation are more likely to successfully integrate cloud solutions into their operations.

The researchers concluded that cloud computing can significantly contribute to the growth and competitiveness of MSMEs in the Philippines. However, successful adoption requires supportive regulatory environments, organizational preparedness, and recognition of the strategic advantages offered by cloud-based technologies. The findings contribute to the growing body of knowledge on technology adoption and provide valuable insights for businesses considering cloud computing implementation.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

This study is relevant to the present research because the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics is built on cloud computing technology. Similar to the findings of Matias and Hernandez (2021), the proposed system aims to leverage cloud-based services to improve accessibility, reduce operational costs, centralize business data, and enhance decision-making capabilities for water refilling station operators.

Jun R Grimaldo, Chin Uy, Ronaldo A Manalo, Ronald R Fernandez (2025) in their study titled "Factors Affecting Online Technology Adoption of Small Businesses: A Case of Third-Class Municipality in the Philippines," the rapid advancement of online technologies has significantly transformed business operations and customer interactions [16]. However, many small businesses, particularly those located in less developed areas, face challenges in adopting digital technologies due to limited resources, inadequate infrastructure, and lack of technical skills. These limitations hinder their ability to improve operational efficiency, expand market reach, and remain competitive in an increasingly digital business environment.

The study utilized Rogers' Diffusion of Innovation Theory to examine the factors influencing the adoption of online technologies among small businesses in a third-class municipality in Marinduque, Philippines. The researchers investigated several variables, including relative advantage, organizational readiness, security concerns, cost implications, managerial characteristics, and intention to adopt online technology. Through a mixed-methods approach and census sampling, the study analyzed how these factors affect business owners' decisions regarding technology adoption.

The findings revealed that organizational readiness, perceived costs, and managerial characteristics significantly influence the intention of small businesses to adopt online technologies. Businesses with sufficient technological resources, supportive management, and a positive perception of technology benefits were more likely to implement digital solutions. On the other hand, concerns regarding costs, infrastructure limitations, and lack of technical expertise were identified as major barriers to adoption.

Furthermore, the researchers emphasized that online technologies should be viewed as strategic assets that can enhance business performance rather than as additional burdens. The adoption of digital tools enables businesses to improve communication, streamline operations, increase productivity, and better respond to customer needs. The study also highlighted the importance of preparing organizations for technological change through training, awareness programs, and infrastructure development.

The study concluded that successful technology adoption among small businesses depends on both organizational and managerial factors. By improving readiness and recognizing the benefits of digital technologies, small businesses can enhance their competitiveness and long-term sustainability. The findings provide valuable insights for policymakers, business owners, and organizations seeking to promote digital transformation among small enterprises in the Philippines.

This study is relevant to the present research because the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics involves the adoption of cloud-based technology to improve business operations. Similar to the findings of Grimaldo et al., the successful implementation of the proposed system depends on organizational readiness, management support, and the willingness of business owners to embrace digital transformation. The study supports the need for technological solutions that can improve efficiency, decision-making, and competitiveness among small businesses such as water refilling stations.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Line spacing: Multiple 1.15 li

2.3 Related Literature

An Integrated Model of Information Systems Adoption in Small Businesses

A study conducted by *James Y.L. Thong* (1999) titled "An Integrated Model of Information Systems Adoption in Small Businesses" examined the factors that influence the adoption of information systems (IS) among small businesses [17]. Drawing from technological innovation theories, the study developed an integrated model that identified decision-maker characteristics, information system characteristics, organizational characteristics, and environmental characteristics as the primary determinants of IS adoption. To validate the proposed model, the researcher conducted a questionnaire survey involving 166 small businesses.

The findings revealed that certain characteristics of business leaders significantly influence the decision to adopt information systems. Specifically, CEOs who demonstrated innovativeness and possessed a higher level of information systems knowledge were more likely to support and implement technological solutions within their organizations. The study also found that innovation-related factors, such as relative advantage, compatibility, and complexity of the information system, play an important role in influencing adoption decisions. Businesses were more likely to adopt information systems when they perceived clear benefits and when the technology was compatible with their existing operations.

In addition, organizational characteristics such as business size and employees' level of information systems knowledge were found to significantly influence the adoption and utilization of information systems. The study concluded that while CEO and innovation characteristics affect the decision to adopt information systems, the extent of actual system usage is primarily determined by organizational factors. Interestingly, environmental competition was found to have no direct influence on the adoption of information systems among small businesses.

The study highlights the importance of organizational readiness, management support, and technological compatibility in the successful implementation of information systems. These findings suggest that businesses are more likely to embrace digital technologies when management understands the benefits of the system and when employees possess the necessary skills to utilize the technology effectively.

This study is relevant to the present research because the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics involves the adoption of an information system within a small business environment. Similar to Thong's findings, the success of the proposed system may depend on management support, organizational readiness, employee knowledge, and the perceived benefits of the technology. The study provides a theoretical foundation for understanding how water refilling station owners and staff may adopt and utilize a cloud-based operations management system.

Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance

Tim Mather, Subra Kumaraswamy, and Shahed Latif (2009) in their book "Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance", cloud computing provides organizations with significant benefits in terms of accessibility, scalability, and cost reduction [9]. However, the authors emphasized that businesses must also address security and privacy concerns when storing and managing data in cloud environments. As organizations increasingly rely on cloud-based services, protecting sensitive information becomes a critical aspect of successful cloud adoption.

The authors discussed key cloud security principles, including confidentiality, integrity, and availability of data. They explained that organizations should implement proper security measures such as identity and access management (IAM), authentication, authorization, and auditing mechanisms to ensure that only authorized users can access cloud

Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

resources. The study also highlighted the importance of compliance with security standards and regulations to maintain the privacy and protection of business information.

Furthermore, the book emphasized that cloud security frameworks help organizations reduce risks associated with unauthorized access, data breaches, and system vulnerabilities. By implementing effective security practices, businesses can increase trust in cloud services and ensure the reliability of their information systems.

This literature is relevant to the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics because the system stores operational and sales data in a cloud environment. Similar to the study, H2Ops requires secure user authentication, controlled access, and proper data management to protect business information. The concepts discussed by Mather, Kumaraswamy, and Latif support the importance of integrating security measures into cloud-based systems to ensure data confidentiality, reliability, and safe access for administrators and staff.

Using Enterprise Systems to Realize Digital Business Strategies

Formatted: Line spacing: Multiple 1.15 li

According to *Sanjay Mathrani, Anuradha Mathrani, and Dennis Viehland* (2013) in their study "Using Enterprise Systems to Realize Digital Business Strategies" published in the Journal of Enterprise Information Management, enterprise systems (ES) play a significant role in helping organizations achieve their digital business strategies [11]. The researchers explained that organizations invest in enterprise systems to integrate and share information from different departments and sources, allowing them to improve operational efficiency and organizational effectiveness.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

The study examined how enterprise system data are transformed into useful knowledge that supports business decision-making and strategic planning. Through interviews with enterprise system vendors, consultants, and organizations that implemented enterprise systems, the researchers found that integrated information systems help businesses collect, process, and analyze data more effectively. This enables managers to gain valuable insights that support operational improvements and business growth.

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Furthermore, the study emphasized that enterprise systems provide real-time access to business information, allowing organizations to make informed decisions based on accurate and updated data. The researchers concluded that the successful use of enterprise systems contributes to better knowledge management, improved business processes, and the realization of organizational goals through digital transformation.

This literature is related to the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics because the system also focuses on integrating operational and sales data into a centralized platform. Similar to the study, H2Ops transforms business data into meaningful information through automated analytics, reports, and dashboards that support decision-making, operational monitoring, and business strategy development for water refilling station owners.

Decision Support Systems

Formatted: Line spacing: Multiple 1.15 li

According to *Peter G.W. Keen* (1980) in his study "Decision Support Systems: A Research Perspective", Decision Support Systems (DSS) are computer-based systems designed to assist managers and decision-makers in analyzing information and making better business decisions. The study explains that DSS combines data, analytical tools, and technology to support the decision-making process rather than replace human judgment [12]. By providing access to relevant information and analytical models, DSS enables managers to evaluate alternatives, solve problems, and make informed decisions more effectively.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Keen emphasized that the primary purpose of a DSS is to improve the quality of decision-making by providing timely and accurate information. The system allows users to access data, analyze business situations, and generate insights that support planning and management activities. Furthermore, DSS helps organizations respond more efficiently to operational challenges and changing business conditions.

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

The study also highlighted that decision support systems bridge the gap between traditional Management Information Systems (MIS) and management science techniques by integrating data processing and analytical capabilities. Through this integration, organizations can transform raw data into meaningful information that supports strategic and operational decision-making.

Cloud-Based E-Business System

Formatted: Line spacing: Multiple 1.15 li

According to *Ni Made Satvika Iswari, Harry B. Santoso, and Zainal A. Hasibuan* (2018) in their study "Cloud-Based E-Business System for Small and Medium Enterprises: A Review of the Literature" presented at the Third International Conference on Informatics and Computing (ICIC), Small and Medium Enterprises (SMEs) play an important role in economic growth and employment generation [18]. However, many SMEs still struggle with limited productivity and difficulties in adopting Information Technology systems due to lack of expertise, resources, and infrastructure.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

The study explained that cloud-based e-business systems provide an effective solution for SMEs because they reduce the need for expensive infrastructure and technical maintenance. Through cloud computing, businesses can access digital services, manage operations efficiently, and improve organizational productivity without large operational costs. The researchers also highlighted that cloud-based systems support better data management, accessibility, and business integration.

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Furthermore, the study identified important factors affecting the adoption of cloud-based systems, including business characteristics, technology readiness, and operational needs. The researchers proposed that cloud-based e-business solutions help SMEs improve business processes, enhance competitiveness, and support long-term business growth.

Cloud Computing Services

Formatted: Line spacing: Multiple 1.15 li

According to *Hamid R. Motahari-Nezhad, Bryan Stephenson, and Sharad Singhal* (2009) in their article titled "Outsourcing Business to Cloud Computing Services: Opportunities and Challenges" cloud computing has significantly transformed the way organizations manage their business operations [19]. The authors explained that businesses have increasingly adopted internet-based technologies to improve efficiency, reduce operational costs, and enhance competitiveness in a rapidly evolving business environment. Through cloud computing, organizations can access software applications, computing resources, and data storage services without the need to invest heavily in physical infrastructure and maintenance.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

The authors highlighted that traditional business operations often require organizations to maintain their own information technology systems, including servers, software, and technical support personnel. These requirements can be costly and difficult to manage, particularly for small and medium-sized enterprises (SMEs). Cloud computing offers an alternative solution by allowing businesses to outsource information technology services to third-party providers, enabling them to focus more on their core business activities while reducing technology-related expenses.

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Furthermore, the study emphasized several benefits of cloud computing, including improved business agility, scalability, flexibility, and operational efficiency. Organizations can quickly adapt to changing business demands, expand their technological capabilities when needed, and access business information from any location with an internet connection. These advantages contribute to faster decision-making, improved collaboration, and enhanced organizational performance.

Despite these benefits, the authors also discussed challenges associated with cloud computing adoption. These challenges include concerns regarding data security, privacy protection, service reliability, and dependence on external service providers. The study stressed the importance of implementing appropriate security measures and selecting reliable cloud service providers to ensure the protection and integrity of business information.

The article concluded that cloud computing services provide organizations with a cost-effective and flexible approach to managing business operations. By leveraging cloud technologies, businesses can improve efficiency, reduce infrastructure costs, and gain greater access to information that supports strategic decision-making and long-term growth.

This literature is relevant to the present study because the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics utilizes cloud computing technology to centralize business data, monitor operations in real time, and generate analytical reports. Similar to the concepts presented by Motahari-Nezhad et al. (2009), the proposed system aims to improve operational efficiency, accessibility, and decision-making through cloud-based services.

Business Activity Monitoring

Formatted: Line spacing: Multiple 1.15 li

According to *Velibor Božić* (2024) in his work titled "Business Activity Monitoring" Business Activity Monitoring (BAM) is a management approach that integrates several technologies, including business intelligence, business process management, balanced scorecards, and integrated application solutions, to provide organizations with real-time information for decision-making. BAM is designed to collect, analyze, and present business data from various internal and external sources, enabling managers to monitor organizational performance and respond quickly to operational issues and opportunities.Božić emphasized that the primary purpose of Business Activity Monitoring is to support real-time decision-making by providing timely, accurate, and relevant information about business activities. Unlike traditional reporting systems that rely on historical data, BAM continuously monitors ongoing operations and delivers immediate insights into business performance. This allows managers to identify problems as they occur, evaluate their impact, and implement corrective actions without delay.

The study further explained that BAM systems enable organizations to combine data from multiple sources and transform them into meaningful information through real-time analysis. These systems provide interactive dashboards, graphical reports, and performance indicators that present data in a clear and intuitive manner. As a result, users can quickly access information, identify trends, and understand the causes of operational issues. The visualization capabilities of BAM systems improve transparency and make complex business information easier to interpret.

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Furthermore, the author highlighted that Business Activity Monitoring contributes to organizational efficiency by enhancing visibility into business processes and improving managerial control. By continuously tracking key performance indicators (KPIs), organizations can measure performance against objectives, monitor operational progress, and ensure that business activities align with strategic goals. This capability supports proactive management and enables faster responses to changing business conditions.

The study concluded that Business Activity Monitoring is an essential component of modern business management because it improves decision-making speed, operational transparency, and organizational responsiveness. Through real-time monitoring and analysis, businesses can gain valuable insights into their operations and achieve better performance outcomes.

This literature is highly relevant to the present study because the proposed H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics incorporates real-time monitoring, dashboard reporting, and sales analytics features. Similar to the concepts presented by Velibor Božić (2024), the proposed system aims to provide water refilling station operators with immediate access to operational data, key performance indicators, and business insights that support efficient management and informed decision-making.

Formatted: Line spacing: Multiple 1.15 li

2.4 Synthesis of the Reviewed Literature

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The reviewed foreign and local studies collectively establish a compelling case for the development and implementation of integrated operations management systems within the small and medium enterprise (SME) sector, specifically within the water refilling industry. The synthesis of these studies reveals several convergent themes that directly inform the design, implementation, and expected predictive outcomes of H2Ops for High Island V.

Theme 1: The Imperative of Digital Transformation for SME Sustainability

Both foreign and local studies emphasize that technology adoption is no longer a luxury but a fundamental requirement for business survival. Foreign literature highlights how cloud-based systems allow small businesses to achieve the same level of operational oversight as large corporations without the need for massive capital investment in hardware. Local studies, specifically in the Philippine context, focus on how water refilling stations (WRS) must adhere to strict Department of Health (DOH) standards, necessitating more accurate record-keeping than manual logbooks can provide. The convergence of these findings suggests that for High Island V to remain competitive in a saturated market, transitioning to a digital platform like H2Ops is essential to maintain service quality and organizational agility.

Theme 2: Eradicating the Deficiencies of Manual Record-Keeping

The reviewed literature consistently identifies manual processes as the primary source of operational failure in SMEs. Common issues cited include "data leakage," mathematical errors, and the physical vulnerability of paper records. In the specific context of water refilling stations, the literature points out the unique risk of environmental damage—where humidity and water exposure frequently ruin handwritten logs. Researchers stress that automated systems mitigate these risks by enforcing standardized data entry and performing instantaneous calculations. These findings directly validate the problem identification of this study, confirming that the "analog" methods currently used by High Island V are a bottleneck to its efficiency and financial transparency.

Theme 3: Deep Integration of Sales and Inventory Systems

A recurring theme in recent studies is the shift from "siloed" applications to integrated ecosystems. The literature suggests that a Point of Sale (POS) system is only half as effective if it is not directly linked to a real-time inventory database. For a water refilling business, this means that the sale of a gallon must automatically trigger a deduction in water volume, seals, and caps. The reviewed studies demonstrate that integrated systems reduce the "reconciliation lag" and prevent inventory shrinkage. This theme is central to the H2Ops design, which seeks to unify branch sales with a centralized cloud inventory, ensuring that data flows seamlessly from the transaction point to the management dashboard without manual intervention.

Theme 4: Predictive Analytics and Data-Driven Proactivity

An emerging and highly relevant theme in modern literature is the use of predictive modeling, such as Simple Linear Regression, to move businesses from reactive to proactive management. Foreign studies illustrate how historical sales data, when processed through mathematical algorithms, can accurately forecast future demand. While many local SMEs still rely on "gut feeling" for ordering supplies, the literature suggests that data-driven forecasting significantly reduces waste and prevents stockouts. This theme provides the theoretical justification for the H2Ops forecasting module, confirming that analyzing the last 30 days of sales is a statistically sound method for predicting the upcoming week's requirements for High Island V.

Theme 5: Cloud Computing as a Solution for Multi-Branch Management

The literature highlights cloud computing as the definitive solution for businesses struggling with geographic dispersion. Studies on "Cloud-SaaS" (Software as a Service) models show that centralizing data in an online database allows owners to monitor remote branches in real-time. This is particularly relevant for High Island V's city and

provincial branches. The reviewed research indicates that cloud-based systems eliminate the "Information Lag" associated with physical report transport, allowing for a "Single Source of Truth." The consensus among researchers believe that web-based architectures provide the most cost-effective and scalable way for multi-branch SMEs to maintain centralized control.

Theme 6: Security, Accountability, and the Audit Trail

Security is a paramount concern in the reviewed literature, moving beyond simple password protection to include Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA). Studies indicate that manual logbooks offer zero accountability, as entries can be altered without detection. Digital systems, however, provide an "Audit Trail" that logs every user action. The literature emphasizes that in a business where cash handling is frequent, such as a water refilling station, having a secure system with Email OTP (One-Time Password) significantly reduces the risk of internal fraud and ensures that sensitive financial data is accessible only to authorized administrators.

Theme 7: Usability and the Human Factor in System Adoption

Finally, multiple studies emphasize that the technical success of a system is dependent on its usability. The "Technology Acceptance Model" (TAM) frequently appears in the literature, suggesting that if a system is not "perceived as easy to use," staff will resist its adoption. For High Island V, where staff may have varying levels of technical proficiency, the literature advocates for "Responsive Web Design"—ensuring the system works intuitively on any device. This theme informs the H2Ops design philosophy of prioritizing a clean, mobile-friendly interface and minimal training requirements to ensure a smooth transition from paper to digital.

Synthesis Conclusion

Overall, the development of H2Ops, a cloud-based operations management system featuring sales tracking, inventory monitoring, and predictive analytics, is strongly supported by the collective findings of foreign and local research. The reviewed literature establishes that integrated, cloud-hosted systems are the most effective way to eliminate the errors inherent in manual record-keeping while providing the real-time visibility necessary for multi-branch oversight.

By replacing handwritten logs with an automated platform, H2Ops aims to boost the operational efficiency of High Island V, ensuring that data is not only accurate but also actionable. The integration of Simple Linear Regression for sales forecasting aligns with the modern trend of using business intelligence to optimize resource allocation and prevent stockouts.

The convergence of these themes suggests that the benefits of the H2Ops system are robust and grounded in proven software engineering and business management principles. Whether addressing the physical risks of water-damaged paper logs or the strategic need for remote branch monitoring, the literature confirms that an integrated, user-centered, and data-driven approach is the definitive pathway for the modernization and sustained growth of High Island V in the digital age.

References

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

[1] L. C. &. R. Bruggemann, "Welcome to the United Nations," 30 July 2021. [Online]. Available: https://sdgs.un.org › goals.

Formatted: Font: 10 pt

[2] D. O. Health, "Philippine National Standards for Drinking Water of 2017," 23 June 2017. [Online]. Available: http://www.doh.gov.ph;.

[3] F. R. J. &. T. W. Jr, "Enterprise resource planning (ERP)," vol. 25, no. 2, pp. 357-363, 12 December 2006.

[4] R. S. Sandhu, "Role-based Access Control," vol. 46, pp. 237-286, 1998.

[5] K. A. M. MD, "Simple Linear Regression," vol. 46, no. 1, pp. 87-93, Jan 2004.

[6] A. L.-P. &. R. P. Clyde Holsapple, "A unified foundation for business analytics," 2014. [Online].

[7] D. D. &. S. Ram, "Journal of Business Analytics," 2018. [Online]. [Accessed 23 Aug 2018].

[8] K. S. &. R. Wajgi, "Data analysis and visualization of sales data," 2016. [Online]. [Accessed 06 Oct 2016].

[9] S. K. &. S. L. Tim Mather, "Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance," 2009. [Online].

[10] D. P. &. A. Borisov, "Case Study on Inventory Management Improvement," *Information Technology and Management Science,* vol. 18, no. 1, 2015.

[11] A. M. &. D. V. Sanjay Mathrani, "Using enterprise systems to realize digital business strategies," *Enterprise Information Management,* vol. 26, no. 4, 2013.

[12] P. G. Keen, Decision Support Systems, 1980.

[13] "The Impact of Digital Transformation in the Philippines," in *Digital Library*, 2024, pp. 107-113.

[14] K. A. O. M. L. M. B. T. J. A. B. A. G. D. F. G. R. J. Harold Lance A. Espinosa, "ResearchGate," Jan 2022. [Online].

[15] J. B. M. &. A. A. Hernandez, "Cloud Computing Adoption Intention by MSMEs in the Philippines," *IMI,* vol. 22, no. 3, 2019.

[16] C. U. R. A. M. R. R. F. Jun R Grimaldo, "Factors Affecting Online Technology Adoption," 18 May 2024. [Online]. Available: ISSN: 2304-1013 .

[17] J. Y. Thong, "An Integrated Model of Information Systems Adoption in Small Businesses," *Management Information Systems,* vol. 15, no. 4, pp. 187-214, 1999.

[18] H. B. S. Z. A. H. Ni Made Satvika Iswari, "IEEEEXPLORE," 2018. [Online]. [Accessed 1 Aug 2019].

[19] B. S. a. S. S. Hamid R Motahari-Nezhad, "labs.hpe.com," 2009. [Online].

[20] C. U. R. A. M. R. R. F. Jun R. Grimaldo, "Factors Affecting Online Technology Adoption," '. [Online].

Formatted
Formatted
Formatted

Formatted: Justified, Line spacing: Multiple 1.15 li

Formatted: Line spacing: Multiple 1.15 li

CHAPTER 3

METHODOLOGY

This chapter provides a rigorous and comprehensive exposition of the research and development methodology employed by the researchers to design, develop, deploy, and evaluate H2Ops: A Responsive Cloud-Based Operations Management System for High Island V. It outlines the systematic framework used to move from the identification of multi-branch operational bottlenecks to the delivery of a mathematically-driven, cloud-hosted solution. The methodology encompasses the developmental research design, the iterative Agile development lifecycle, the technical architecture of the platform, and the validation techniques used to ensure the platform meets the complex needs of a multi-branch water refilling enterprise.

3.1 Research Design

This study utilizes the Developmental Research Design. Developmental research involves the systematic study of designing, developing, and evaluating programs, processes, and products that meet identified needs within a specific operational or organizational context. It is a structured and iterative approach that moves from problem identification through solution design to system validation, ensuring that the resulting output is both technically sound and practically relevant to the environment in which it will be deployed.

Developmental research as a methodological approach occupies a distinctive position in the research methodology spectrum, bridging the gap between pure theoretical research and applied implementation studies. Unlike experimental research, which seeks to establish causal relationships between variables under strictly controlled conditions, developmental research focuses on creating and refining practical solutions to real-world problems. Unlike purely descriptive research, which documents existing conditions without intervention, developmental research actively intervenes to improve those conditions through the creation of new tools, processes, or systems. In the context of High Island V, this design is used to intervene in an inefficient manual system by introducing a custom-built digital management ecosystem.

The developmental research approach is classified into two primary categories. Type 1 developmental research focuses on the design, development, and evaluation of specific products, tools, or systems. This type employs research methods such as case studies, in-depth interviews, field observations, and document analysis during the design and development phases, followed by evaluation methods including surveys, usability testing, and expert review during the validation phase. Type 2 developmental research focuses on the development, use, and validation of theoretical models or frameworks, often employing literature reviews and Delphi studies.

The present study aligns with Type 1 developmental research, as it focuses on the design, development, and evaluation of a specific technological product: H2Ops, a responsive, cloud-based operations management system. The methodology encompasses several phases that correspond to the natural progression of system development: needs analysis, design specification, development implementation, and evaluation validation. Each phase builds upon the previous, ensuring that the resulting system is grounded in actual operational requirements and validated against real-world performance criteria.

The application of this design begins with an in-depth needs analysis of the current operational practices across High Island V's various city and provincial branches. This initial phase involved a thorough examination of the existing workflows, manual recording methods, and physical reporting mechanisms. By investigating existing operational bottlenecks—specifically the mathematical errors in sales logs, the time-intensive nature of report consolidation, and the "Information Lag" between remote branches—the researchers were able to define the precise functional requirements needed for the H2Ops system.

Formatted: Font: 12 pt
Formatted: Centered, Line spacing: 1.5 lines
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: (Default) Times New Roman, 13
Formatted: Centered, Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: (Default) Times New Roman, 13
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

This phase ensures that the system is not a generic template but a bespoke tool tailored to the water refilling industry. Following the analysis phase, the developmental process focuses on designing and engineering the technological solution. This phase encompasses the planning of the system architecture, the design of the user interface (UI) to ensure it is mobile-responsive for delivery riders, and the development of the core functional modules, including the predictive forecasting engine. By automating manual workflows related to sales and inventory, the developed system provides real-time data updates, enhances container tracking accuracy, and reduces the operational errors that have historically undermined the reliability of High Island V's records.

The evaluation phase assesses the system's usability, functionality, and effectiveness in supporting multi-branch management decision-making through feedback gathered from administrators and staff. This evaluation is not merely a technical verification of code but an empirical assessment of the system's practical value in the water refilling context. The evaluation criteria—functionality, usability, accuracy, security, and efficiency—provide a comprehensive framework for assessing system quality from multiple perspectives, ensuring that the transition from a manual to a digital system is both sustainable and beneficial for the business.

3.2 System Development Methodology

The development of H2Ops follows the Agile Methodology, a process defined by continuous iteration, cross-functional collaboration, and rapid response to change. Unlike traditional linear models such as the Waterfall approach, which proceeds sequentially through distinct phases with limited opportunity for revision, Agile breaks the project into small, manageable increments known as Sprints. This allows the researchers to progressively build and refine the system based on frequent feedback and evolving requirements from High Island V's stakeholders.

The Agile methodology was selected because traditional predictive approaches often fail in dynamic business environments where requirements may shift during development. The Agile values—individuals and interactions over processes and tools, and responding to change over following a rigid plan—are particularly relevant to High Island V, where the transition from paper to digital is a significant cultural shift for the staff. The development process evolved through five repeated phases:

Phase 1: Requirement Analysis and Backlog Management

In this phase, user stories and system requirements are identified and prioritized in a Product Backlog. This ensures that development efforts are aligned with the actual needs of High Island V's owners and staff. Requirements are expressed as user stories, such as: "As an Admin, I want to see a chart of all branch sales so that I can identify which location is growing." These stories capture the operational value that each feature is intended to deliver. Each requirement is evaluated for technical feasibility (such as the integration of the Simple Linear Regression algorithm) and operational impact to establish development priorities.

Additional requirements are identified based on the existing workflow of High Island V, including sales recording, inventory monitoring, branch management, reporting, and user access. The researchers organize these requirements according to their importance and complexity so that essential functions can be developed first. This approach ensures that the system remains focused on the actual problems encountered by the business rather than including unnecessary features.

Phase 2: Iterative Design and Development

During this phase, design and coding occur concurrently. Database schemas (MySQL) and backend routes (Flask) are developed and refined as the system grows. This concurrent approach allows the developers to discover design issues during implementation—such as a conflict in inventory deduction logic—and adjust accordingly before the code is finalized. The system is built incrementally; early sprints focused on core authentication and branch management,

Formatted: Indent: First line: 0", Line spacing: Multiple 1.15 li

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

while later sprints added the Point of Sale (POS), the predictive forecasting module, and the automated reporting tools. This provides early user feedback on working software rather than abstract designs.

Each sprint produces a functional portion of H2Ops that can be reviewed and improved before additional features are developed. The researchers ensure that newly developed modules are compatible with existing components, particularly the relationship between sales transactions, inventory records, and analytics. This incremental approach reduces development risks and allows technical problems to be addressed before they affect the entire system.

Phase 3: Continuous Testing

Testing is not treated as a final step but as an ongoing process integrated throughout the development cycle. Features are verified as they are built to identify and resolve bugs early. This includes Unit Testing (testing individual Python functions like the regression calculation), Integration Testing (ensuring the POS correctly updates the Inventory tables), and Security Testing (verifying that the Email OTP effectively blocks unauthorized access). By testing each feature as it is developed, the team can correct defects when they are most easily addressed, close to the time of their introduction.

Testing also helps ensure that the system produces accurate results from the data entered by users. The researchers verify that sales transactions are correctly recorded, inventory quantities are properly updated, and reports reflect the information stored in the database. Usability testing is also considered to determine whether High Island V staff can understand and operate the system effectively with minimal difficulty.

Phase 4: Deployment and Review

At the end of each sprint, a functional increment of the system is delivered and reviewed. This allows High Island V stakeholders to monitor progress in real-time. The deployment of sprint increments to a staging environment enables branch managers to interact with the software, providing feedback that directly influences subsequent development priorities. This regular review cycle ensures that the development process remains aligned with the unique operational "rhythm" of the water refilling business.

During the review, the researchers demonstrate the completed features and allow users to provide comments regarding functionality, usability, and compatibility with their existing procedures. Any identified issues or requested improvements are documented and considered for the following sprint. This process allows H2Ops to gradually adapt to the actual workflow of High Island V rather than relying entirely on assumptions made during the initial planning stage.

Phase 5: Feedback and Retrospective

Based on feedback gathered from the owners and staff, the researchers identify areas for improvement and adjust the backlog for the next sprint. The retrospective phase involves reflection on the development process itself—what worked well, what caused delays (such as database configuration), and what changes should be made to the team's working methods. This ensures that the development process continuously improves alongside the product, resulting in a system that is both technically robust and perfectly aligned with user expectations.

The feedback collected during each iteration also helps determine whether the developed features successfully address the operational problems identified during the earlier stages of the research. Improvements may involve modifying the user interface, adjusting system processes, correcting database relationships, or refining analytics and reporting functions. Through repeated cycles of development, testing, review, and feedback, the researchers can progressively improve H2Ops until the system meets the defined requirements and objectives of the study.

3.3 System Architecture

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The H2Ops platform utilizes a Three Tier Client-Server Architecture designed to facilitate secure multi-tenant data management and a clean separation of concerns. By isolating the user interface and presentation layer (client-side) from the database storage and business logic processing (server-side), this architecture ensures high system responsiveness, centralized security control, and seamless scalability as High Island V expands to new physical branches.

The architecture allows authorized users from different branches to access the system through a web browser while maintaining centralized processing and data storage. User requests are sent to the server, where authentication, business rules, transaction processing, and data retrieval are performed before the appropriate information is returned to the client. This structure allows H2Ops to maintain a consistent source of operational information while supporting the different access requirements of administrators, branch managers, and staff.

3.3.1 Server-Side Architecture

The server-side serves as the centralized intelligence engine of the H2Ops system. Its primary responsibility is to process incoming client requests, execute complex business logic, communicate with the relational database, and generate dynamic responses. The server-side architecture also ensures that important business processes are handled centrally rather than directly by individual client devices. This provides greater control over data validation, user permissions, transaction processing, and system security. Centralizing these processes also makes the system easier to maintain because updates to the application's logic can be managed from the server without requiring major changes to every device accessing the system. To maintain a professional and manageable codebase, the server architecture is comprised of the following technical components:

Flask Web Framework: The system is built using the Flask micro-framework, which manages URL routing and coordinates the various modules of the application. To ensure modularity, the researchers utilized Flask Blueprints, which organize the system into logical segments: Auth, Sales, Inventory, Customers, Reports, and Users. This modular approach allows for easier debugging and independent development of each system feature. The modular structure also allows the researchers to modify or improve individual components without unnecessarily affecting unrelated parts of the system. This is particularly useful for the Agile development approach, where features are developed and refined through multiple iterations.

Object-Relational Mapping (SQLAlchemy): To bridge the gap between Python's object-oriented structure and the relational database, SQLAlchemy is employed. This abstraction layer translates Python classes defined in the models.py file into SQL tables. Beyond simplifying database operations—such as querying, joining, and updating tables—this layer provides a critical security benefit by protecting the application from SQL injection vulnerabilities. Through SQLAlchemy, the application can manage relationships between important records such as users, branches, products, sales, and inventory. This helps maintain consistency between the application and database while reducing the complexity of writing and managing individual database queries.

Forecasting Engine: A core innovation of the H2Ops platform is the predictive module located in forecasting.py. This engine retrieves the past 30 days of historical sales records for a specific branch and applies a Simple Linear Regression algorithm. This process allows the system toproject a 7-day revenue trendline, providing management with empirical data to support future business decisions. The forecasting results are presented as supplementary information for management and are based on the historical data available in the system. The accuracy of the forecast depends on the quality, completeness, and volume of recorded sales data. Therefore, the forecasting feature is intended to support decision-making rather than replace the judgment of business owners and managers.

Security and Middleware Measures: The server incorporates several layers of protection to safeguard sensitive business data:

Bcrypt Encryption: This library is used to cryptographically hash user passwords with random salt values before they are stored in the database. This ensures that even in the event of a data breach, plaintext credentials remain unreadable.

Role-Based Access Control (RBAC): Access to sensitive endpoints is strictly controlled via session variables (e.g., session['active_branch_id']) and custom Python decorators. For example, administrative functions in users.py are restricted solely to accounts with high-level privileges. This access-control mechanism ensures that users can only perform actions appropriate to their assigned roles. It also helps prevent unauthorized users from accessing sensitive business information or modifying important operational records.

Email OTP Dispatcher: Integrated via Flask-Mail and SMTP protocols, this component enforces multi-factor authentication. It generates a random 6-digit verification code, stores it in a temporary OTP table with a 5-minute expiration timestamp, and dispatches it to the user's registered email for identity verification. The additional verification step strengthens the authentication process by requiring users to provide information beyond their regular login credentials. This provides an additional layer of protection for accounts containing sensitive sales and operational information.

Relational Database Management System (RDBMS): The persistent storage layer is powered by MySQL (facilitated via XAMPP during the local development phase) or PostgreSQL (for remote cloud deployment). The database follows a normalized structure, containing tables for Branch, User, Customer, Product, Sale, SaleItem, InventoryLog, OTP, and PasswordReset. This ensures structural data integrity through the strict application of primary-key and foreign-key constraints. The relational structure allows H2Ops to efficiently organize and retrieve information while maintaining relationships between different operational records. For example, sales transactions can be associated with specific branches, users, products, and customers. This centralized database structure supports the system's reporting and analytics functions by providing organized and consistent data for processing.

3.3.2 Client-Side Architecture

The client-side represents the presentation layer of the H2Ops platform, serving as the primary interface through which authorized staff and administrators interact with the system. Because H2Ops is a web-based solution, it is designed for cross-platform accessibility; users can access the system via standard web browsers on desktops, tablets, or smartphones without the need for localized software installation.

The frontend is constructed using a modern web stack to ensure a responsive and interactive user experience:

HTML5: This serves as the semantic foundation of the application. It establishes the structural framework for all system components, including the administrative dashboards, secure login forms, sales registers, and inventory management logs.

CSS3: Is utilized to govern the visual presentation, styling, and typography of the platform. By implementing responsive grid systems and Flexbox layouts, the researchers ensure that the interface automatically adapts to various screen dimensions, maintaining usability across both mobile and desktop devices.

JavaScript (ES6): JavaScript handles the dynamic interactivity of the client-side. It manages asynchronous UI updates—such as real-time price calculations during product selection in the Point of Sale (POS) module—and smooth interface transitions. This ensures a fluid user experience by reducing the need for constant page reloads.

Chart.js: To support managerial decision-making, Chart.js is integrated into the administrator's dashboard. This library parses JSON-formatted analytics datasets provided by the server to render interactive line graphs, bar charts, and historical sales comparisons. These visualizations allow stakeholders to identify trends at a glance.

Formatted: Font: 10 pt

Formatted: Font: 10 pt

Jinja2 Templating Engine: Jinja2 acts as the bridge between the Flask backend and the frontend. This server-side engine compiles variables, conditional logic, and loops into standard HTML before it is served to the client. This allows the system to display dynamic, context-aware content, such as branch-specific product lists or real-time low-stock alerts tailored to the logged-in user's permissions.

3.3.3 Architectural Benefits

The adoption of a Client-Server Architecture provides several critical strategic advantages for High Island V:

Centralized Data Management: This architecture eliminates the risks associated with "data silos" or localized spreadsheets. Every transaction performed at any branch is synchronized directly with the central database, providing management with a "single source of truth" for all business operations. This centralized approach makes it easier for the owner and authorized managers to access consistent and updated information. Sales, inventory, customer, and delivery records can be managed within the same system, reducing duplicate records and minimizing discrepancies between branch reports. It also supports the automated reporting and sales analytics features of H2Ops because the system can process information from a centralized database.

Computational Efficiency: By offloading resource-intensive tasks—such as complex mathematical forecasting and large-scale database queries—to the server-side, the client interface remains lightweight and fast. This allows the platform to remain fully operational even on lower-powered mobile devices or tablets used on the shop floor. This approach is particularly beneficial for staff members who primarily use the system for recording sales and checking operational information. The device does not need to perform all calculations locally because the server processes the required operations and returns the results to the user. This helps maintain a responsive user experience while supporting the system's analytical functions.

Enhanced Security Isolation: A significant security benefit of this design is that all sensitive database schemas and validation logic are housed on the secure server. Since verification happens server-side, malicious users cannot bypass security protocols or data validation by simply modifying local client scripts or browser code.

The architecture also supports the implementation of authentication and Role-Based Access Control (RBAC), ensuring that users can only access functions appropriate to their assigned roles. This is important for H2Ops because the system handles sensitive operational information such as sales records, inventory data, user accounts, and business reports.

Streamlined Maintenance and Scalability:

The centralized nature of the architecture simplifies the deployment of software updates. When the researchers or administrators deploy a patch or a new feature to the cloud server, the update is instantly available across all branches. This ensures system-wide uniformity without requiring manual updates on individual cashier machines. The architecture also provides a foundation for future expansion of H2Ops. If High Island V adds additional branches or users, the centralized server can accommodate additional system activity without requiring a completely different architecture. Future improvements, such as additional reporting features, expanded analytics, or new operational modules, can also be integrated into the existing system while maintaining the separation between the client, application, and database components.

3.4 Tools and Technologies Used

The development of H2Ops utilizes a set of web-based tools and technologies selected for their reliability, compatibility, security, and suitability for building a multi-branch operations management and monitoring system. Each technology has been deliberately chosen not only for its individual capabilities but also for its ability to integrate

Formatted: Indent: First line: 0.25", Line spacing: Multiple 1.15 li

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

seamlessly with other components of the system, forming a cohesive, secure, and stable development stack that supports the full range of functional requirements identified for H2Ops.

Backend Technology: Python and Flask

Python is used to handle all server-side operations of the system. In conjunction with Flask, a lightweight and modular web framework, it manages user requests, coordinates routing between various blueprints, facilitates communication between the user interface and the database, and processes all backend calculations. Python is widely recognized in backend engineering and is highly suited for building dynamic, data-driven applications like H2Ops. This makes it a practical and dependable choice for managing the server-side logic that powers the system's core features, including real-time sales transactions, inventory monitoring, multi-branch tenant isolation, SMTP-based email OTP dispatching, and automated forecasting projections.

Python's widespread adoption in web development ensures extensive documentation, mature libraries, and abundant learning resources that support ongoing development and maintenance. The integration of Flask-SQLAlchemy simplifies database access and schema management through an Object-Relational Mapping (ORM) layer, protecting the system from SQL injection vulnerabilities. Flask's execution model and blueprint structure align well with the stateless nature of HTTP and the modular design requirements of the POS, sales, inventory, and reporting modules.

Frontend Technologies: HTML, CSS, JavaScript, and Jinja2

HTML, CSS, and JavaScript are used to develop the user-facing interface of the system. These technologies work together to create a visually organized, responsive, and accessible interface that allows High Island V administrators, managers, and cashiers to navigate and interact with the system efficiently across desktops, tablets, and smartphones. HTML5 provides the structural foundation of each page, CSS3 governs the visual presentation and responsive layout grids to ensure a clean design, and JavaScript handles dynamic client-side interactions—such as real-time total calculations in the Point-of-Sale (POS) cashier interface—enhancing overall responsiveness.

The integration of Jinja2, Flask's native template engine, allows the backend to compile database variables, loops, and conditional statements into HTML templates dynamically before serving them to the client browser, mitigating Cross-Site Scripting (XSS) risks. Additionally, Chart.js is integrated to parse JSON analytics data retrieved from the server, rendering interactive line charts and bar graphs that display sales trends and forecasting projections directly on the administrator's dashboard.

Database Technology: MySQL and PostgreSQL

-H2Ops utilizes a relational database management system to store, organize, and manage all operational data. During local development and testing, MySQL is used via XAMPP to host the `h2ops_db` database, providing a fast and lightweight relational database environment. For cloud-based staging and production environments, PostgreSQL is utilized, offering enterprise-grade reliability, scalability, and robust concurrency control. The database schema stores critical tables including branch records, user profiles, customers, products, sales details, inventory logs, and security OTP codes, maintaining strict data integrity through foreign key constraints and transactional boundaries.

The database's relational model is highly aligned with High Island V's multi-branch operational structure, where entities such as branches, products, sales, and inventory logs have well-defined relationships. Through SQLAlchemy's ORM abstraction, the system can communicate seamlessly with both MySQL and PostgreSQL without requiring modifications to the backend code. Database transactions ensure that operations—such as committing a sale and simultaneously deducting the corresponding stock of containers, seals, and caps—complete successfully or roll back entirely, preventing inconsistent inventory states.

Development Tools

-Visual Studio Code (VS Code) is used as the primary code editor for writing and managing the source code of H2Ops. It offers a lightweight yet feature-rich environment, supporting syntax highlighting, extension-based debugging, and git integration to streamline the development workflow. XAMPP is used as the local server environment, enabling developers to run and test the local MySQL database and Apache administrative interfaces. Additionally, Python's native Virtual Environments (`venv`) are utilized to isolate project-specific dependencies, ensuring a clean and reproducible runtime environment that prevents package version conflicts.

Browser: Google Chrome

-Google Chrome is used as the primary web browser for testing, debugging, and validating the system's responsive user interface and client-side scripts. As a widely adopted web browser, Chrome provides a standardized testing environment equipped with powerful developer tools, which allows the development team to inspect layouts, profile performance, verify JavaScript behavior, and ensure that the H2Ops dashboard performs consistently across cashiers' and administrators' devices.

Technology Selection Rationale

-These tools were selected because they are open-source, widely documented, and highly compatible with one another, providing a secure, stable, and cost-effective foundation for developing a responsive operations management system. Collectively, they represent an accessible stack that does not require expensive licensing, making it appropriate for the resource constraints of a multi-branch business like High Island V.

The use of widely adopted tools ensures that future developers or technical personnel will be able to maintain, update, or expand H2Ops with ease as the business requirements of High Island V evolve. The minimal licensing and maintenance overhead associated with Python, Flask, MySQL, and PostgreSQL ensures that the system provides a low total cost of ownership compared to proprietary enterprise resource planning (ERP) platforms.

3.5 Data Gathering Techniques

Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

The researchers utilized a combination of qualitative and quantitative data gathering techniques to establish a comprehensive understanding of the operational workflows, administrative pain points, and technical requirements of High Island V. To ensure the reliability and validity of the gathered requirements, three distinct data gathering techniques were implemented: interviews, direct observations, and document analysis. This multi-method approach enabled the triangulation of operational facts and minimized subjective bias in the system design. The information collected through these techniques served as the foundation for identifying the system requirements of H2Ops. The gathered data were analyzed to determine which business processes could be automated and which information should be included in the system's database, dashboards, reports, and sales analytics. The results also helped the researchers determine the appropriate user roles, system permissions, and features needed by the business.

3.5.1 Technique 1: Interview

Semi-structured interviews were conducted with key stakeholders of High Island V, including the business owners or administrators, branch managers, and cashiers. This allowed the researchers to gather qualitative insights regarding daily routines, administrative challenges, security concerns, and expectations for the proposed H2Ops system. The interviews provided the researchers with direct information from individuals who are involved in the actual operation of the water refilling stations. Their responses helped identify difficulties that may not be immediately visible through system observation alone. The information gathered was also used to determine the features that would provide the greatest operational benefit, particularly in sales recording, inventory monitoring, reporting, and branch management.

Interview Protocol Design

The interview protocol was structured into six distinct sections to systematically capture the operational realities at each level of the organization.

Section 1: Operational Overview focused on understanding the organizational hierarchy, roles of cashiers and managers, communication channels between provincial and urban branches, and the general business flow of the water refilling stations.

Section 2: Sales Recording Practices investigated how daily transactions such as walk-ins, refills, container purchases, and deliveries are processed, how subtotals and changes are computed, how receipts are issued through manual carbon-copy or pre-printed logs, and how cash register balances are reconciled at the end of each shift.

Section 3: Inventory Management Practices explored how the stock of purified water, empty containers such as 5-gallon round or slim bottles, caps, and plastic seals is tracked; how damages, losses, or container returns are recorded; and how restocking thresholds are determined.

Section 4: Reporting and Analysis examined how branch managers compile daily and weekly sales summaries, how reports are transmitted to the central administrators, how long this transmission takes, and how administrators analyze performance trends or make sales forecasts.

Section 5: Operational Challenges gathered data on specific friction points, such as mathematical discrepancies in cash reconciliation, undocumented inventory loss or shrinkage, long delays in provincial branch report delivery, and security concerns regarding remote access.

Section 6: System Expectations solicited feedback on user interface preferences, willingness to adopt a digital platform, concerns about internet connectivity issues, and requirements for administrative controls.

Interview Implementation

The interviews were conducted face-to-face and through video conferencing to accommodate stakeholders in remote provincial locations. A total of eight participants, including one owner or administrator, two branch managers, and five branch cashiers, were interviewed. Each session was recorded with the participants' consent, and key responses were transcribed and coded to extract recurring themes. These coded themes directly informed the system's user stories and access control requirements.

3.5.2 Technique 2: Observation

To validate the findings from the interviews and identify unarticulated operational habits, the researchers conducted direct, non-participatory observations of the daily workflows at selected High Island V branches.

Observation Protocol Design - The observation was guided by a structured protocol to ensure consistency across different operational environments.

Temporal Coverage - Observations were scheduled during peak hours, from 10:00 AM to 2:00 PM, when customer traffic is highest, as well as during shift handovers and end-of-day cash reconciliation. This coverage spanned a total of three business days per branch.

Spatial Coverage - The researchers observed activities at both urban or flagship and provincial or remote branches, focusing on the layout of the checkout counter, the proximity of inventory storage, and the communication between cashiers and delivery personnel.

Observation Dimensions - The observation focused on three main dimensions: the duration of tasks, such as the time taken to process a refill and record it; the physical movement of inventory; and the tools used to execute processes.

Specific Observation Targets - Inventory Management Activities involved observing how empty gallons returned by customers are inspected, counted, and stored, and when and how caps and seals are applied to filled containers. Report

Preparation involved observing the cashier writing manual receipts, totaling sales on a physical calculator, and filling out the daily logbook, as well as the manager's process of compiling these logs into weekly summaries. Workaround Identification involved noting informal practices used by cashiers to speed up service during peak hours, such as writing sales on scrap paper before transferring them to logbooks, or verbal agreements on container deposits without formal records.

Observation Documentation - The researchers compiled detailed field notes, process flow diagrams, and temporal logs. These notes highlighted the high cognitive load placed on cashiers during peak hours and the physical bottlenecks in tracking inventory items like caps and seals, confirming the necessity of automated calculations and inventory deductions in H2Ops.

3.5.3 Technique 3: Document Analysis

To supplement qualitative data with objective business history, the researchers analyzed the physical and digital documents currently used by High Island V to manage operations. Document Categories Analyzed Sales Records included manual carbon-copy receipt books, delivery logbooks, and cashier log sheets tracking walk-in refilling transactions. Inventory Records included physical stock cards for caps, seals, and containers, and delivery transit receipts for inventory transferred from the central warehouse to provincial branches. Financial Records included monthly expense summaries, cash reconciliation reports, and records of delivery driver collections and outstanding customer balances. Administrative Documents included employee attendance logs, shifting schedules, and existing operational guidelines or instructions for cashiers.

Document Analysis Procedures

The researchers reviewed historical records spanning a three-month period to analyze data structures and mathematical integrity. They traced data entries from the cashier's receipt to the branch logbook, and finally to the administrative summaries. This analysis revealed a high frequency of transposition errors in manual calculations and highlighted the lack of standardized structures for tracking container deposits.

Baseline Data Establishment

The quantitative data extracted from these documents served as the baseline for H2Ops data schemas and validation test cases. For instance, the historical sales data was used to test the mathematical accuracy of the simple linear regression forecasting model during the system testing phase.

3.5.4 Triangulation and Validation

To ensure that the software requirements derived from the data gathering process were highly accurate, credible, and free from individual investigator bias, the researchers implemented a rigorous triangulation and validation framework.

Triangulation Procedures - The data gathered from interviews, observations, and document analysis were cross-analyzed through three triangulation methods. Convergence Analysis identified points of agreement across all three sources. For example, if cashiers reported reporting delays during the interview, were observed struggling to compile daily sheets during observation, and document analysis revealed missing logs from provincial branches, then the need for a real-time cloud dashboard was confirmed.

Divergence Analysis investigated discrepancies between data sources. For instance, when interviews indicated that inventory audits occurred weekly, but document analysis showed missing weeks and observation revealed that cashiers relied on physical estimations rather than exact counts, the researchers recognized a critical gap in automated inventory deduction controls. Complementarity Analysis used one technique to explain findings from another. The qualitative explanations from cashiers regarding why they bypassed certain guidelines during interviews were mapped directly to the physical counter bottlenecks observed by the researchers.

Validation Criteria

The validity of the developmental research findings was evaluated using the Guba and Lincoln framework for qualitative and system development credibility. Credibility was established through prolonged engagement, spending several weeks analyzing the business workflow; member checking, presenting system wireframes and process models back to the owners and cashiers to verify accuracy; and peer debriefing. Transferability was ensured by providing a thick description of the multi-branch water refilling business context. While H2Ops is designed specifically for High Island V, the detailed operational parameters allow the methodology and architecture to be transferred to other small-to-medium multi-branch retail enterprises.

Dependability was achieved through a detailed audit trail of all design decisions, requirement adjustments, and sprint logs in the Agile development process. This allows future researchers to audit the development lifecycle step by step. Confirmability was maintained by ensuring that system designs, database schemas, and regression algorithms are strictly derived from raw empirical data, including the baseline documents and observed workflows, rather than the subjective preferences of the development team.

3.6 Testing Procedures

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The validation of H2Ops follows a structured three phase testing process to make sure the system is stable and useful before it is fully used. This process starts from testing small parts of the system and continues until the whole system is tested in a real working setup similar to the actual operations of High Island V. Each phase has its own purpose to confirm that the system works properly and meets the needs of its users. This kind of step by step testing is a common practice in software development. Starting from small tests and moving to full system testing helps find errors early and prevents bigger problems later. Careful testing is important so the system works correctly and safely before it is used in real business operations.

3.6.1 Phase 1 Unit Testing

This phase focuses on testing each part of the system one by one. Examples include backend functions and user interface features. Each part is tested alone to make sure it works correctly before combining it with other parts of the system. For systems that use databases, it is important to test database functions, login processes, and calculation functions to make sure data is correct and secure. Testing tools can also be used to repeat tests when the system is updated. Unit testing for H2Ops includes several areas. Database Testing checks if data is saved, updated, deleted, and retrieved correctly. It also checks if related data stays consistent across tables like users, customers, products, sales, and inventory. It ensures that changes either complete fully or do not happen at all to avoid errors. Calculation Testing checks if the system correctly computes values such as total sales, discounts, delivery fees, profit, and inventory updates when items are sold. Authentication Testing checks if the login system works properly. It ensures correct users can log in, wrong passwords are rejected, and user roles like staff and admin have proper access limits. It also checks session handling to keep accounts secure. Input Validation Testing checks if the system rejects wrong or invalid data. It ensures numbers are used where needed, dates are correct, and inputs like product codes and emails follow the correct format.

3.6.2 Phase 2 System Testing

This phase tests the whole system working together. It checks if all parts function properly as one system under normal and heavy use. It also checks if the system stays stable when many users are using it at the same time. System testing for H2Ops includes several types. Workflow Testing checks if full processes work correctly from login to completing a transaction and generating reports. It makes sure data flows correctly from one step to another. Load Testing checks how the system performs when many users are using it at the same time, such as multiple cashiers processing sales in different branches. It measures speed and checks if the system slows down under pressure. Data

Accuracy Testing checks if updates in one part of the system correctly reflect in other parts. For example, when a sale is made, inventory should automatically update. Security Testing checks if unauthorized users are blocked and if user roles are followed correctly. It also checks if data is protected during use. Stress Testing checks how the system behaves under extreme conditions, such as very high transaction volume or large amounts of data. This helps identify system limits.

3.6.3 Phase 3 User Acceptance Testing

This phase involves real users such as managers and staff testing the system in a controlled environment. The goal is to check if the system is easy to use and fits actual work needs. User Acceptance Testing is focused more on real use rather than technical performance. It checks if users can complete tasks easily and understand the system. This phase includes several activities. Test Scenarios require users to perform normal tasks such as processing orders, updating inventory, and generating reports. Each task has a target result to check if it is done correctly. Observation involves watching users while they use the system. It helps identify confusion, mistakes, or difficulties when navigating the system. Feedback Collection gathers user opinions through surveys or interviews. It checks how users feel about the system, how easy it is to use, and what improvements they suggest.

3.6.4 Overall Quality Assurance

The three testing phases work together to ensure the system is reliable and ready for use. Early testing finds small problems before they become bigger issues. Final testing ensures the system works well in real conditions and meets user needs. Good quality assurance checks three main things. First is correctness of the system. Second is performance under real use. Third is user satisfaction. For systems like H2Ops, testing also ensures that data is accurate, the system is fast enough, and users trust the results. This helps make sure the system is useful in real business operations. Overall, this testing process ensures that H2Ops is ready, stable, and helpful for daily operations of High Island V.

To make the study clear, the following terms are defined in simple meaning.

Business Analytics refers to tools and methods used to collect and study business data to help managers make decisions. In this study, it focuses on sales, inventory, and operations data.

Business Intelligence refers to the overall process of turning data into useful information for decision making in an organization.

Dashboard refers to a screen in the system that shows important information like sales and inventory in a simple visual form such as charts and tables. H2Ops refers to the system developed in this study that helps manage operations of High Island V water stations across multiple branches.

Inventory refers to the materials used in the water station such as bottles, caps, and seals.

Inventory Tracking refers to the process of monitoring stock levels and recording changes when items are added, used, or adjusted.

Manual Record Keeping refers to the old method of writing records by hand or using basic spreadsheets before the system was used.

Micro Small and Medium Enterprises refers to small businesses with limited staff and resources like water refilling stations.

Multi Branch Environment refers to a business setup where operations are located in different places but managed together in one system.

Operational Efficiency refers to how well a business uses its time, money, and resources to produce results with less waste.

Point of Sale refers to the place or system where customers pay for goods or services.

Real Time Monitoring refers to the ability of the system to show updated information immediately after transactions happen.

Sales Transaction refers to a record of a customer purchase including items bought, price, and payment.

Stockout refers to a situation where items are no longer available in inventory.

Water Refilling Station refers to a business that cleans and sells drinking water to customers.

Web Based System refers to a system that runs on the internet and can be accessed using a browser without installing software.

3.7 Evaluation Criteria

Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

The evaluation of H2Ops is conducted based on the following five core criteria to ensure that the system meets both technical standards and the operational needs of High Island V. Each criterion has been carefully selected to reflect the specific functional requirements and operational priorities identified during the needs analysis phase of this study, providing a comprehensive and structured basis for assessing the system's overall readiness and effectiveness for deployment in a real water refilling station environment.

The selection of these five criteria—functionality, usability, accuracy, security, and efficiency—draws upon established frameworks for information systems evaluation. In alignment with the literature reviewed in Chapter 2, evaluating a small business information system requires a multi-dimensional approach that balances technical performance with organizational and human factors (Thong, 1999). Information systems development for small enterprises, such as water refilling stations, emphasizes the necessity of assessing functionality and accuracy to solve operational delays (Espinosa et al., 2022), while cloud-based system implementation requires a strict focus on security risks (Mather et al., 2009) and system usability (Grimaldo et al., 2025). Furthermore, assessing how enterprise data is processed into actionable dashboard analytics to support management decisions (Mathrani et al., 2013) highlights the role of Decision Support Systems (DSS) in operational evaluation (Keen, 1980).

3.7.1 Criterion 1: Functionality

This criterion assesses whether the system successfully executes all intended features, including the Point of Sale (POS) module, real-time sales and inventory monitoring, automated report generation, and dashboard data visualization, ensuring that all components operate without failure. Evaluators verify that each feature performs as intended under normal operating conditions and that no critical functions are absent or producing incorrect outputs during the testing phase. Functionality evaluation involves systematic testing of each system feature against its specified requirements, verification that feature outputs match expected results, and confirmation that all features are accessible to authorized users. The functionality criterion ensures that H2Ops delivers the complete set of capabilities that High Island V requires for operational management.

In the context of water refilling operations, Espinosa et al. (2022) highlighted that manual record-keeping limitations in retail water stations restrict management's ability to coordinate operations. They argued that functionality testing must verify that digital platforms automate routine transaction processing, inventory management, and reporting to establish effective operational control. The role of database-driven systems in consolidating functional components into a single central platform to support business strategy is further supported by Mathrani et al. (2013), who note that integrated platforms are essential for dynamic data sharing. Keen (1980) also stresses that the core functionality of decision support tools should bridge raw transactions with analytical models to guide planning.

The functionality evaluation for H2Ops encompasses verification that the Point of Sale module correctly processes transactions (such as refills, deliveries, and new container purchases), calculates totals, records payments, and generates receipt formats; confirmation that the inventory tracking system accurately records stock movements, updates quantities in real-time, and generates low-stock alerts; validation that the reporting module produces correct summaries of sales, inventory, and branch performance; and confirmation that the dashboard displays accurate, current operational data in an accessible visual format.

3.7.2 Criterion 2: Usability

This criterion measures how easily and intuitively High Island V administrators, managers, and cashiers can navigate the system, access relevant data, and perform their assigned tasks with minimal training or technical difficulty. Feedback gathered from staff and management during the evaluation phase is used to assess the clarity of the interface, the ease of navigation, and the overall user experience provided by the system in the context of daily water refilling operations. Usability evaluation employs both objective measures such as task completion time, error rates, and navigation path efficiency and subjective measures such as user satisfaction ratings, perceived ease of use, and confidence in system operation. The usability criterion ensures that H2Ops is not merely technically capable but practically accessible to its intended users.

The relevance of usability is strongly highlighted in small business technology adoption literature. Thong (1999) established that CEO and employee information systems knowledge, alongside innovation characteristics like low compatibility and high complexity, are critical determinants of technology adoption. Similarly, Grimaldo et al. (2025) noted that a lack of technical skills and operational readiness represents a major barrier to digital transformation in small enterprises, making user-friendly design a prerequisite for successful implementation. Furthermore, Matias and Hernandez (2021) identified that organizational readiness and ease of access are key drivers for MSMEs adopting cloud technologies, demonstrating that systems must be intuitive to build trust and sustain user engagement.

The usability evaluation for H2Ops encompasses measurement of task completion times for common operations (recording a sale, checking inventory, generating a report), assessment of error rates during data entry and navigation, evaluation of user satisfaction through structured questionnaires, and observation of user interactions to identify points of confusion or difficulty.

3.7.3 Criterion 3: Accuracy

This criterion evaluates the reliability and correctness of the data processed and displayed by the system, including sales records, inventory levels, and generated reports, to ensure that the information presented accurately reflects actual operational conditions. Evaluators compare system-generated data against actual operational records to identify any discrepancies and confirm that H2Ops consistently produces reliable and trustworthy outputs.

Accuracy evaluation involves data verification tests that compare system outputs to manually calculated results, reconciliation tests that verify consistency between related data elements, and historical validation that confirms system reports match known operational outcomes. The accuracy criterion ensures that management decisions based on H2Ops data are founded on correct information.

Data accuracy is fundamental to the value proposition of any business analytics or monitoring system. Espinosa et al. (2022) observed that traditional manual methods in water refilling stations frequently lead to transaction inaccuracies, inventory discrepancies, and inconsistent reports, emphasizing that information systems must provide absolute data integrity to support data-driven decision-making. Mathrani et al. (2013) also reinforce this by showing that enterprise systems must present real-time, accurate business data to ensure that managers gain valid insights for operational improvements. Furthermore, Keen (1980) notes that the effectiveness of decision support systems depends on their ability to deliver timely, correct information to managers.

The accuracy evaluation for H2Ops encompasses comparison of system-generated sales totals against manually calculated figures from baseline records, verification that inventory deductions (caps, seals, containers) correctly reflect actual consumable consumption, confirmation that financial reports accurately aggregate transaction data across branches, and validation that dashboard displays and sales forecasts reflect current rather than cached or outdated information.

3.7.4 Criterion 4: Security

This criterion ensures that all operational data stored and transmitted by the system is adequately protected through secure user authentication, access controls, and safe communication between the client and server, preventing unauthorized access or data breaches. The evaluation examines the effectiveness of the system's login mechanisms, role-based access controls, and data transmission protocols to confirm that the system meets acceptable standards of data protection.

Security evaluation includes penetration testing that attempts to identify vulnerabilities, access control verification that confirms users can access only authorized functions, and data protection assessment that verifies encryption and secure handling of sensitive information. The security criterion ensures that branch operational data remains confidential and protected against unauthorized access.

Security is a crucial element for cloud-based configurations. Mather, Kumaraswamy, and Latif (2009) state that cloud computing platforms require rigorous security controls, including identity and access management (IAM), secure authentication, authorization, and audit mechanisms, to maintain the confidentiality, integrity, and availability of sensitive business data. Additionally, Grimaldo et al. (2025) identified that security concerns and data privacy risks are major barriers influencing small business owners' decisions to adopt online technology, showing that security must be evaluated and verified to build trust.

The security evaluation for H2Ops encompasses verification that password-based authentication correctly validates user credentials, confirmation that role-based access controls restrict users to functions appropriate for their responsibilities (RBAC), assessment that data transmission between client and server is encrypted in production, verification that session management prevents unauthorized access from unattended devices, and validation of the two-factor authentication (email OTP) login layer.

3.7.5 Criterion 5: Efficiency

This criterion assesses the system's overall performance, including the speed at which data is processed, updated, and displayed on the dashboard, ensuring that the system remains responsive and reliable during regular multi-branch operations. The evaluation examines the system's response times and overall stability under typical usage conditions to confirm that H2Ops can sustain consistent performance throughout the demands of the daily operations. Efficiency evaluation involves performance testing that measures response times for typical operations, load testing that verifies performance under expected user volumes, and stability testing that confirms reliable operation over extended periods. The efficiency criterion ensures that H2Ops supports rather than impedes operational workflow speed.

System efficiency is critical for ensuring that technology supports rather than hinders operational workflows. Espinosa et al. (2022) emphasize that water refilling stations require digital automation to eliminate the operational delays and low productivity associated with manual management. By using cloud computing, Matias and Hernandez (2021) argue that

MSMEs can access scalable computational power on-demand, reducing the need for costly physical infrastructure while enhancing flexibility. Iswari, Santoso, and Hasibuan (2018) also emphasize that cloud-hosted applications reduce IT maintenance overhead and improve SME operational efficiency, ensuring that web-based systems can process transaction logs with minimal latency.

The efficiency evaluation for H2Ops encompasses measurement of response times for common operations (loading the dashboard, processing a sale, generating a report, switching branches, and running the forecasting algorithm), verification that the system remains responsive when multiple cashiers access it simultaneously across branches, confirmation that real-time updates appear on dashboards within acceptable time limits, and assessment of system stability over extended operational periods.

3.7.6 Comprehensive Evaluation Framework

Taken together, these five evaluation criteria form a comprehensive and balanced framework for assessing the overall quality and operational suitability of the H2Ops system. By measuring the system against standards of functionality, usability, accuracy, security, and efficiency, this study ensures that the evaluation process goes beyond a purely technical assessment to encompass the practical and human dimensions of system performance within a real business setting.

The use of multi-dimensional evaluation frameworks for information systems is well-supported in the literature. As demonstrated in Chapter 2, assessing the readiness and success of small business technologies requires examining technological, organizational, and environmental readiness factors (Matias & Hernandez, 2021; Grimaldo et al., 2025; Thong, 1999). Research in the water refilling sector (Espinosa et al., 2022) indicates that operational success is achieved when systems successfully resolve manual inefficiencies (functionality and accuracy) while keeping technical demands manageable (usability). Additionally, security frameworks in cloud environments (Mather et al., 2009) and data consolidation methods (Mathrani et al., 2013) support the inclusion of security and efficiency as critical technical benchmarks.

The results of this evaluation determine the extent to which H2Ops successfully addresses the operational challenges of High Island V and provide valuable insights and recommendations that can guide future improvements to the system and inform similar development efforts aimed at supporting the digital transformation of small and medium enterprises in the water refilling industry.

SUPPORTING DIAGRAMS

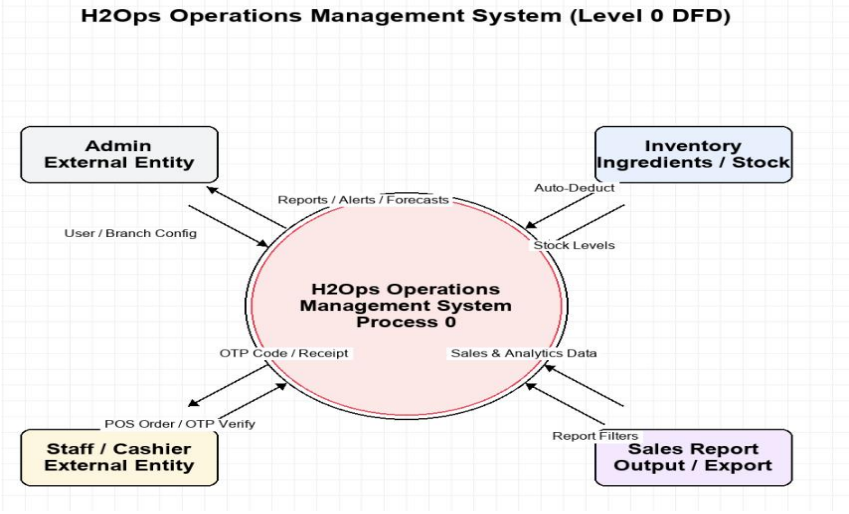
This Level 0 Diagram illustrates the H2Ops Operations Management System (Process 0), serving as the high-level architectural blueprint for High Island V's multi-branch water refilling station platform. The diagram captures the essential interactions and data flows between the central system and four critical external entities: Admin, Staff / Cashier, Inventory, and Sales Report. The system functions as an integrated operational hub that isolates branch tenancies, secures remote and local logins with multi-factor authentication (email OTP), automates point-of-sale transaction processing, maintains real-time stock visibility, and delivers actionable business intelligence and predictive forecasting.

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li



Central Process — H2Ops Operations Management System (Process 0)

Formatted: Font: 10 pt

The **H2Ops Operations Management System** stands as the nucleus of the entire operational ecosystem. It functions as the intelligent orchestrator that receives, processes, transforms, and distributes data across all connected components. This central process embodies the "brain" of the organization, executing five primary functions:

1. Multi-Tenant Separation & Authentication:** Segregates operational records between branches using unique identifiers (`branch_id`) and secures access through role-based privileges and SMTP-dispatched 6-digit email OTP verification.

2. Point-of-Sale (POS) & Transaction Management:** Captures storefront transactions (purified water, alkaline water, container purchases, caps, seals, and delivery options), computes totals, applies discounts, and records payment methods.

3. Automated Inventory Control:** Automatically adjusts stock levels for consumables (water volume, caps, seals, and containers) in real time upon sale confirmation, preventing manual reconciliation bottlenecks and triggering low-stock notifications.

4. Predictive Sales Forecasting:** Processes the past 30 days of historical sales records through a native Simple Linear Regression algorithm to calculate 7-day revenue projections for strategic planning.

5. Analytics & Report Generation:** Aggregates multi-branch operational datasets and translates them into interactive dashboards and filterable summaries.

The system maintains bidirectional communication with every external entity, ensuring data consistency, operational transparency, and closed-loop feedback across the business.

Admin — External Entity

The Admin represents the managerial and supervisory layer of the organization—typically owners, managers, or system administrators who require elevated privileges and cross-branch visibility.

Outgoing Data Flow: Branch / User & Product / Price Configuration →

The Admin initiates system setup and ongoing maintenance by transmitting configurations for branches (names, locations), user accounts (usernames, roles, emails, assigned branches), and system settings.

The Admin defines product details (purified water, alkaline water, caps, seals, empty containers), unit prices, initial stock levels, and low-stock threshold rules.

Incoming Data Flow: Reports / Alerts / Forecasts ←

The system proactively pushes performance dashboards, transaction audit logs, and critical alerts (such as low-stock warnings) back to the Admin.

The system delivers 7-day sales forecasts computed from historical data to guide administrative decisions on logistics and stocking.

Staff / Cashier — External Entity

The Staff / Cashier represents the frontline operational personnel who interact directly with customers and the POS cashier interface.

Outgoing Data Flow: POS Order Input / OTP Verify →**

Cashiers input customer transactions through the POS interface, transmitting order details including product types, quantities, customer details, payment methods, and delivery instructions.

Staff members enter manual inventory adjustments (stock restocks or logs for damages and losses).

Staff submit the 6-digit OTP code received in their email to complete secure login authentication.

Incoming Data Flow: OTP Code / Receipt / Alerts ←

The system triggers a verification email with a 6-digit OTP code to the cashier's verified address during login.

The system returns an order confirmation receipt with a unique transaction ID, itemized totals, payment status, and automatic stock deduction confirmation.

* The system returns real-time warnings for insufficient stock or low-inventory alerts.

Inventory — Ingredients / Stock

The Inventory entity represents the physical stock repository—purified water volumes, empty 5-gallon containers (round and slim), caps, seals, and purification consumables.

Incoming Data Flow: Auto-Deduct / Manual Adjustment Logs ←

The system automatically decrements inventory counts for caps, seals, and water containers in real time upon cashier order confirmation.

The system logs manual adjustments from staff restocks or damage logs.

Outgoing Data Flow: Stock Levels →

The Inventory entity continuously reports current stock levels, enabling the central process to perform availability checks before transaction confirmation and trigger low-stock alerts.

Sales Report — Output / Export

The Sales Report entity represents the analytical and documentation layer—the destination for processed business intelligence that supports monitoring, evaluation, and external reporting requirements.

Incoming Data Flow: Sales & Analytics Data / Forecasts ←

The system compiles multi-branch sales records, financial performance tables, and forecasting trends, formatting them for analytical consumption or export to external files.

Outgoing Data Flow: Report Filters → Users specify temporal and spatial parameters (date ranges, branch selection, cashier, product type) to filter transaction logs into targeted reports.

Key Design Characteristics

1. **Multi-Tenant Separation:**Operational data is strictly segregated using `branch_id`, ensuring branch cashiers only access localized records while administrators maintain cross-branch auditing.

2**. Automated Inventory Deduction:**Stock levels are automatically updated in real-time, reducing human error and ensuring inventory records synchronize with actual consumption.

3. **Closed-Loop Feedback:** Every external entity both provides input and receives output, creating balanced, interactive relationships.

4. **Centralized Security:** Identity verification is handled via secure password hashing (Bcrypt), role-based endpoint protection, and SMTP-dispatched OTP verification.

5. **Decision Support:**Linear regression algorithms translate raw transaction history into predictive 7-day forecasts, elevating the system from a transaction processor to a business intelligence platform.

Entity Relationship Diagram (ERD)

3.5 Entity Relationship Diagram (ERD)

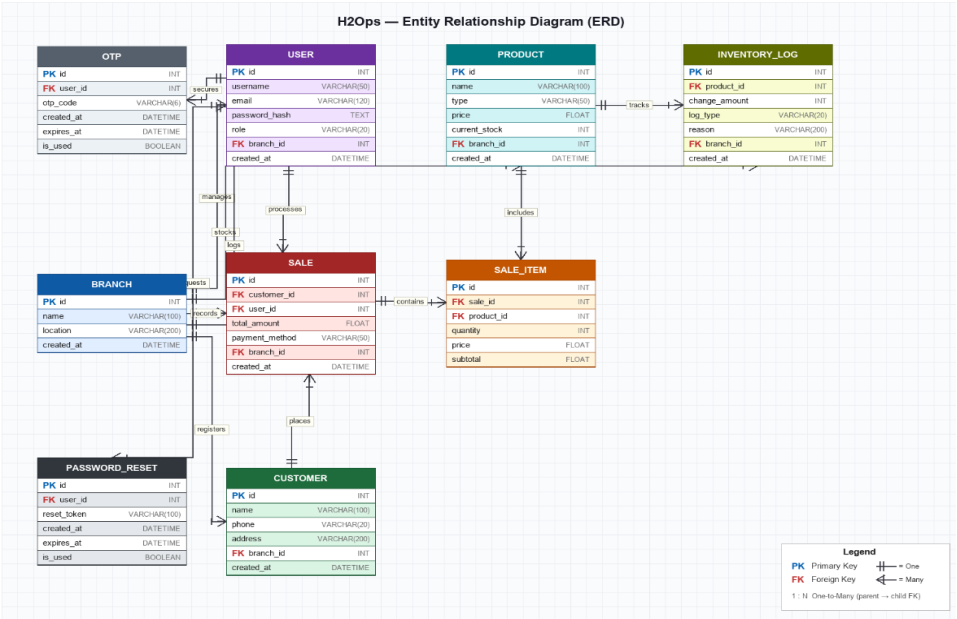
This diagram is an Entity-Relationship Diagram (ERD) for the H2Ops Operations Management System. It outlines how data is structured, stored, and shared between branches, users, customers, products, sales, inventory logs, and security verification logs within the multi-branch operational environment of High Island V.

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li



Entities and Their Attributes

BRANCH (Branch Records): Stores the operational information of each physical branch location of High Island V (such as flagship urban stations or remote provincial water refilling facilities). Key attributes include id (Primary Key), name (unique identifier), location (physical address), and created_at (timestamp).

USER (System Users): Stores the account credentials and access privilege levels of all system administrators, branch managers, and cashiers. Key attributes include id (Primary Key), username (unique identifier), email (for verification and OTP dispatch), password_hash (secured via Bcrypt), role (defines permissions, e.g., 'admin' or 'staff'), branch_id (Foreign Key referencing the user's assigned branch), and created_at.

CUSTOMER (Registered Customers): Records the contact details and branch assignments of residential, commercial, or delivery customers. Key attributes include id (Primary Key), name (customer name), phone (contact number), address (delivery address), branch_id (Foreign Key linking the customer to the registering branch), and created_at.

PRODUCT (Inventory Items): Manages the catalogue of retail items, water categories, and operational consumables stocked at each branch. Key attributes include id (Primary Key), name (product name), type (e.g., 'Purified Water', 'Alkaline Water', 'Seal', 'Cap', or empty containers), price (unit price), current_stock (real-time quantity available), branch_id (Foreign Key identifying which branch stocks the item), and created_at.

SALE (Sales Transactions): Records storefront transactions processed at the cashier terminal. Key attributes include id (Primary Key), customer_id (Foreign Key referencing the customer; nullable for walk-in purchases), user_id (Foreign Key referencing the cashier who processed the order), total_amount (computed transaction total), payment_method (e.g., 'Cash' or digital wallets), branch_id (Foreign Key identifying the branch where the sale occurred), and created_at.

Formatted: Font: 10 pt

SALE_ITEM (Transaction Line Items): Represents the individual itemized products contained in each sales transaction. Key attributes include id (Primary Key), sale_id (Foreign Key referencing the parent sale), product_id (Foreign Key referencing the product purchased), quantity (number of units purchased), price (the unit price at the time of purchase), and subtotal (computed as quantity multiplied by price).

INVENTORY_LOG (Stock Adjustments): Tracks historical stock movements, audits manual adjustments, and monitors consumable consumption logs. Key attributes include id (Primary Key), product_id (Foreign Key referencing the product), change_amount (positive for restocks, negative for sales or damage losses), log_type (e.g., 'IN' or 'OUT'), reason (e.g., 'Restock', 'Sale', or 'Damage'), branch_id (Foreign Key linking the log to the branch), and created_at.

OTP (One-Time Password Verification): Manages temporary security verification codes for multi-factor user authentication. Key attributes include id (Primary Key), user_id (Foreign Key linking the OTP to the requesting user), otp_code (6-digit alphanumeric or numeric code), created_at (generation time), expires_at (expiration deadline), and is_used (boolean status flag).

PASSWORD_RESET (Reset Request Logs): Manages password recovery tokens requested by users. Key attributes include id (Primary Key), user_id (Foreign Key referencing the user account), reset_token (secure unique string), created_at (generation time), expires_at (expiration timestamp), and is_used (boolean status flag).

Relationships and Cardinalities

BRANCH → USER (1:N): A branch has one or many assigned users (administrators, managers, cashiers). This is a one-to-many relationship, represented by the branch_id Foreign Key in the USER table.

BRANCH → CUSTOMER (1:N): A branch registers and services one or many local customers. This is a one-to-many relationship, tracked by the branch_id Foreign Key in the CUSTOMER table.

BRANCH → PRODUCT (1:N): A branch stocks and manages one or many product items (various water types and consumables). This is a one-to-many relationship, linked by the branch_id Foreign Key in the PRODUCT table.

BRANCH → SALE (1:N): A branch records one or many sales transactions. This is a one-to-many relationship, managed by the branch_id Foreign Key in the SALE table.

BRANCH → INVENTORY_LOG (1:N): A branch generates one or many inventory logs to record stock restocks or damages. This is a one-to-many relationship, established by the branch_id Foreign Key in the INVENTORY_LOG table.

USER → OTP (1:N): A user receives one or many generated One-Time Passwords (OTPs) over time to secure login sessions. This is a one-to-many relationship, linked by the user_id Foreign Key in the OTP table.

USER → PASSWORD_RESET (1:N): A user requests one or many password reset tokens for account recovery. This is a one-to-many relationship, linked by the user_id Foreign Key in the PASSWORD_RESET table.

USER → SALE (1:N): A user (cashier) processes one or many storefront sales. This is a one-to-many relationship, established by the user_id Foreign Key in the SALE table.

CUSTOMER → SALE (1:N): A customer places one or many sales transactions. This is a one-to-many relationship, where a single customer record can be linked to multiple purchases (represented by the customer_id Foreign Key in the SALE table; this relationship is optional/nullable to allow walk-in transactions without customer profiles).

SALE → SALE_ITEM (1:N): A sale contains one or many transaction line items. This is a one-to-many relationship, where the deletion of a parent sale cascades to delete all associated line items (linked via the sale_id Foreign Key in the SALE_ITEM table).

PRODUCT → SALE_ITEM (1:N): A product is included in one or many sale line items. This is a one-to-many relationship, matching products to transaction lines via the product_id Foreign Key in the SALE_ITEM table.

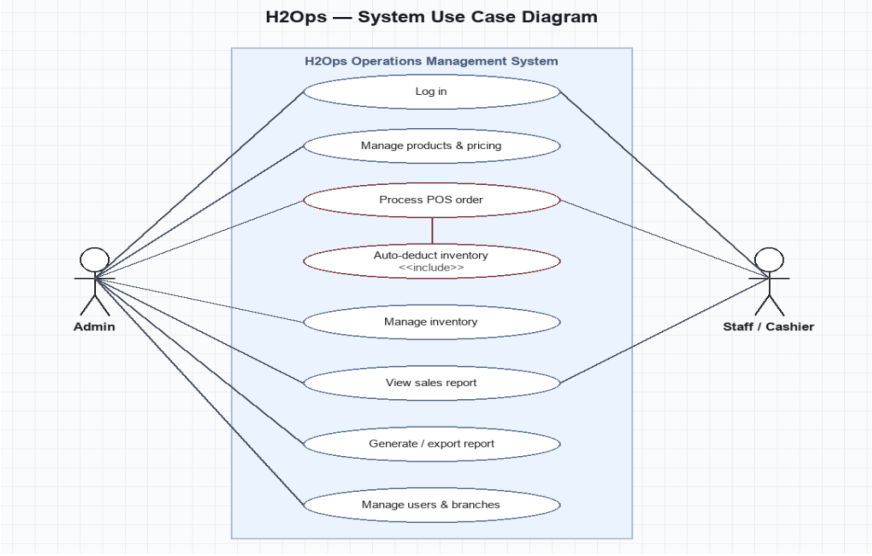
PRODUCT → INVENTORY_LOG (1:N): A product tracks its stock history through one or many inventory logs. This is a one-to-many relationship, where inventory actions reference specific catalog items via the product_id Foreign Key in the INVENTORY_LOG table.

The database schema of H2Ops, composed of nine entities and twelve defined relationships, establishes a comprehensive and logically structured data management foundation that directly supports High Island V's multi-branch operational objectives. The careful design of entity relationships ensures referential integrity across all interconnected tables, minimizing the risk of data inconsistencies that could compromise the accuracy of sales records, multi-tenant branch data isolation, inventory levels, and generated report summaries.

By aligning the database structure with the actual operational workflows of High Island V, the schema enables H2Ops to process, store, and retrieve business data in a manner that is both highly efficient and reflective of real-world water refilling station operations. Automatic cascade-delete rules on transaction line items prevent orphaned database records, while indexable branch and user foreign keys isolate localized operations and ensure fast reporting. Ultimately, this structural architecture provides management with a reliable, normalized, and well-organized data environment upon which sound and informed business decisions can be consistently made.

Use Case Diagram

This Use Case Diagram illustrates the functional relationships and access controls established within the H2Ops Operations Management System. It shows the interactions between the system and its primary users, namely the Admin and the Staff or Cashier. The diagram explains how tasks are assigned, how system operations are secured, and how business information is generated to support the workflows of High Island V.



Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

Role Based Access Control and System Boundaries

Formatted: Font: 10 pt

The role based access control (RBAC) design of H2Ops is intended to maintain operational security while reducing the risk of unauthorized data modification and accidental system misuse. The Admin has full access to system configuration, user accounts, product management, pricing, inventory adjustments, and reports. Meanwhile, the Staff or Cashier has limited access focused on sales transactions and login authentication.

The use of role based access control is supported by several studies. Laudon and Laudon (2020) explain that RBAC is an important security mechanism that allows users to access only the functions related to their responsibilities. O'Brien and Marakas (2018) state that assigning permissions based on user roles simplifies system administration and improves security. Dela Peña and Santos (2021) implemented a similar structure in their food service POS system where administrators managed system settings, user accounts, and reports, while staff members focused on sales processing and inventory inquiries. Arig et al. (2025) also separated system access by restricting staff to transaction processing and giving administrators complete control. Bienvenido (2023) found that clear role differentiation improved security and reduced errors caused by unauthorized access. Cruz, Dela Cruz, and Santos (2022) also used separate admin and staff roles in their sales and inventory system to improve efficiency and security.

Primary Actors

The H2Ops system has two primary actors, the Admin and the Staff or Cashier. Each actor is assigned specific functions based on their responsibilities in High Island V. Sommerville (2015) explains that identifying actors is important in understanding how users interact with a system. Pressman (2014) states that use case diagrams must clearly separate different user types and their permissions. Laudon and Laudon (2020) also emphasize that understanding user roles helps align the system with organizational workflows. Dela Peña and Santos (2021), Tanaman et al. (2023), and Bienvenido (2023) similarly identified administrators and staff as primary users in their systems.

1. Admin

The Admin represents the business owner or branch manager of High Island V. This user has complete access to the system and is responsible for supervising all operations.

According to **O'Brien and Marakas (2018),** administrators are responsible for managing system operations, user accounts, security settings, and maintenance activities. Laudon and Laudon (2020) describe administrators as users who maintain system integrity and generate reports for decision making. Previous studies by Cruz, Dela Cruz, and Santos (2022), Dela Peña and Santos (2021), and Arig et al. (2025) also assigned administrators full control over sales data, inventory records, user management, and system configuration.

2. Staff or Cashier

The Staff or Cashier represents branch employees who are given limited access to perform daily operational tasks. Their responsibilities include processing sales transactions, recording inventory adjustments, and logging into the system. Pressman (2014) explains that limiting user access reduces accidental data errors and simplifies system usage. Laudon and Laudon (2020) add that staff should only access functions related to their job responsibilities. O'Brien and Marakas (2018) also state that restricting access is a common security practice. Studies by Bienvenido (2023), Dela Peña and Santos (2021), and Arig et al. (2025) support limiting staff functions to transaction processing and basic inventory tasks.

Detailed Use Cases

1. Log In (Admin and Staff or Cashier)

This function allows users to securely access the system. User identities are verified before access is granted to their respective dashboards. H2Ops also uses multi factor authentication through email OTP for additional security. O'Brien

and Marakas (2018) describe authentication as the first layer of system security. Laudon and Laudon (2020) emphasize that login procedures should balance security and usability. Previous studies by Cruz, Dela Cruz, and Santos (2022), Dela Peña and Santos (2021), and Arig et al. (2025) also required user authentication before granting access.

2. Manage Products and Pricing (Admin)

This function allows the Admin to create, update, delete, and categorize products such as purified water, alkaline water, empty containers, caps, and seals. The Admin also sets product prices. **Dela Peña and Santos (2021)** identified product management as an important administrative function. **Laudon and Laudon (2020)** explain that accurate product management is necessary for reliable transaction records. **O'Brien and Marakas (2018)** add that administrative control prevents unauthorized modifications.

3. Process POS Order (Admin and Staff or Cashier)

This function allows users to record sales transactions, calculate totals, process payments, and confirm container selections. According to **Dela Peña and Santos (2021),** POS order processing is the central function of a retail system. **Laudon and Laudon (2020)** explain that transaction processing must be accurate and efficient. **O'Brien and Marakas (2018), Arig et al. (2025),** and **Bienvenido (2023)** also identified transaction processing as a key operational activity.

4. Auto Deduct Inventory (System Triggered by Process POS Order)

Once a transaction is completed, the system automatically deducts the corresponding quantities of water, caps, seals, and containers from inventory. Dela Peña and Santos (2021) explain that automated inventory updates reduce manual work. Laudon and Laudon (2020) and O'Brien and Marakas (2018) state that automation improves data accuracy and efficiency. Tanaman et al. (2023) and Few (2013) also emphasize the importance of automatic updates for real time reporting.

5. Manage Inventory (Admin)

This function allows the Admin to monitor stock levels, record restocks, review inventory logs, and set low stock alerts. Laudon and Laudon (2020) describe inventory management as a critical business process. O'Brien and Marakas (2018) explain that inventory systems help prevent stock shortages and overstocking. Similar functions were implemented by Cruz, Dela Cruz, and Santos (2022), Tanaman et al. (2023), Dela Peña and Santos (2021), and Bienvenido (2023).

6. View Sales Report (Admin and Staff or Cashier)

This function enables users to view sales summaries. The Admin can access complete analytics, charts, and forecasts, while the Staff or Cashier can view daily sales summaries. Chen, Chiang, and Storey (2012) explain that reporting converts raw data into useful information. Few (2013) emphasizes the importance of dashboards for quick decision making. Laudon and Laudon (2020), Dela Peña and Santos (2021), and Cruz, Dela Cruz, and Santos (2022) also recognize reporting as an essential business function.

7. Generate or Export Report (Admin)

This function allows the Admin to export reports such as transaction histories, forecasts, and inventory logs into formats like Excel. Laudon and Laudon (2020) explain that export functions support information sharing and compliance requirements. O'Brien and Marakas (2018), Alomari (2025), Davenport and Harris (2017), Cruz, Dela Cruz, and Santos (2022), and Tanaman et al. (2023) also highlight the importance of report generation and data sharing.

8. Manage Users and Branches (Admin)

This function allows the Admin to create branches, assign locations, create user accounts, update permissions, and reset login credentials. O'Brien and Marakas (2018) explain that user management is important for maintaining security. Laudon and Laudon (2020) emphasize that access permissions must match organizational roles. Pressman (2014), Dela Peña and Santos (2021), Arig et al. (2025), and Bienvenido (2023) also support limiting these functions to administrators only.

System Design Rationale and Theoretical Alignment

The use cases in H2Ops demonstrate a structured approach to role based access control and automation. These functions support business analytics, data visualization, and digital transformation principles. By separating administrative tasks from operational tasks, the system improves security and increases cashier efficiency. Features such as automated inventory deduction, real time sales reporting, and user role separation directly address the operational problems identified at High Island V. Together, these functions reduce manual work, minimize cash discrepancies, improve reporting accuracy, and provide administrators with reliable information for better decision making across all branches.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

H2Ops is a web-based and mobile-responsive operations management and business intelligence application designed specifically to centralize, secure, and automate the daily operational workflows of multi-branch water refilling station enterprises like High Island V. The system assists business owners and managers in monitoring real-time branch performance, tracking customer accounts, and managing product inventory levels. By transforming raw, manually recorded sales logs and inventory records into dynamic visual reports and predictive analytics, H2Ops enables informed, data-driven operational decision-making.

Unlike conventional Point-of-Sale (POS) systems that operate in local isolation and focus strictly on recording transactions, H2Ops integrates a cloud-hosted multi-tenant architecture and an automated forecasting engine to link daily front-end transactions directly with backend inventory control and demand prediction. The application processes walk-in and delivery sales across remote city and provincial branches, executes automatic inventory deduction triggers, generates transactional audit logs, and renders interactive dashboards.

The system aims to provide growing multi-branch enterprises with a scalable and secure business intelligence solution that mitigates human error, eliminates paper-based ledger discrepancies, prevents inventory shrinkage, and optimizes supply chain planning through mathematical forecasting.

The major objectives of the developed system are as follows:

- Automate the entry, calculation, and secure cloud storage of storefront sales and delivery orders
- Centralize multi-branch sales, inventory, and user data into a single multi-tenant database.
- Implement real-time inventory control with automated stock deduction triggers to prevent shrinkage
- Secure user logins using password hashing (Bcrypt) and Multi-Factor Authentication (Email OTP).
- Enforce Role-Based Access Control (RBAC) to isolate branch staff views while keeping global oversight for admins.
- Forecast future water refilling demand and target sales using a 30-day historical Simple Linear Regression model.
- Generate interactive visual dashboards (using Chart.js).

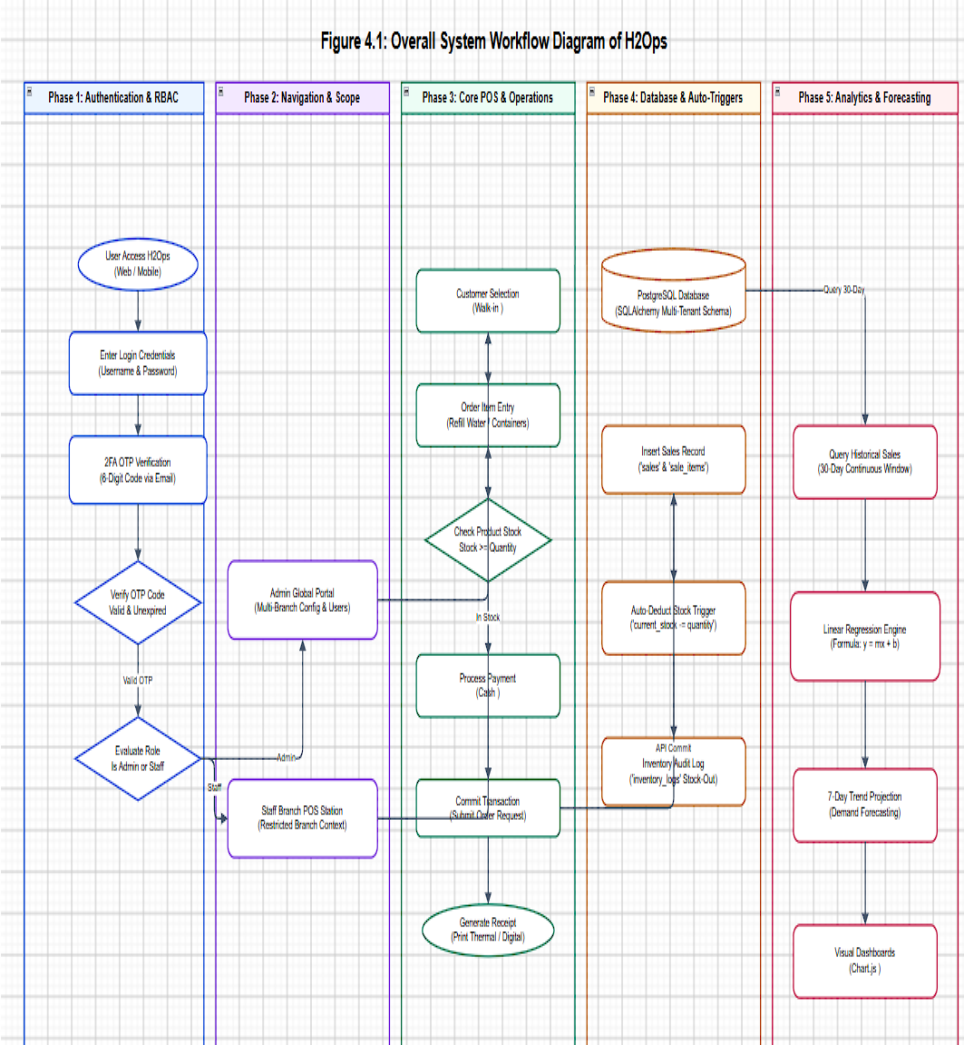
The development of H2Ops follows modern software engineering principles, incorporating a responsive presentation layer (HTML5, CSS3, JS ES6, Jinja2), a modular business logic layer (Python Flask), secure Object-Relational Mapping (SQLAlchemy ORM), and a robust database engine (MySQL/PostgreSQL). The completed platform consists of several interconnected modules that work collaboratively to capture, process, secure, analyze, and visualize multi-branch business data.

Formatted: Font: 12 pt
Formatted: Centered, Line spacing: 1.5 lines
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Font: 10 pt

Figure 4. 11 illustrates the overall workflow of the developed system.

Formatted: Font: 10 pt

Formatted: Font: 10 pt



The workflow begins with secure user authentication, requiring credentials followed by a two-factor authentication (2FA) verification code dispatched via email. Once authenticated, the system evaluates the user's role: administrators are routed to a global multi-branch portal to manage branches, users, products, and system-wide reports, while cashiers are restricted to their assigned branch POS station.

Within the POS environment, staff select the customer, enter the order details, and commit the transaction. This action automatically verifies product stock availability, calculates the change, and generates a customer receipt. Concurrently, the backend processes the sale in the PostgreSQL database, automatically executing triggers to deduct inventory counts (`current_stock -= quantity`) and record stock-out audit entries in the inventory logs. For strategic planning, the accumulated transaction data is queried over a 30-day historical window. The forecasting engine then applies a Simple

Formatted: Font: 10 pt

Linear Regression model to project a 7-day sales demand trendline. This resulting analysis is presented through interactive dashboards and graphical summaries, enabling business owners to monitor branch performance and plan inventory replenishment dynamically.

4.3 System Development Methodology

Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

The development of the H2Ops platform adopted the Agile Software Development Methodology, an iterative approach that emphasizes continuous collaboration, incremental development, and regular evaluation throughout the software development life cycle (SDLC). Agile was selected because it accommodates evolving operational requirements and promotes frequent feedback from stakeholders (the owners, branch managers, and cashiers of High Island V), ensuring that the resulting application aligns with the multi-branch operational needs of the water refilling business.

The Agile methodology enabled the development team to divide the project into manageable 8 to 9-week iterations, commonly referred to as sprints. At the end of each sprint, a functional component of the system was produced, tested, and reviewed before proceeding to the next iteration. This process minimized development risks and facilitated the early detection of database constraint errors, security leaks, or performance bottlenecks. The development process consisted of the following phases:

Requirements Gathering

The first phase involved collecting information from prospective users through interviews, direct field observations, and document analysis of physical sales ledgers at urban and provincial High Island V branches. Business owners, managers, and cashiers were consulted to determine the common challenges encountered in manual transaction logs, delayed branch consolidations, inventory shrinkage, and guesswork-based stock replenishment. The gathered requirements served as the basis for defining the functional and non-functional requirements of the system, formulated as a product backlog of user stories.

System Planning

After identifying the project requirements, the researchers established the project scope, multi-branch boundaries, system specifications, development schedule, and resource allocation. The planning stage also included feasibility analysis to determine the technical, operational, and economic viability of deploying a cloud-hosted platform (using Python, Flask, and PostgreSQL) to replace the existing paper-based system.

System Design

During this phase, the overall client-server architecture of H2Ops was designed. Relational database structures, process flows, user interface layouts, and security boundaries were prepared. Unified Modeling Language (UML) Use Case diagrams, Entity Relationship Diagrams (ERDs), and wireframes were developed to provide a blueprint for software construction, ensuring clean separation of concerns and multi-tenant branch data isolation

Software Development

The coding phase involved implementing the designed modules using the selected programming languages and frameworks. Individual system functionalities were developed incrementally, including Bcrypt-encrypted credentials entry, SMTP-dispatched 6-digit Email OTP 2FA middleware, point-of-sale (POS) cashier register, automated inventory deduction database triggers, and a Python-coded Simple Linear Regression engine ($y = mx + b$) for 7-day sales and demand trendline projections

Testing

Each completed module underwent unit testing (e.g., verifying the mathematical accuracy of the regression slope and intercept calculations) before being integrated into the main application. Integration testing ensured that transactions committed in the POS correctly updated the sales tables, automatically decremented the product stock, and logged audit trails in the inventory log tables. User Acceptance Testing (UAT) was later conducted with selected cashiers and administrators of High Island V to validate whether the user interface was responsive, accessible, and aligned with daily workflows.

Deployment

Following successful testing and verification, the application was deployed in the target environment. This included setting up the Flask serverless environment on Vercel, migrating schemas to a remote PostgreSQL cloud database cluster, configuring SMTP mail delivery credentials, and onboarding administrators and cashiers to initiate active operation.

1. Cloud Deployment Infrastructure

The deployment of the H2Ops system was engineered to guarantee high availability, security, and low-latency access across all remote city and provincial branches of High Island V. The production setup consists of the following cloud infrastructure components:

Application Hosting : The Python Flask application is deployed as serverless functions on Vercel. This provides automatic scaling to handle transaction traffic spikes during peak hours, global CDN caching for fast page loading, and enforced SSL/TLS encryption for secure HTTPS transactions.

Database Tier: (Managed Cloud PostgreSQL Database):** To secure critical business data, the relational schema is hosted on a managed remote PostgreSQL cluster. This ensures automated daily database backups, strict transaction isolation, and fast query execution for historical sales lookups and forecasting equations.

Communication Tier: (SMTP Gateway Integration):** Flask-Mail is configured with secure SMTP credentials to interface with an enterprise mail gateway. This ensures that 6-digit Multi-Factor Authentication (MFA) OTP codes are generated, signed, and dispatched to cashiers' emails within seconds of a login request.

Environment Segregation: Application secrets, database connection URIs, and mail server passwords are segregated into system-level environment variables (using a `.env` configuration file), preventing hardcoded credentials in the source code repository.

2. UI/UX Coordination & Responsive Presentation: The user interface and user experience (UI/UX) of H2Ops were meticulously coordinated with the deployment architecture to ensure maximum accessibility and responsiveness on frontline devices:

Multi-Device Adaptability (Responsive Web Design): Recognizing that cashiers frequently operate using touch-screen tablets or mobile phones at the physical storefront counter while managers monitor metrics on desktop computers, the frontend utilizes a responsive fluid layout. Using CSS media queries, the layout reflows dynamically, transforming multi-column administrative dashboards into stacked, thumb-friendly interfaces on mobile viewports.

Point-of-Sale Efficiency: The POS cashier interface is optimized to minimize data-entry friction. It features touch-friendly selection cards for products and automatic recalculation scripts that compute subtotals, taxes, and change client-side (via JavaScript) before committing requests to the server

Dynamic Analytics Rendering (Chart.js Integration): The administrative dashboard coordinates directly with the backend forecasting routes. When the server returns JSON payloads containing 30-day sales arrays and 7-day regression coordinates, Chart.js dynamically renders interactive line graphs and bar charts on the client-side canvas, complete with hover tooltips and animated transitions.

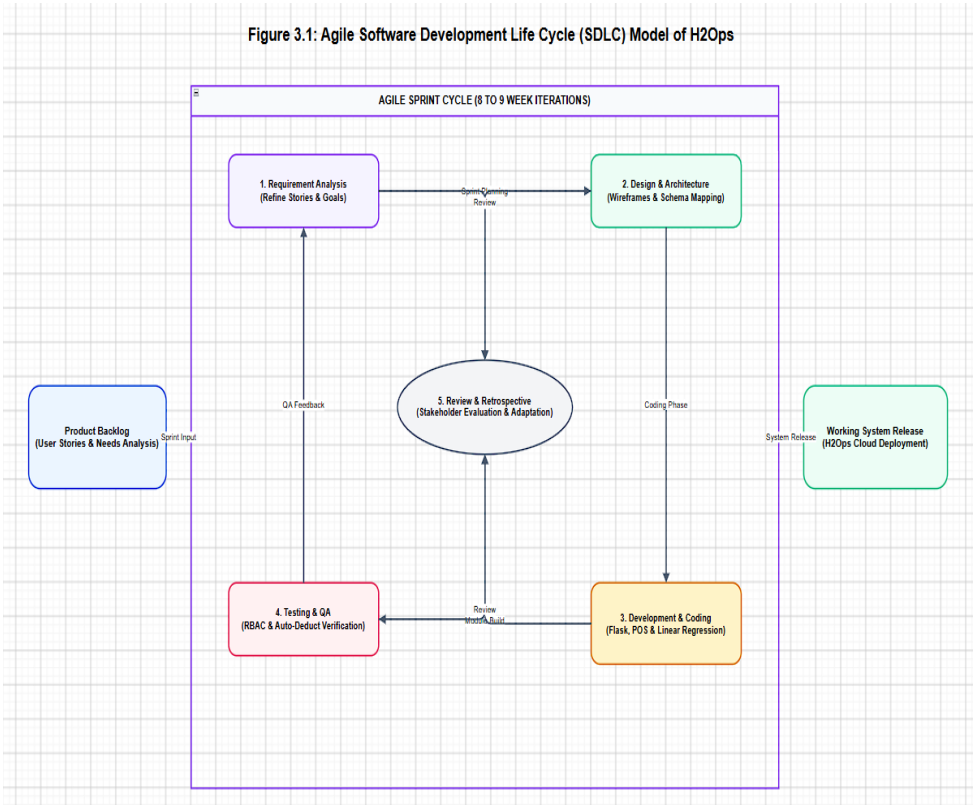
Real-Time System Feedback: To reduce cognitive load, H2Ops utilizes clear, color-coded visual alerts. For example, successful sales transactions trigger emerald-green toast notifications, while stock levels falling below defined thresholds trigger amber low-stock badges next to inventory items, instantly alerting cashiers.

Maintenance

The final phase involves monitoring system performance, correcting identified issues, implementing software updates, and enhancing system features based on user feedback. Maintenance ensures the long-term reliability and usability of the developed application.

Figure 4. 22 presents the Agile Software Development Life Cycle adopted in the study.

Figure 3.1: Agile Software Development Life Cycle (SDLC) Model of H2Ops

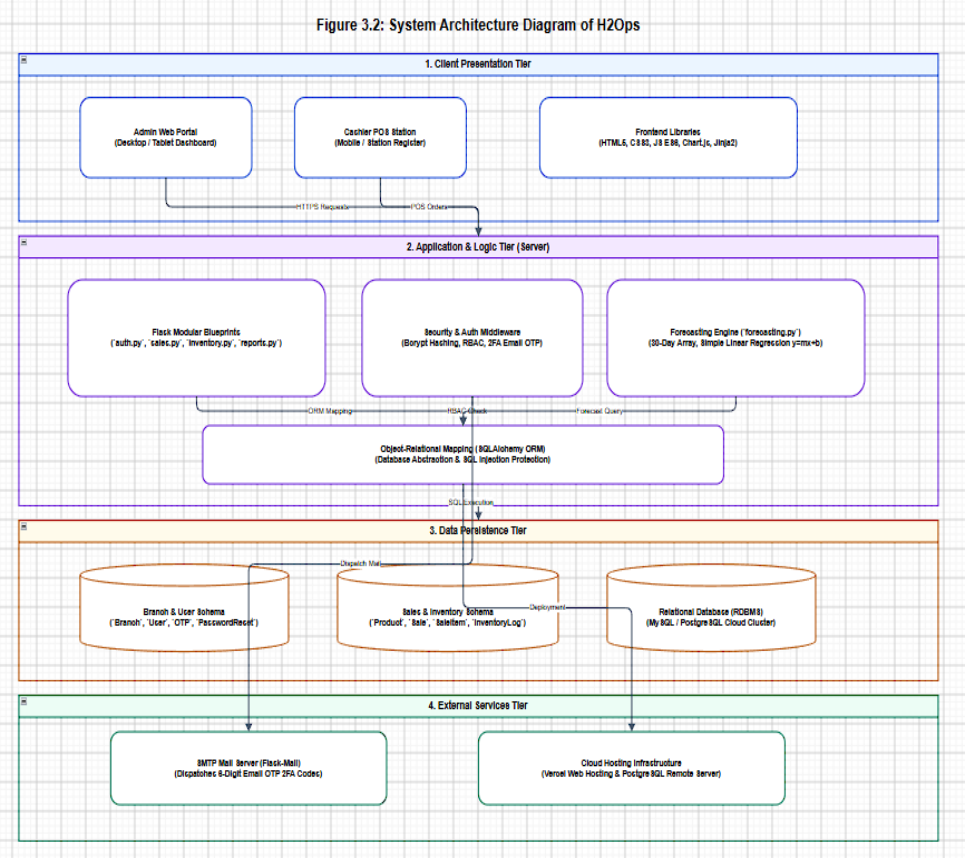


4.4 System Architecture

The H2Ops platform utilizes a **Three-Tier Client-Server Architecture**, which separates the presentation layer (client tier), application layer (business logic tier), and data layer (persistence tier). This architectural approach improves system maintainability, scalability, and security by organizing software components according to their respective responsibilities.

Figure 4. 33 illustrates the architecture of the developed system.

Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: (Default) Times New Roman, 10
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Font: 10 pt



The architecture consists of the following components:

Formatted: Font: (Default) Times New Roman, 10

The presentation layer serves as the responsive interface between the users and the H2Ops application. It is responsible for rendering dashboards, capturing operational inputs, displaying real-time stock levels, and presenting analytical reports through interactive visualizations. Authorized branch cashiers and administrators may access the application securely using desktop computers, laptops, tablets, or smartphones through standard, modern web browsers.

Major interface components include:

Formatted: Font: (Default) Times New Roman, 10 pt

- Secure Login & 2FA Page:
- Administrative Dashboard:
- Cashier Point-of-Sale:
- Inventory & Stock Management Module:

● Analytics & Forecasting

Application Layer (Business Logic Tier)

The application layer contains the core server-side logic responsible for routing, validating input data, evaluating roles, performing analytical computations, and executing database triggers.

Its primary responsibilities include:

● User Authentication & OTP Dispatch:

● Transaction Processing:

● Automated Inventory Triggering:

● Role-Based Access Control (RBAC):

● Sales & Demand Analysis:

● Linear Regression Forecasting Engine:

The Application server acts as the secure intermediary between the presentation layer and the database.

Data Layer

The data layer stores all multi-tenant operational records in a centralized relational database. It is responsible for maintaining relational data integrity, enforcing primary/foreign-key constraints, and supporting fast data retrieval through SQLAlchemy Object-Relational Mapping (ORM).

Stored information includes:

● Branch Configurations:

● User Accounts:

● Products & Pricing:

● Customer Records:

● MFA OTP Records:

This architecture ensures secure communication between the application server and the database while restricting unauthorized direct access to sensitive business records.

4.5 Hardware and Software Requirements

The successful implementation and optimal deployment of H2Ops require appropriate hardware and software resources. These specifications ensure that the application performs efficiently during multi-branch data processing, automated point-of-sale (POS) cashier transactions, real-time inventory tracking, security authentication, report generation, and predictive sales analytics computation.

4.5.1 Hardware Requirements

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Table 4. 1↓ Hardware Specifications for H2Ops.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 / AMD Ryzen 3 (Dual-Core, 2.0 GHz)	Intel Core i5 / AMD Ryzen 5 or higher (Quad-Core+, 2.5 GHz or faster)
Memory (RAM)	4 GB DDR4	8 GB DDR4 or higher
Storage	128 GB SSD / 256 GB HDD	256 GB SSD / 512 GB NVMe SSD or higher
Monitor / Display	1366 × 768 HD Resolution	1920 × 1080 Full HD Display or higher
Internet Connection	10 Mbps Stable Broadband / Mobile Data	50 Mbps or higher High-Speed Fiber Internet
Mobile / Client Device	Android 8.0 / iOS 13 or higher (Smartphones/Tablets)	Android 12+ / iOS 15+ with a modern HTML5 mobile web browser

The recommended hardware configuration provides sufficient processing capability for executing backend analytical operations (such as the Simple Linear Regression algorithm for 7-day sales forecasting), managing concurrent user sessions across multiple High Island V branch locations, and rendering real-time dynamic dashboards without significant latency or processing delays.

4.5.2 Software Requirements

Table 4.2 presents the software requirements.

Software / Technology	Purpose / Functional Role in H2Ops
Windows 10 / 11 / Ubuntu Linux	**Operating System:** Serves as the primary operating system environment for host development, local testing, and production deployment.

Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

Visual Studio Code (VS Code)	**Source Code Editor:** Integrated Development Environment (IDE) used for writing Python modules, Flask web routes, HTML5/CSS3 templates, and JavaScript code.
Python (v3.10+)	**Backend Programming Language:** Powers core system logic, server-side data processing, arithmetic calculations, and predictive linear regression algorithms.
Flask (v3.0.0)	**Backend Web Framework:** Lightweight Python micro-framework handling web routing, request dispatching, session state, and template rendering via Jinja2.
SQLAlchemy (v3.1.1)	**Object-Relational Mapper (ORM):** Translates database objects into Python classes, automating SQL database queries and multi-tenant schema relations (branch_id).
MySQL (XAMPP / MariaDB) / PostgreSQL	**Database Management System (DBMS):** Relational database storing multi-branch user credentials, customer profiles, POS sales records, and inventory stock logs.
HTML5 / CSS3 / JavaScript (ES6+)	**Frontend Technologies:** Used to build a responsive, intuitive user interface for cashier POS terminals and administrative management dashboards.
Chart.js	**Data Visualization Library:** JavaScript charting library used to dynamically render interactive sales trends, revenue charts, and projected 7-day forecast graphs.
Flask-Bcrypt & Flask-Mail	**Security & Authentication Extensions:** Implements password hashing via Bcrypt, secure session management, and Email One-Time Password (OTP) Multi-Factor Authentication.
Google Chrome / Mozilla Firefox	**Web Browser:** Modern web-standard browsers used by cashiers and administrators to access the web application across desktop and mobile devices.
XAMPP / Gunicorn	**Local & Production Web Server:** XAMPP provides local Apache/MySQL testing servers, while Gunicorn acts as the production WSGI application server.

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Git & GitHub	**Version Control System:** Manages source code repositories, commit histories, version tracking, and collaborative development.

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The selection of these software tools and frameworks was based on their stability, open-source accessibility, high security standards, extensive community support, and widespread adoption in modern responsive web application development.

4.6 Database Design

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

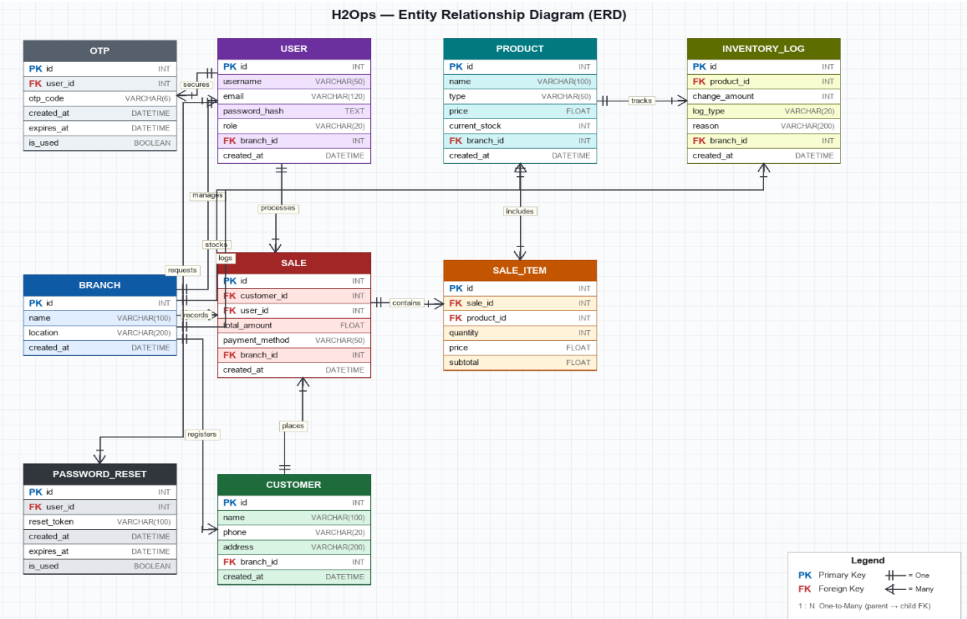
Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The database serves as the operational foundation of H2Ops by organizing, securing, and centralizing transaction records, user accounts, customer profiles, product inventories, audit logs, and analytical data across multiple branch locations. A well-structured database design ensures high-performance data retrieval, minimizes data redundancy, enforces multi-tenant branch data isolation, and maintains strict transactional integrity throughout the operation of the system.

The researchers employed a relational database model implemented using MySQL (via XAMPP in local development) and PostgreSQL (in cloud production), managed seamlessly through SQLAlchemy Object-Relational Mapping (ORM). Database normalization techniques up to Third Normal Form (3NF) were systematically applied to eliminate data duplication, establish clear primary-foreign key relationships, and maintain consistency among related entities.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the H2Ops system.



The primary entities of the H2Ops database include:

- Branches
- Users
- Customers
- Products
- Sales
- Sale Items
- Inventory Logs
- One-Time Passwords (OTP)
- Password Resets

Users Table (`user`)

This table stores user account credentials, roles, and branch assignments.

Field	Description
id	Primary Key
username	Login Username
email	User Email Address
password_hash	Encrypted Password (Bcrypt)
role	User Access Level (Admin / Staff)
branch_id	Foreign Key (`branch`)
created_at	Account Creation Date

Customers Table (`customer`)

This table contains customer contact details used for subscriber order management and delivery tracking.

Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

Field	Description
id	Primary Key
name	Complete Name
phone	Contact Phone Number
address	Delivery Address
branch_id	Foreign Key (`branch`)
created_at	Registration Date

Products Table (`product`)

This table stores product inventory information, item types, unit pricing, and current stock levels.

Field	Description
id	Primary Key
name	Product Name (e.g., Purified Water, Alkaline Water)
type	Product Category/Type (e.g., Purified, Alkaline, Seal, Cap)
price	Selling Price per Unit
current_stock	Available Inventory Stock
branch_id	Foreign Key (`branch`)
created_at	Item Registration Date

Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

Sales Table (`sale`)

This table records completed Point-of-Sale (POS) cashier transactions and total amounts.

Field	Description
id	Primary Key
customer_id	Foreign Key (`customer`) (optional for walk-in customers)
user_id	Foreign Key (`user`) (Cashier)
total_amount	Total Purchase Amount
payment_method	Payment Method (e.g., Cash, GCash)
branch_id	Foreign Key (`branch`)
created_at	Date and Time Purchased

Sale Items Table (`sale_item`)

This table records the individual items and quantities included in each sales transaction.

Field	Description
id	Primary Key
sale_id	Foreign Key (`sale`)
product_id	Foreign Key (`product`)
quantity	Quantity Purchased

Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

price	Unit Price at Time of Sale
subtotal	Item Line Subtotal (`quantity × price`)

Inventory Logs Table (`inventory_log`)

This table stores immutable stock-in additions and automated stock-out deduction logs for auditing and loss prevention.

Field	Description
id	Primary Key
product_id	Foreign Key (`product`)
change_amount	Stock Change Quantity (+ for IN, – for OUT)
log_type	Action Type (IN / OUT)
branch_id	Foreign Key (`branch`)
created_at	Date and Time Logged

OTP Table (`otp`)

This table manages 6-digit verification codes generated for Email Multi-Factor Authentication.

Field	Description
id	Primary Key
user_id	Foreign Key (`user`)

Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

otp_code	6-Digit Verification Code
created_at	Generation Timestamp
expires_at	Expiration Timestamp
is_used	Verification Status Flag

Password Resets Table (`password_reset`)

This table stores secure cryptographic reset tokens used during account password recovery.

Field	Description
id	Primary Key
user_id	Foreign Key (`user`)
reset_token	Unique Cryptographic Token
created_at	Generation Timestamp
expires_at	Expiration Timestamp
is_used	Token Redemption Status Flag

The relationships among these tables allow the application to trace customer transactions across branches, maintain automatic real-time inventory adjustments, log cashier accountability, calculate product demand trends, generate 7-day Simple Linear Regression sales forecasts, and output comprehensive financial and operational reports. Proper indexing and foreign key constraints maintain referential integrity and optimize database query performance, ensuring that the system can efficiently support both high-speed point-of-sale operations and computational backend data analytics.

Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusions and recommendations of the study entitled **H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics**. It serves as the final chapter of the research, summarizing the significant findings obtained from the design, development, implementation, testing, and evaluation of the proposed system. The conclusions presented in this chapter are based on the results gathered throughout the research process and determine whether the objectives of the study were successfully achieved.

The primary objective of this study was to develop a cloud-based operations management system that addresses the operational challenges commonly encountered by water refilling stations, particularly in managing sales transactions, inventory, customer information, delivery records, and business reports. The researchers aimed to replace manual and paper-based processes with an integrated information system capable of automating routine business operations while providing real-time access to operational data and automated sales analytics. Through the implementation of H2Ops, the researchers sought to improve operational efficiency, enhance the accuracy of business records, and support informed decision-making among business owners and managers.

The findings discussed in this chapter are supported by the results of system implementation, software testing, and user evaluation conducted during the course of the study. These findings demonstrate the effectiveness of the developed system in simplifying business processes,

minimizing human errors, improving data management, and providing timely and accurate reports for operational monitoring. The evaluation also assessed the functionality, usability, reliability, efficiency, and overall acceptability of the proposed system from the perspective of its intended users.

In addition to presenting the conclusions of the study, this chapter provides recommendations for future researchers, software developers, and water refilling station owners who may adopt or further improve the proposed system. These recommendations focus on possible system enhancements, additional functionalities, implementation strategies, and future research opportunities that can further increase the effectiveness, scalability, and sustainability of H2Ops. The chapter also highlights the contribution of the study to the field of information technology and demonstrates how cloud computing and automated sales analytics can support the digital transformation of small and medium-sized business enterprises, particularly within the water refilling station industry.

5.2 Summary of Findings

The development and evaluation of **H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics** produced positive results that successfully addressed the objectives of the study. Based on the implementation and evaluation of the proposed system, the following findings were obtained:

- The developed system successfully automated the recording and management of daily business operations, including sales transactions, inventory monitoring, customer information, delivery records, and report

Formatted: Font: 12 pt
Formatted: Centered, Line spacing: 1.5 lines
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li
Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

generation. This significantly reduced the dependence on manual record-keeping and minimized human errors commonly associated with paper-based documentation.

- The cloud-based architecture enabled centralized storage of operational data, allowing authorized users to access updated information anytime and anywhere through an internet-connected device. This feature improved communication and coordination, particularly for businesses operating multiple branches.
- The Automated Sales Analytics module successfully transformed recorded sales transactions into meaningful business information through graphical dashboards and statistical reports. Business owners were able to monitor sales performance, identify sales trends, evaluate branch performance, and analyze product demand using real-time operational data.
- The Inventory Management module effectively monitored stock availability and automatically updated inventory records after every completed sales transaction. This reduced the possibility of inaccurate inventory counts and improved stock management.
- The Customer Management and Delivery Management modules organized customer records and delivery schedules into a centralized database, allowing employees to retrieve customer information more efficiently while improving service quality.
- System testing, including Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT), confirmed that all system modules functioned according to their intended purpose. The respondents evaluated the system positively in terms of functionality, usability, reliability, efficiency, and user satisfaction.
- The implementation of H2Ops demonstrated that integrating cloud computing and automated sales analytics into the daily operations of water refilling stations improves operational efficiency, simplifies report generation, supports managerial decision-making, and enhances overall business management.
-

5.3 Conclusion

Based on the findings of the study, the researchers conclude that H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics is a reliable, efficient, and practical solution for improving the operational management of water refilling station businesses.

The developed system successfully addressed the limitations of manual business operations by automating sales recording, inventory monitoring, customer management, delivery scheduling, report generation, and business analytics. Through automation, the system minimized manual computations, reduced data redundancy, improved record accuracy, and accelerated report preparation. These improvements contributed to more organized and efficient business operations.

The cloud-based architecture further enhanced the system by providing centralized data management and real-time accessibility. Business owners and managers can monitor operational activities remotely, access updated reports instantly, and oversee multiple branches without relying on manual consolidation of records. This capability improves operational transparency and supports timely decision-making.

One of the major contributions of H2Ops is its Automated Sales Analytics feature. Instead of merely storing transactional data, the system converts business information into meaningful reports, charts, and dashboards that help management evaluate sales performance, identify trends, monitor product demand, and make informed business decisions. This supports data-driven management practices and enables owners to respond more effectively to changing business conditions.

Formatted: Line spacing: Multiple 1.15 li, Outline numbered + Level: 1 + Numbering Style: Bullet + Aligned at: 0.25" + Tab after: 0.5" + Indent at: 0.5"

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The evaluation results indicate that users found the system functional, user-friendly, reliable, and effective in supporting their daily operations. The positive feedback obtained during User Acceptance Testing demonstrates that the developed application meets the needs and expectations of its intended users.

Therefore, the researchers conclude that the objectives of the study were successfully achieved. H2Ops serves as an effective cloud-based operations management system that improves business efficiency, enhances data management, supports decision-making through automated sales analytics, and contributes to the digital transformation of water refilling station businesses.

5.4 Recommendations

Formatted: Font: (Default) Times New Roman, 13

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Although the proposed system successfully achieved its intended objectives, the researchers recommend several improvements that may further enhance the capabilities of H2Ops.

System Improvements

Future developers may incorporate additional features that further improve business operations and user convenience. These include:

- Implement automated notifications for low inventory levels, delayed deliveries, overdue customer balances, and system maintenance reminders.
- Integrate SMS and email notification services to automatically inform customers about delivery schedules, order confirmations, promotional offers, and payment reminders.
- Develop barcode or QR code scanning functionality to improve transaction speed and inventory management.
- Include online payment integration to support digital payment methods such as GCash, Maya, bank transfers, and other electronic payment platforms.
- Expand the Sales Analytics module by incorporating predictive analytics capable of forecasting future sales trends based on historical business data.
- Implement two-factor authentication (2FA), stronger password encryption, and additional cybersecurity measures to further protect business information and customer records.
- Enhance the report generation module by allowing users to export reports in PDF, Excel, and CSV formats for easier documentation and business reporting.

Future Research Directions

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

The researchers recommend that future studies continue improving and expanding the capabilities of H2Ops by incorporating emerging technologies and additional system functionalities that can further enhance business operations and decision-making. As technology continues to evolve, future researchers have the opportunity to build upon the foundation established by this study and develop more advanced solutions for water refilling station businesses.

One possible enhancement is the integration of Artificial Intelligence (AI) and Machine Learning (ML). These technologies can analyze historical sales data to forecast future sales trends, predict customer demand, and recommend optimal inventory levels. Predictive analytics would enable business owners to make proactive decisions, reduce inventory shortages or overstocking, and improve overall business planning. AI can also be used to generate recommendations for product demand, identify seasonal sales patterns, and support more accurate business forecasting.

Future researchers may also develop a mobile application version of H2Ops that allows users to monitor business operations through smartphones and tablets. A mobile application would enable owners and managers to access reports, monitor inventory, review sales transactions, and manage deliveries even while away from the business establishment. Incorporating offline functionality would also be beneficial for areas with unstable internet connectivity, allowing transactions to be synchronized automatically once a connection becomes available.

Additional features such as online payment integration, barcode or QR code scanning, SMS or email notifications, and customer loyalty programs may also be included in future versions of the system. These enhancements can improve customer convenience, increase operational efficiency, and provide a better overall user experience. Future studies may also integrate supplier management and equipment maintenance monitoring to make H2Ops a more comprehensive operations management system.

Lastly, future researchers are encouraged to implement and evaluate the system in a larger number of water refilling stations across different municipalities or provinces. Conducting long-term studies involving more respondents would provide a broader evaluation of the system's effectiveness, usability, reliability, and impact on business productivity. Such studies can also assess how cloud-based management systems contribute to the long-term growth and sustainability of water refilling station businesses.

Implementation Strategies

To maximize the effectiveness and long-term success of H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics, proper implementation strategies should be established before and after the system is deployed. These strategies will help ensure that users can effectively utilize the system and that the organization can fully benefit from its features.

Before full implementation, business owners should conduct orientation and training sessions for administrators, managers, and employees who will use the system. The training should focus on system navigation, sales transaction recording, inventory monitoring, customer management, delivery scheduling, and report generation. Providing sufficient training will help users become familiar with the system, minimize errors during daily operations, and improve user confidence in adopting the new technology.

Business owners should also establish standardized procedures for data entry and system usage. Since the accuracy of reports and sales analytics depends on the quality of the recorded data, employees should consistently follow the prescribed procedures when entering customer information, sales transactions, inventory updates, and delivery records. Implementing standardized workflows will help maintain data accuracy, consistency, and reliability across all branches of the business.

To ensure continuous system performance, organizations should perform regular database backups, software updates, and preventive maintenance. Routine maintenance helps protect important business information from data loss, security threats, and unexpected system failures while ensuring that the application continues to operate efficiently. Monitoring system performance and addressing technical issues immediately can also reduce downtime and maintain business continuity.

Finally, business owners should encourage users to provide regular feedback regarding the system's performance and usability. User suggestions can help identify areas that require improvement and guide future system enhancements. Continuous evaluation and periodic updates will allow H2Ops to adapt to the changing needs of water refilling station businesses while ensuring that it remains reliable, secure, and effective in supporting daily business operations.

Formatted: Font: 13 pt

Formatted: Line spacing: 1.5 lines

Formatted: Font: 10 pt

Formatted: Line spacing: Multiple 1.15 li

Formatted: Line spacing: Multiple 1.15 li

5.5 Impact of the Study

Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

The development of H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics provides significant contributions to both the field of information technology and the water refilling station industry. The study demonstrates how cloud computing and business process automation can be utilized to address the operational challenges commonly experienced by small and medium-sized enterprises. By integrating sales management, inventory monitoring, customer management, delivery management, and automated sales analytics into a single cloud-based platform, the system promotes more efficient and organized business operations.

For water refilling station owners and managers, H2Ops serves as an effective tool for improving operational efficiency and decision-making. The system minimizes manual record-keeping, reduces human errors, and automates the generation of reports, allowing management to access accurate and real-time business information. The Automated Sales Analytics feature also provides meaningful insights into sales performance, inventory movement, and business trends, enabling owners to make informed decisions that support business growth and sustainability.

Employees also benefit from the implementation of the system through simplified workflows and automated processes. Routine tasks such as recording sales, updating inventory, managing customer information, and preparing reports become faster and more accurate, reducing workload while improving overall productivity. The centralized cloud database further enhances collaboration by ensuring that authorized users can access updated information anytime and from different locations.

From an academic perspective, this study contributes to the growing body of knowledge on cloud-based information systems, business process automation, and sales analytics. It serves as a valuable reference for future researchers, students, and software developers who wish to develop similar systems for other industries or improve the existing application. Furthermore, the study supports the digital transformation of local businesses by demonstrating that affordable and practical information systems can help small enterprises improve their competitiveness, operational efficiency, and long-term business performance.

5.6 Summary

Formatted: Font: 13 pt
Formatted: Line spacing: 1.5 lines
Formatted: Font: 10 pt
Formatted: Line spacing: Multiple 1.15 li

This chapter presented the overall conclusions and recommendations derived from the development and evaluation of H2Ops: Cloud-Based Operations Management System for Water Refilling Stations with Automated Sales Analytics. The study demonstrated that the proposed system successfully addressed the operational challenges experienced by water refilling station businesses through the integration of cloud computing technology and automated sales analytics. By replacing manual record-keeping with an automated and centralized management system, H2Ops improved the efficiency, accuracy, and reliability of daily business operations.

The findings revealed that the system effectively automated essential business processes, including sales transaction recording, inventory monitoring, customer information management, delivery scheduling, report generation, and business performance monitoring. The implementation of a centralized cloud database enabled real-time access to operational information, allowing authorized users to monitor business activities from different locations while ensuring data consistency and security. Furthermore, the Automated Sales Analytics module provided valuable insights into sales performance, product demand, and operational trends through dashboards and graphical reports, enabling business owners to make informed and data-driven decisions.

The researchers concluded that the objectives of the study were successfully achieved. The system was found to be functional, reliable, user-friendly, and capable of improving the operational efficiency of water refilling station businesses. Results from system testing and user evaluation also confirmed that the developed application met the

requirements and expectations of its intended users. The integration of cloud technology, automation, and analytical reporting demonstrated the potential of information systems to enhance business management and support the digital transformation of small and medium-sized enterprises.

Finally, this chapter presented several recommendations for future development and implementation. These include integrating additional features such as predictive analytics, mobile application support, online payment integration, enhanced security mechanisms, and notification services to further improve the functionality of the system. Future researchers are encouraged to continue exploring innovative technologies that can enhance business process automation and decision support within the water refilling station industry. Overall, this study serves as a valuable reference for future researchers, developers, and entrepreneurs by demonstrating how cloud-based operations management systems can improve business performance, streamline daily operations, and contribute to the long-term growth and sustainability of water refilling station businesses.

Formatted: Indent: First line: 0.25", Line spacing: Multiple 1.15 li